

Loading packages

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
##   filter, lag  
  
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)  
library(caret)
```

```
## Loading required package: lattice
```

```
library(randomForest)
```

```
## randomForest 4.7-1.1  
  
## Type rfNews() to see new features/changes/bug fixes.  
  
##  
## Attaching package: 'randomForest'  
  
## The following object is masked from 'package:ggplot2':  
##  
##   margin  
  
## The following object is masked from 'package:dplyr':  
##  
##   combine
```

```
library(pROC)
```

```
## Type 'citation("pROC")' for a citation.  
  
##  
## Attaching package: 'pROC'  
  
## The following objects are masked from 'package:stats':  
##  
##   cov, smooth, var
```

```
library(corrplot)
```

```
## corrplot 0.92 loaded
```

```
library(mltools)
library(rockchalk)
```

```
##
## Attaching package: 'rockchalk'
```

```
## The following object is masked from 'package:mltools':
##
##      skewness
```

```
## The following object is masked from 'package:dplyr':
##
##      summarize
```

Loading database

```
data2 <- read.csv("Area_01_Filtered_N0unit.csv", sep=";", header= TRUE)
```

```
summary(data2)
```

```
##      trngcv      bio14      slope      rugg500s
## Min.   : 8.713   Min.   :0.01801   Min.   : 0.2666   Min.   : 0.02999
## 1st Qu.: 9.242   1st Qu.:0.11977   1st Qu.: 1.9476   1st Qu.: 1.41555
## Median : 9.323   Median :0.16467   Median : 3.3912   Median : 2.87604
## Mean   : 9.343   Mean   :0.17485   Mean   : 4.8679   Mean   : 3.28088
## 3rd Qu.: 9.414   3rd Qu.:0.21515   3rd Qu.: 7.5301   3rd Qu.: 4.88870
## Max.   :10.125   Max.   :0.42712   Max.   :17.2472   Max.   :10.10635
##      pil_twim      pil_twicv      pil_topocv      pil_slps
## Min.   : 5.611   Min.   :0.03184   Min.   :0.00000   Min.   : 0.08751
## 1st Qu.: 7.941   1st Qu.:0.12586   1st Qu.:0.00000   1st Qu.: 1.09978
## Median : 8.988   Median :0.16938   Median :0.02253   Median : 3.08139
## Mean   : 8.994   Mean   :0.17768   Mean   :0.06768   Mean   : 4.12518
## 3rd Qu.:10.064   3rd Qu.:0.22529   3rd Qu.:0.13598   3rd Qu.: 5.90168
## Max.   :13.544   Max.   :0.40829   Max.   :0.35341   Max.   :19.05097
##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
## Min.   :0.1719   Min.   :0.01924   Min.   :0.0000   Min.   :0.00000
## 1st Qu.:0.3731   1st Qu.:0.14779   1st Qu.:0.0000   1st Qu.:0.00000
## Median :0.4711   Median :0.22176   Median :0.0000   Median :0.00000
## Mean   :0.5192   Mean   :0.24437   Mean   :0.3047   Mean   :0.05822
## 3rd Qu.:0.6152   3rd Qu.:0.34663   3rd Qu.:0.0000   3rd Qu.:0.00000
## Max.   :1.6098   Max.   :0.71240   Max.   :2.8579   Max.   :2.59788
##      minfertf      lf7rup      hstructn      geolrngaggn
## Min.   :2.000   Min.   :1.000   Min.   : 0.000   Min.   : 0.0115
## 1st Qu.:2.000   1st Qu.:2.000   1st Qu.: 1.043   1st Qu.: 2.5880
## Median :2.000   Median :3.000   Median : 1.043   Median : 2.5880
```

```
## Mean :2.222 Mean :3.192 Mean : 1.470 Mean :185.9895
## 3rd Qu.:2.000 3rd Qu.:4.000 3rd Qu.: 1.043 3rd Qu.:300.0000
## Max. :3.000 Max. :7.000 Max. :15.429 Max. :900.0000
## elevationm wr_unrn solpawhcn slopern
## Min. :443.9 Min. : 0.00 Min. : 82.0 Min. : 0.00162
## 1st Qu.:558.3 1st Qu.:17.86 1st Qu.: 96.0 1st Qu.: 0.01991
## Median :575.0 Median :17.86 Median : 96.0 Median : 0.05780
## Mean :583.5 Mean :17.25 Mean :100.4 Mean : 0.18529
## 3rd Qu.:611.7 3rd Qu.:17.86 3rd Qu.: 96.0 3rd Qu.: 0.16345
## Max. :726.1 Max. :72.00 Max. :157.0 Max. :10.06944
## MIN_AGE_MA HubDist month_collection ROCKTYPE1
## Min. :2454 Min. : 1.294 Length:634 Length:634
## 1st Qu.:2454 1st Qu.: 16.807 Class :character Class :character
## Median :2454 Median : 21.831 Mode :character Mode :character
## Mean :2473 Mean : 41.478
## 3rd Qu.:2494 3rd Qu.: 78.344
## Max. :2597 Max. :104.012
## FORMATION HubName true_troglofauna LATITUDE
## Length:634 Length:634 Min. :0.0000 Min. : -22.64
## Class :character Class :character 1st Qu.:0.0000 1st Qu.: -22.22
## Mode :character Mode :character Median :1.0000 Median : -22.16
## Mean :0.6625 Mean : -22.20
## 3rd Qu.:1.0000 3rd Qu.: -22.12
## Max. :1.0000 Max. : -22.04
## LONGITUDE
## Min. :117.0
## 1st Qu.:117.7
## Median :117.8
## Mean :117.7
## 3rd Qu.:117.9
## Max. :118.0
```

```
dim(data2)
```

```
## [1] 634 29
```

```
sapply(data2,class)
```

```
##      trngcv      bio14      slope      rugg500s
##      "numeric"      "numeric"      "numeric"      "numeric"
##      pil_twim      pil_twicv      pil_topocv      pil_slps
##      "numeric"      "numeric"      "numeric"      "numeric"
##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
##      "numeric"      "numeric"      "numeric"      "numeric"
##      minfertf      lf7rup      hstructn      geolrngaggn
##      "integer"      "integer"      "numeric"      "numeric"
##      elevationm      wr_unrn      solpawhcn      slopern
##      "numeric"      "numeric"      "integer"      "numeric"
##      MIN_AGE_MA      HubDist month_collection      ROCKTYPE1
##      "integer"      "numeric"      "character"      "character"
##      FORMATION      HubName true_troglofauna      LATITUDE
##      "character"      "character"      "integer"      "numeric"
##      LONGITUDE
##      "numeric"
```

```
#Converting "true_stygofauna", "true_troglofauna" to a factor
data2$true_troglofauna <- factor(data2$true_troglofauna)
```

```
#Converting categorical variables to factor
```

```
data2$HubName <- factor(data2$HubName)
data2$FORMATION <- factor(data2$FORMATION)
data2$ROCKTYPE1 <- factor(data2$ROCKTYPE1)
#data2$UNITNAME <- factor(data2$UNITNAME)
data2$month_collection <- factor(data2$month_collection)

sapply(data2,class)
```

```
##          trngcv          bio14          slope          rugg500s
##      "numeric"      "numeric"      "numeric"      "numeric"
##      pil_twim      pil_twicv      pil_topocv      pil_slps
##      "numeric"      "numeric"      "numeric"      "numeric"
##      pil_slpcv      pil_elr3cv      mrvbf          mrrtf
##      "numeric"      "numeric"      "numeric"      "numeric"
##      minfertf      lf7rup          hstructn      geolrngaggn
##      "integer"      "integer"      "numeric"      "numeric"
##      elevationm      wr_unrn      solpawhcn      slopern
##      "numeric"      "numeric"      "integer"      "numeric"
##      MIN_AGE_MA      HubDist month_collection      ROCKTYPE1
##      "integer"      "numeric"      "factor"      "factor"
##      FORMATION      HubName true_troglofauna      LATITUDE
##      "factor"      "factor"      "factor"      "numeric"
##      LONGITUDE
##      "numeric"
```

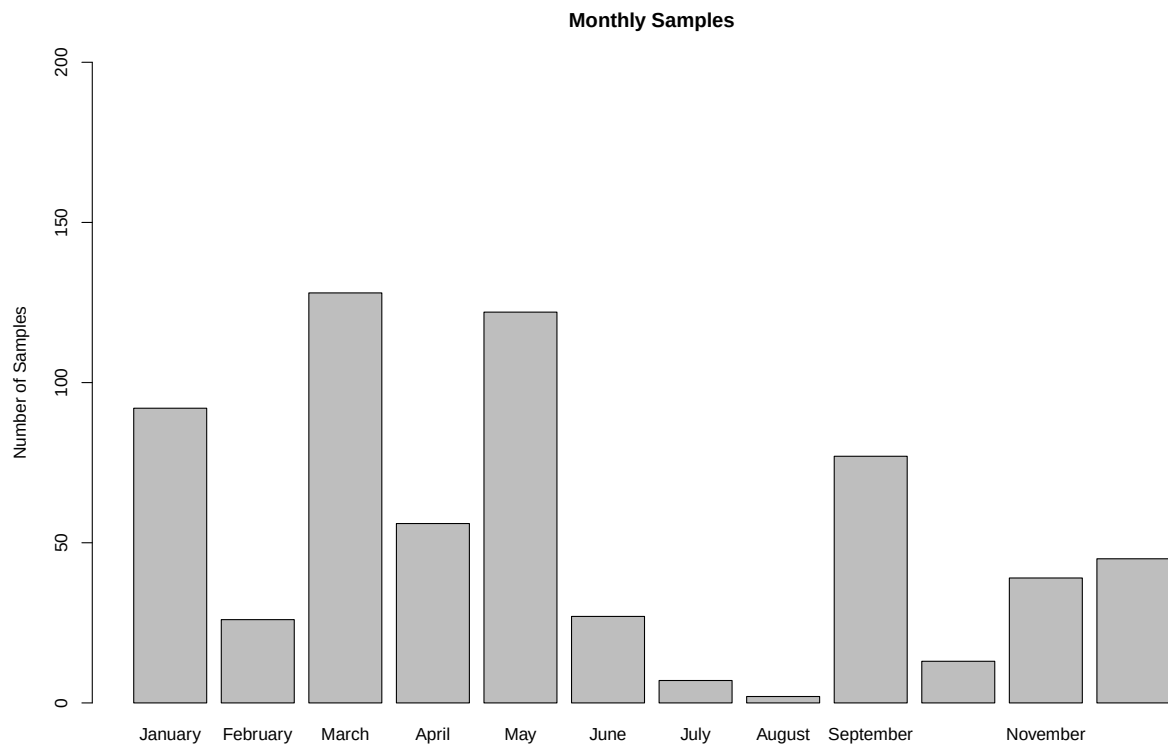
```
summary(data2$month_collection)
```

```
##      April      August      December      February      January      July      June      March
##          56          2          45          26          92          7          27         128
##      May      November      October      September
##         122          39          13          77
```

```
data2$month_collection<-factor(data2$month_collection, levels=c("January", "February", "March","April",
summary(data2$month_collection)
```

```
##      January      February      March      April      May      June      July      August
##          92          26          128          56          122          27          7          2
##      September      October      November      December
##          77          13          39          45
```

```
plot(data2$month_collection,main="Monthly Samples",ylim=c(0,200),ylab="Number of Samples")
```



```
m1<-which(data2$month_collection == "January");length(m1)
```

```
## [1] 92
```

```
m2<-which(data2$month_collection == "February");length(m2)
```

```
## [1] 26
```

```
m3<-which(data2$month_collection == "March");length(m3)
```

```
## [1] 128
```

```
m4<-which(data2$month_collection == "April");length(m4)
```

```
## [1] 56
```

```
m5<-which(data2$month_collection == "May");length(m5)
```

```
## [1] 122
```

```
m6<-which(data2$month_collection == "June");length(m6)
```

```
## [1] 27
```

```
m7<-which(data2$month_collection == "July");length(m7)
```

```
## [1] 7
```

```
m8<-which(data2$month_collection == "August");length(m8)
```

```
## [1] 2
```

```
m9<-which(data2$month_collection == "September");length(m9)
```

```
## [1] 77
```

```
m10<-which(data2$month_collection == "October");length(m10)
```

```
## [1] 13
```

```
m11<-which(data2$month_collection == "November");length(m11)
```

```
## [1] 39
```

```
m12<-which(data2$month_collection == "December");length(m12)
```

```
## [1] 45
```

```
tm<-c(m1,m2,m3,m4,m5,m6,m7,m8,m9,m10,m11,m12);length(tm)
```

```
## [1] 634
```

```
ltm<-c(length(m1),length(m2),length(m3),length(m4),length(m5),length(m6),length(m7),length(m8),length(m9),length(m10),length(m11),length(m12))
```

```
## [1] 42
```

```
set.seed(78945)
```

```
sm1<-sample(m1, 50, replace = FALSE);length(sm1);head(sm1)
```

```
## [1] 50
```

```
## [1] 110 102 27 370 86 377
```

```
sm3<-sample(m3, 50, replace = FALSE);length(sm3);head(sm3)
```

```
## [1] 50
```

```
## [1] 261 40 372 503 279 578
```

```
sm4<-sample(m4, 50, replace = FALSE);length(sm4);head(sm4)
```

```
## [1] 50
```

```
## [1] 155 449 156 135 38 57
```

```
sm5<-sample(m5, 50, replace = FALSE);length(sm5);head(sm5)
```

```
## [1] 50
```

```
## [1] 438 141 127 472 126 298
```

```
sm9<-sample(m9, 50, replace = FALSE);length(sm9);head(sm9)
```

```
## [1] 50
```

```
## [1] 321 195 308 355 339 243
```

```
utm<-c(sm1,m2,sm3,sm4,sm5,m6,m7,m8,sm9,m10,m11,m12);length(utm)
```

```
## [1] 409
```

```
data1<-data2[utm,];dim(data1)
```

```
## [1] 409 29
```

```
summary(data1)
```

```
##      trngcv      bio14      slope      rugg500s
##  Min.   : 8.713   Min.   :0.01801   Min.   : 0.2666   Min.   : 0.02999
##  1st Qu.: 9.234   1st Qu.:0.12241   1st Qu.: 1.9117   1st Qu.: 1.33417
##  Median : 9.317   Median :0.17267   Median : 3.3563   Median : 2.77642
##  Mean   : 9.324   Mean   :0.18034   Mean   : 4.8591   Mean   : 3.24606
##  3rd Qu.: 9.410   3rd Qu.:0.22232   3rd Qu.: 7.1721   3rd Qu.: 4.97943
##  Max.   :10.026   Max.   :0.42288   Max.   :17.2472   Max.   :10.10635
##
##      pil_twim      pil_twicv      pil_topocv      pil_slps
##  Min.   : 5.749   Min.   :0.03184   Min.   :0.00000   Min.   : 0.08751
##  1st Qu.: 7.965   1st Qu.:0.12336   1st Qu.:0.00000   1st Qu.: 1.07585
##  Median : 9.006   Median :0.16938   Median :0.01735   Median : 2.73185
##  Mean   : 9.007   Mean   :0.17782   Mean   :0.06433   Mean   : 3.96138
##  3rd Qu.:10.122   3rd Qu.:0.23025   3rd Qu.:0.13316   3rd Qu.: 5.71830
##  Max.   :13.544   Max.   :0.40758   Max.   :0.35341   Max.   :15.25064
##
##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
##  Min.   :0.1719   Min.   :0.03157   Min.   :0.000   Min.   :0.00000
##  1st Qu.:0.3758   1st Qu.:0.13822   1st Qu.:0.000   1st Qu.:0.00000
##  Median :0.4696   Median :0.22558   Median :0.000   Median :0.00000
```

```

## Mean :0.5165 Mean :0.24364 Mean :0.306 Mean :0.06968
## 3rd Qu.:0.6153 3rd Qu.:0.33502 3rd Qu.:0.000 3rd Qu.:0.00000
## Max. :1.6098 Max. :0.71240 Max. :2.858 Max. :2.59788
##
## minfertf lf7rup hstructn geolrngaggn
## Min. :2.000 Min. :1.000 Min. : 0.000 Min. : 0.0115
## 1st Qu.:2.000 1st Qu.:2.000 1st Qu.: 1.043 1st Qu.: 2.5880
## Median :2.000 Median :3.000 Median : 1.043 Median : 2.5880
## Mean :2.208 Mean :3.227 Mean : 1.647 Mean :171.7449
## 3rd Qu.:2.000 3rd Qu.:4.000 3rd Qu.: 1.043 3rd Qu.:300.0000
## Max. :3.000 Max. :7.000 Max. :15.429 Max. :900.0000
##
## elevationm wr_unrn solpawhcn slopern
## Min. :443.9 Min. : 0.00 Min. : 82.0 Min. : 0.00162
## 1st Qu.:559.7 1st Qu.:17.86 1st Qu.: 96.0 1st Qu.: 0.02093
## Median :577.0 Median :17.86 Median : 96.0 Median : 0.06396
## Mean :585.3 Mean :16.78 Mean :101.3 Mean : 0.21262
## 3rd Qu.:614.1 3rd Qu.:17.86 3rd Qu.: 96.0 3rd Qu.: 0.16934
## Max. :726.1 Max. :72.00 Max. :157.0 Max. :10.06944
##
## MIN_AGE_MA HubDist month_collection
## Min. :2454 Min. : 1.294 January : 50
## 1st Qu.:2454 1st Qu.: 16.613 March : 50
## Median :2454 Median : 21.793 April : 50
## Mean :2477 Mean : 39.334 May : 50
## 3rd Qu.:2494 3rd Qu.: 76.847 September: 50
## Max. :2597 Max. :104.012 December : 45
## (Other) :114
## ROCKTYPE1
## sedimentary carbonate : 26
## sedimentary other chemical or biochemical:308
## sedimentary siliciclastic : 75
##
##
##
## FORMATION
## Brockman Iron Formation :273
## Marra Mamba Iron Formation : 35
## Mount McRae Shale and Mount Sylvia Formation: 75
## Wittenoom Formation : 26
##
##
##
## HubName
## anticline, exposed :235
## exposed : 21
## strike-slip, exposed, showing relative dextral displacement: 10
## syncline, exposed :143
##
##
## true_troglofauna LATITUDE LONGITUDE
## 0:147 Min. :-22.64 Min. :117.0

```



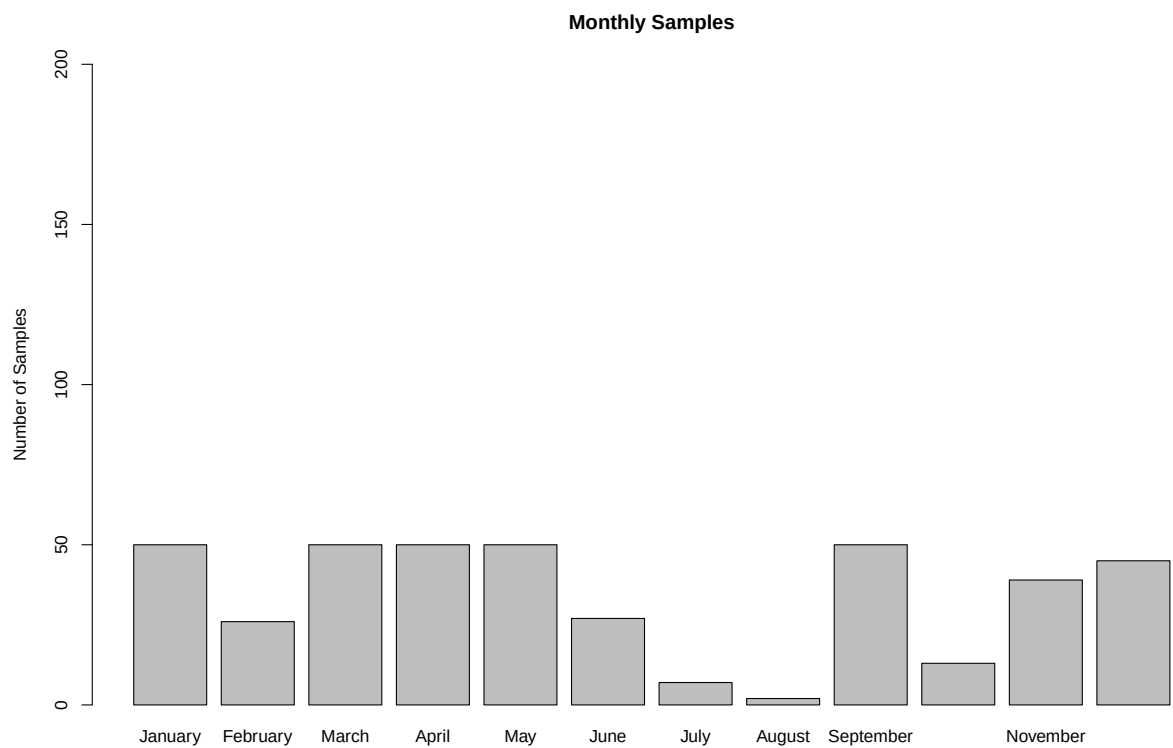
```
## 1:262      1st Qu.: -22.23  1st Qu.: 117.7
##          Median : -22.16  Median : 117.8
##          Mean   : -22.21  Mean   : 117.7
##          3rd Qu.: -22.12  3rd Qu.: 117.9
##          Max.    : -22.04  Max.    : 118.0
##
```

```
data2<-data1
```

```
summary(data2$month_collection)
```

```
##   January  February    March    April    May    June    July    August
##         50         26         50         50         50         27         7         2
## September  October  November  December
##         50         13         39         45
```

```
plot(data2$month_collection,main="Monthly Samples",ylim=c(0,200),ylab="Number of Samples")
```



```
levels(data2$true_troglofauna);levels(data2$HubName);levels(data2$FORMATION);levels(data2$ROCKTYPE1);le
```

```
## [1] "0" "1"
```

```
## [1] "anticline, exposed"
```

```
## [2] "exposed"
```

```
## [3] "strike-slip, exposed, showing relative dextral displacement"
## [4] "syncline, exposed"

## [1] "Brockman Iron Formation"
## [2] "Marra Mamba Iron Formation"
## [3] "Mount McRae Shale and Mount Sylvia Formation"
## [4] "Wittenoom Formation"

## [1] "sedimentary carbonate"
## [2] "sedimentary other chemical or biochemical"
## [3] "sedimentary siliciclastic"

## [1] "January" "February" "March" "April" "May" "June"
## [7] "July" "August" "September" "October" "November" "December"
```

```
summary(data2)
```

```
##      trngcv      bio14      slope      rugg500s
## Min.   : 8.713   Min.   :0.01801   Min.   : 0.2666   Min.   : 0.02999
## 1st Qu.: 9.234   1st Qu.:0.12241   1st Qu.: 1.9117   1st Qu.: 1.33417
## Median : 9.317   Median :0.17267   Median : 3.3563   Median : 2.77642
## Mean   : 9.324   Mean   :0.18034   Mean   : 4.8591   Mean   : 3.24606
## 3rd Qu.: 9.410   3rd Qu.:0.22232   3rd Qu.: 7.1721   3rd Qu.: 4.97943
## Max.   :10.026   Max.   :0.42288   Max.   :17.2472   Max.   :10.10635
##
##      pil_twim      pil_twicv      pil_topocv      pil_slps
## Min.   : 5.749   Min.   :0.03184   Min.   :0.00000   Min.   : 0.08751
## 1st Qu.: 7.965   1st Qu.:0.12336   1st Qu.:0.00000   1st Qu.: 1.07585
## Median : 9.006   Median :0.16938   Median :0.01735   Median : 2.73185
## Mean   : 9.007   Mean   :0.17782   Mean   :0.06433   Mean   : 3.96138
## 3rd Qu.:10.122   3rd Qu.:0.23025   3rd Qu.:0.13316   3rd Qu.: 5.71830
## Max.   :13.544   Max.   :0.40758   Max.   :0.35341   Max.   :15.25064
##
##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
## Min.   :0.1719   Min.   :0.03157   Min.   :0.000   Min.   :0.00000
## 1st Qu.:0.3758   1st Qu.:0.13822   1st Qu.:0.000   1st Qu.:0.00000
## Median :0.4696   Median :0.22558   Median :0.000   Median :0.00000
## Mean   :0.5165   Mean   :0.24364   Mean   :0.306   Mean   :0.06968
## 3rd Qu.:0.6153   3rd Qu.:0.33502   3rd Qu.:0.000   3rd Qu.:0.00000
## Max.   :1.6098   Max.   :0.71240   Max.   :2.858   Max.   :2.59788
##
##      minfertf      lf7rup      hstructn      geolrngaggn
## Min.   :2.000   Min.   :1.000   Min.   : 0.000   Min.   : 0.0115
## 1st Qu.:2.000   1st Qu.:2.000   1st Qu.: 1.043   1st Qu.: 2.5880
## Median :2.000   Median :3.000   Median : 1.043   Median : 2.5880
## Mean   :2.208   Mean   :3.227   Mean   : 1.647   Mean   :171.7449
## 3rd Qu.:2.000   3rd Qu.:4.000   3rd Qu.: 1.043   3rd Qu.:300.0000
## Max.   :3.000   Max.   :7.000   Max.   :15.429   Max.   :900.0000
##
##      elevationm      wr_unrn      solpawhcn      slopern
## Min.   :443.9   Min.   : 0.00   Min.   : 82.0   Min.   : 0.00162
## 1st Qu.:559.7   1st Qu.:17.86   1st Qu.: 96.0   1st Qu.: 0.02093
## Median :577.0   Median :17.86   Median : 96.0   Median : 0.06396
```

```

## Mean :585.3 Mean :16.78 Mean :101.3 Mean : 0.21262
## 3rd Qu.:614.1 3rd Qu.:17.86 3rd Qu.: 96.0 3rd Qu.: 0.16934
## Max. :726.1 Max. :72.00 Max. :157.0 Max. :10.06944
##
## MIN_AGE_MA HubDist month_collection
## Min. :2454 Min. : 1.294 January : 50
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## Median :2454 Median : 21.793 April : 50
## Mean :2477 Mean : 39.334 May : 50
## 3rd Qu.:2494 3rd Qu.: 76.847 September: 50
## Max. :2597 Max. :104.012 December : 45
## (Other) :114
## ROCKTYPE1
## sedimentary carbonate : 26
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##
##
## FORMATION
## Brockman Iron Formation :273
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##
##
##
## HubName
## anticline, exposed :235
## exposed : 21
## strike-slip, exposed, showing relative dextral displacement: 10
## syncline, exposed :143
##
##
## true_troglofauna LATITUDE LONGITUDE
## 0:147 Min. :-22.64 Min. :117.0
## 1:262 1st Qu.: -22.23 1st Qu.:117.7
## Median :-22.16 Median :117.8
## Mean :-22.21 Mean :117.7
## 3rd Qu.: -22.12 3rd Qu.:117.9
## Max. :-22.04 Max. :118.0
##
##

```

```
dim(data2)
```

```
## [1] 409 29
```

```
instanceconvert <- colnames(data2[, -c((ncol(data2) - 7):ncol(data2))])
instanceconvert
```

```
## [1] "trngcv" "bio14" "slope" "rugg500s" "pil_twim"
```

```
## [6] "pil_twicv" "pil_topocv" "pil_slps" "pil_slpcv" "pil_elr3cv"
## [11] "mrvmf" "mrrtf" "minfertf" "lf7rup" "hstructn"
## [16] "geolrngaggn" "elevationm" "wr_unrn" "solpawhcn" "slopern"
## [21] "MIN_AGE_MA"
```

```
for (i in instanceconvert)
{
  data2[[i]] <- as.numeric(data2[[i]])
}
```

```
sapply(data2,class)
```

```
##          trngcv          bio14          slope          rugg500s
##          "numeric"          "numeric"          "numeric"          "numeric"
##          pil_twim          pil_twicv          pil_topocv          pil_slps
##          "numeric"          "numeric"          "numeric"          "numeric"
##          pil_slpcv          pil_elr3cv          mrvmf          mrrtf
##          "numeric"          "numeric"          "numeric"          "numeric"
##          minfertf          lf7rup          hstructn          geolrngaggn
##          "numeric"          "numeric"          "numeric"          "numeric"
##          elevationm          wr_unrn          solpawhcn          slopern
##          "numeric"          "numeric"          "numeric"          "numeric"
##          MIN_AGE_MA          HubDist month_collection          ROCKTYPE1
##          "numeric"          "numeric"          "factor"          "factor"
##          FORMATION          HubName true_troglofauna          LATITUDE
##          "factor"          "factor"          "factor"          "numeric"
##          LONGITUDE
##          "numeric"
```

```
names(data2)
```

```
## [1] "trngcv"          "bio14"          "slope"          "rugg500s"
## [5] "pil_twim"          "pil_twicv"          "pil_topocv"          "pil_slps"
## [9] "pil_slpcv"          "pil_elr3cv"          "mrvmf"          "mrrtf"
## [13] "minfertf"          "lf7rup"          "hstructn"          "geolrngaggn"
## [17] "elevationm"          "wr_unrn"          "solpawhcn"          "slopern"
## [21] "MIN_AGE_MA"          "HubDist"          "month_collection" "ROCKTYPE1"
## [25] "FORMATION"          "HubName"          "true_troglofauna" "LATITUDE"
## [29] "LONGITUDE"
```

```
sapply(data2,class)
```

```
##          trngcv          bio14          slope          rugg500s
##          "numeric"          "numeric"          "numeric"          "numeric"
##          pil_twim          pil_twicv          pil_topocv          pil_slps
##          "numeric"          "numeric"          "numeric"          "numeric"
##          pil_slpcv          pil_elr3cv          mrvmf          mrrtf
##          "numeric"          "numeric"          "numeric"          "numeric"
##          minfertf          lf7rup          hstructn          geolrngaggn
##          "numeric"          "numeric"          "numeric"          "numeric"
##          elevationm          wr_unrn          solpawhcn          slopern
##          "numeric"          "numeric"          "numeric"          "numeric"
```

```
##      MIN_AGE_MA      HubDist month_collection      ROCKTYPE1
##      "numeric"      "numeric"      "factor"      "factor"
##      FORMATION      HubName true_troglofauna      LATITUDE
##      "factor"      "factor"      "factor"      "numeric"
##      LONGITUDE
##      "numeric"
```

```
levels(data2$true_troglofauna);levels(data2$HubName);levels(data2$FORMATION);levels(data2$ROCKTYPE1);le
```

```
## [1] "0" "1"
```

```
## [1] "anticline, exposed"
```

```
## [2] "exposed"
```

```
## [3] "strike-slip, exposed, showing relative dextral displacement"
```

```
## [4] "syncline, exposed"
```

```
## [1] "Brockman Iron Formation"
```

```
## [2] "Marra Mamba Iron Formation"
```

```
## [3] "Mount McRae Shale and Mount Sylvia Formation"
```

```
## [4] "Wittenoom Formation"
```

```
## [1] "sedimentary carbonate"
```

```
## [2] "sedimentary other chemical or biochemical"
```

```
## [3] "sedimentary siliciclastic"
```

```
## [1] "January" "February" "March" "April" "May" "June"
```

```
## [7] "July" "August" "September" "October" "November" "December"
```

```
# Get column names with their indexes
```

```
column_indexes <- seq_along(names(data2))
```

```
# Display the column names and their corresponding indexes
```

```
column_indexes_named <- setNames(column_indexes, names(data2))
```

```
print(column_indexes_named)
```

```
##      trngcv      bio14      slope      rugg500s
##      1      2      3      4
##      pil_twim      pil_twicv      pil_topocv      pil_slps
##      5      6      7      8
##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
##      9      10      11      12
##      minfertf      lf7rup      hstructn      geolrngaggn
##      13      14      15      16
##      elevationm      wr_unrn      solpawhcn      slopern
##      17      18      19      20
##      MIN_AGE_MA      HubDist month_collection      ROCKTYPE1
##      21      22      23      24
##      FORMATION      HubName true_troglofauna      LATITUDE
##      25      26      27      28
##      LONGITUDE
##      29
```

```
sapply(data2,class)
```

```
##          trngcv          bio14          slope          rugg500s
##      "numeric"      "numeric"      "numeric"      "numeric"
##      pil_twim      pil_twicv      pil_topocv      pil_slps
##      "numeric"      "numeric"      "numeric"      "numeric"
##      pil_slpcv      pil_elr3cv      mrvbf          mrrtf
##      "numeric"      "numeric"      "numeric"      "numeric"
##      minfertf      lf7rup          hstructn      geolrngaggn
##      "numeric"      "numeric"      "numeric"      "numeric"
##      elevationm      wr_unrn      solpawhcn      slopern
##      "numeric"      "numeric"      "numeric"      "numeric"
##      MIN_AGE_MA      HubDist month_collection      ROCKTYPE1
##      "numeric"      "numeric"      "factor"      "factor"
##      FORMATION      HubName true_troglofauna      LATITUDE
##      "factor"      "factor"      "factor"      "numeric"
##      LONGITUDE
##      "numeric"
```

Convert to numeric

```
instanceconvert <- colnames(data2[, -c((ncol(data2) - 7):ncol(data2))])

for (i in instanceconvert)
{
  data2[[i]] <- as.numeric(data2[[i]])
}

sapply(data2,class)
```

```
##          trngcv          bio14          slope          rugg500s
##      "numeric"      "numeric"      "numeric"      "numeric"
##      pil_twim      pil_twicv      pil_topocv      pil_slps
##      "numeric"      "numeric"      "numeric"      "numeric"
##      pil_slpcv      pil_elr3cv      mrvbf          mrrtf
##      "numeric"      "numeric"      "numeric"      "numeric"
##      minfertf      lf7rup          hstructn      geolrngaggn
##      "numeric"      "numeric"      "numeric"      "numeric"
##      elevationm      wr_unrn      solpawhcn      slopern
##      "numeric"      "numeric"      "numeric"      "numeric"
##      MIN_AGE_MA      HubDist month_collection      ROCKTYPE1
##      "numeric"      "numeric"      "factor"      "factor"
##      FORMATION      HubName true_troglofauna      LATITUDE
##      "factor"      "factor"      "factor"      "numeric"
##      LONGITUDE
##      "numeric"
```

```
str(data2)
```

```
## 'data.frame':    409 obs. of  29 variables:
## $ trngcv      : num  9.89 9.47 9.44 9.32 9.42 ...
## $ bio14       : num  0.254 0.134 0.125 0.139 0.12 ...
## $ slope       : num  9.74 11.57 13.37 1.15 10.27 ...
## $ rugg500s    : num  5.46 6.52 3.05 1.47 3.42 ...
## $ pil_twim     : num  5.75 8.43 9.24 9.98 6.86 ...
## $ pil_twicv    : num  0.0979 0.1694 0.3028 0.1597 0.1373 ...
## $ pil_topocv   : num  0.1106 0.1269 0.1116 0 0.0426 ...
## $ pil_slps     : num  10.968 3.564 4.554 0.951 14.71 ...
## $ pil_slpcv    : num  0.323 0.289 0.454 0.615 0.59 ...
## $ pil_elr3cv   : num  0.2294 0.2977 0.1921 0.36 0.0948 ...
## $ mrvbf        : num  0 0 0 0 0 ...
## $ mrrtf        : num  0 0 0 0.661 0 ...
## $ minfertf     : num  2 2 2 3 2 3 2 2 2 ...
## $ lf7rup       : num  2 2 2 6 3 4 4 1 4 3 ...
## $ hstructn     : num  1.04 1.04 1.04 1.04 1.04 ...
## $ geolrngaggn  : num  300 2.588 2.588 0.0115 2.588 ...
## $ elevationm   : num  726 543 543 562 551 ...
## $ wr_unrn      : num  17.9 17.9 17.9 17.9 17.9 ...
## $ solpawhcn    : num  96 96 96 96 96 96 96 96 96 ...
## $ slopern      : num  0.00341 0.00742 0.08237 0.19806 0.01298 ...
## $ MIN_AGE_MA   : num  2454 2494 2494 2454 2454 ...
## $ HubDist      : num  76.1 21.5 21.7 15.3 21.6 ...
## $ month_collection: Factor w/ 12 levels "January","February",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ ROCKTYPE1    : Factor w/ 3 levels "sedimentary carbonate",...: 2 3 3 2 2 2 3 2 3 2 ...
## $ FORMATION     : Factor w/ 4 levels "Brockman Iron Formation",...: 1 3 3 1 1 1 3 1 3 1 ...
## $ HubName       : Factor w/ 4 levels "anticline, exposed",...: 4 1 1 1 1 1 1 1 1 1 ...
## $ true_troglofauna: Factor w/ 2 levels "0","1": 1 1 2 1 2 1 2 1 2 2 ...
## $ LATITUDE      : num  -22.2 -22.1 -22.1 -22.1 -22.1 ...
## $ LONGITUDE     : num  118 118 118 118 118 ...
```

```
dim(data2)
```

```
## [1] 409 29
```

```
num_rows <- 409

# Randomly select row indices
selected_rows <- sample(nrow(data2), num_rows)

# Create a new dataframe with randomly selected rows
data2 <- data2[selected_rows, ]
```

```
dim(data2)
```

```
## [1] 409 29
```

```
tg <- which(data2$true_troglofauna == 1)
tgg <- data2[tg, "true_troglofauna"]
tgg <- length(tgg)

fg <- which(data2$true_troglofauna == 0)
```

```

fgg <- data2[fg, "true_troglofauna"]
fgg <- length(fgg)

# Data for pie chart
plot_num_ident <- data.frame(Absence = 100 * fgg / (tgg + fgg),
                             Presence = 100 * tgg / (tgg + fgg))

# Convert data to a numeric vector
pie_data <- as.numeric(plot_num_ident)

# Labels for the pie chart (rounded to 1 decimal place)
labels <- c(paste0("Absent\n", round(pie_data[1], 1), "%"),
            paste0("Present\n", round(pie_data[2], 1), "%"))

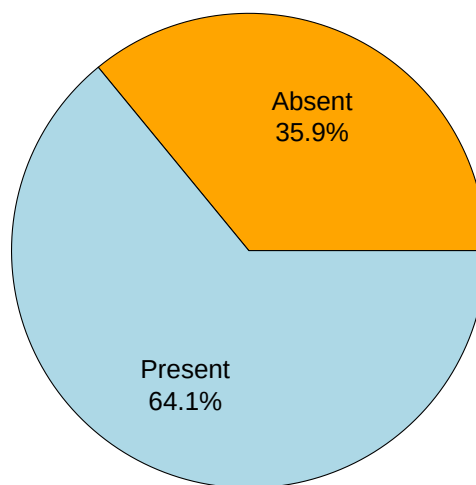
# Create pie chart
pie(pie_data,
    labels = NA, # Remove default labels
    main = "AREA 1",
    col = c("orange", "lightblue"))

# Calculate the midpoints of each pie slice for label positioning
pie_slices <- cumsum(pie_data) - pie_data / 2

# Add the labels inside the pie chart
text(x = 0.5 * cos(2 * pi * pie_slices / sum(pie_data)),
     y = 0.5 * sin(2 * pi * pie_slices / sum(pie_data)),
     labels = labels, cex = 1.5)

```


AREA 1



```
selectcol_Data2 <- data2[, -c((ncol(data2) - 6):ncol(data2))]
dim(selectcol_Data2)
```

```
## [1] 409 22
```

```
names(selectcol_Data2)
```

```
## [1] "trngcv"      "bio14"      "slope"      "rugg500s"   "pil_twim"
## [6] "pil_twicv"   "pil_topocv" "pil_slps"   "pil_slpcv"  "pil_elr3cv"
## [11] "mrvbf"      "mrrtf"      "minfertf"   "lf7rup"     "hstructn"
## [16] "geolrngaggn" "elevationm" "wr_unrn"    "solpawhcn"  "slopern"
## [21] "MIN_AGE_MA" "HubDist"
```

```
correlations<-cor(selectcol_Data2)
correlations
```

```
##           trngcv      bio14      slope      rugg500s      pil_twim
## trngcv      1.00000000 -0.14438838  0.69176889  0.477926942 -0.279056003
## bio14      -0.14438838  1.00000000  0.07029883 -0.128492177 -0.226561962
## slope       0.69176889  0.07029883  1.00000000  0.517511499 -0.387364499
## rugg500s    0.47792694 -0.12849218  0.51751150  1.000000000 -0.138864300
## pil_twim   -0.27905600 -0.22656196 -0.38736450 -0.138864300  1.000000000
## pil_twicv   0.01702041 -0.29376592 -0.09729433  0.005626062  0.367094043
## pil_topocv  0.05959847 -0.10025211  0.04272442  0.288421473  0.035020538
## pil_slps    0.42525435 -0.02594281  0.47393877  0.385615519 -0.612390330
```

## pil_slpcv	-0.03527169	-0.24291583	-0.10400086	0.185707019	0.252591021
## pil_elr3cv	-0.02048230	-0.23334121	-0.09470449	0.449626784	0.499634174
## mrvbf	-0.26619272	-0.03411611	-0.44723384	-0.338185583	0.334840763
## mrrtf	-0.21269176	0.14716955	-0.22818463	-0.150700825	-0.007752769
## minfertf	-0.20113479	-0.33116358	-0.33421583	-0.267348143	0.295346069
## lf7rup	-0.36740253	-0.16925245	-0.48975316	-0.492283947	0.281602595
## hstructn	-0.44239447	0.32588806	-0.20271250	-0.269254064	0.053314883
## geolrngaggn	0.23224359	0.16859387	0.17634243	0.132026204	-0.431930571
## elevationm	0.32827534	0.45904523	0.28126622	0.035784792	-0.297688074
## wr_unrn	-0.06234149	-0.03941867	0.05620525	0.073818209	-0.206746105
## solpawhcn	-0.14671433	0.23912653	-0.17955216	-0.202149466	0.178602521
## slopern	-0.18198480	-0.02970640	-0.15398297	-0.157829835	0.264328275
## MIN_AGE_MA	-0.46854175	0.41961974	-0.18098149	-0.332963519	-0.133647611
## HubDist	0.25900610	-0.18161858	0.19846532	0.154051977	0.030718072
##	pil_twicv	pil_topocv	pil_slps	pil_slpcv	pil_elr3cv
## trngcv	0.017020409	0.05959847	0.42525435	-0.035271690	-0.02048230
## bio14	-0.293765922	-0.10025211	-0.02594281	-0.242915826	-0.23334121
## slope	-0.097294332	0.04272442	0.47393877	-0.104000860	-0.09470449
## rugg500s	0.005626062	0.28842147	0.38561552	0.185707019	0.44962678
## pil_twim	0.367094043	0.03502054	-0.61239033	0.252591021	0.49963417
## pil_twicv	1.000000000	0.13935933	-0.04876311	0.280298038	0.25864734
## pil_topocv	0.139359332	1.000000000	0.13690929	0.212719258	0.21220279
## pil_slps	-0.048763108	0.13690929	1.000000000	0.316376132	-0.29073744
## pil_slpcv	0.280298038	0.21271926	0.31637613	1.000000000	0.25064028
## pil_elr3cv	0.258647345	0.21220279	-0.29073744	0.250640275	1.000000000
## mrvbf	0.143789374	-0.19705951	-0.34099450	0.029400935	0.01624646
## mrrtf	-0.154008477	0.11667742	-0.10285428	-0.065781453	-0.04792797
## minfertf	0.213223063	-0.18208188	-0.25711115	0.033170964	0.12410535
## lf7rup	0.127503291	-0.19397234	-0.38134680	-0.010497415	-0.03479760
## hstructn	-0.198425125	-0.14170688	-0.15945066	-0.096202710	-0.14666667
## geolrngaggn	-0.185108425	0.05077707	0.36090860	-0.140684002	-0.21925198
## elevationm	-0.308216732	-0.06616776	0.21818619	-0.117813496	-0.22561452
## wr_unrn	-0.106794102	-0.07493941	0.11352341	0.002507419	-0.02330359
## solpawhcn	-0.012078107	0.05138687	-0.22465768	-0.047500788	-0.02096902
## slopern	0.031603804	-0.13003649	-0.16139194	0.104391301	0.01645636
## MIN_AGE_MA	-0.195963275	-0.12402301	-0.14535780	-0.114125936	-0.25250500
## HubDist	0.039375610	0.10601700	0.17613904	0.266842636	0.12772916
##	mrvbf	mrrtf	minfertf	lf7rup	hstructn
## trngcv	-0.26619272	-0.212691756	-0.20113479	-0.367402529	-0.44239447
## bio14	-0.03411611	0.147169552	-0.33116358	-0.169252446	0.32588806
## slope	-0.44723384	-0.228184626	-0.33421583	-0.489753156	-0.20271250
## rugg500s	-0.33818558	-0.150700825	-0.26734814	-0.492283947	-0.26925406
## pil_twim	0.33484076	-0.007752769	0.29534607	0.281602595	0.05331488
## pil_twicv	0.14378937	-0.154008477	0.21322306	0.127503291	-0.19842512
## pil_topocv	-0.19705951	0.116677416	-0.18208188	-0.193972339	-0.14170688
## pil_slps	-0.34099450	-0.102854276	-0.25711115	-0.381346802	-0.15945066
## pil_slpcv	0.02940093	-0.065781453	0.03317096	-0.010497415	-0.09620271
## pil_elr3cv	0.01624646	-0.047927972	0.12410535	-0.034797599	-0.14666667
## mrvbf	1.00000000	-0.065557280	0.32959668	0.527999249	0.20005240
## mrrtf	-0.06555728	1.000000000	0.02801221	0.031133482	0.29308847
## minfertf	0.32959668	0.028012213	1.000000000	0.444817211	0.03598896
## lf7rup	0.52799925	0.031133482	0.44481721	1.000000000	0.07011658
## hstructn	0.20005240	0.293088472	0.03598896	0.070116581	1.00000000
## geolrngaggn	-0.21208074	0.029608944	-0.23450476	-0.197103865	-0.03969216

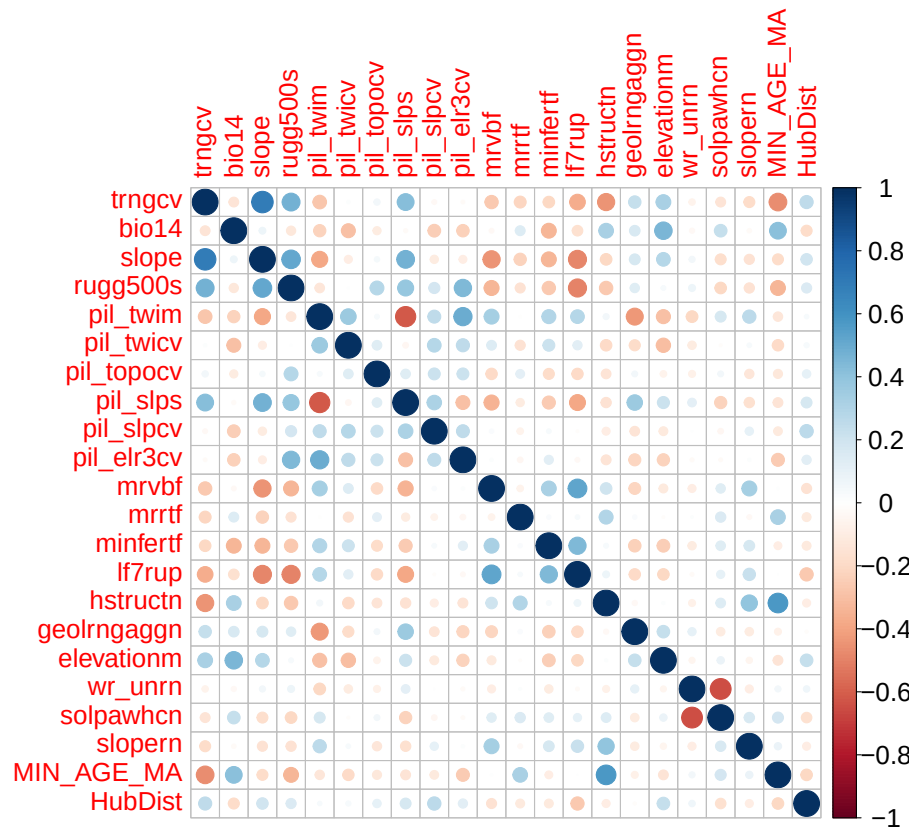
```

## elevationm -0.11948245 -0.010279115 -0.24459358 -0.200140640 0.01344205
## wr_unrn -0.09664253 0.010595057 -0.10583662 -0.022543060 -0.07426712
## solpawhcn 0.13166310 0.159540420 0.13802186 0.101190355 0.14136903
## slopern 0.33641174 -0.034854407 0.17182369 0.228577910 0.39431854
## MIN_AGE_MA 0.00730014 0.323227324 -0.09194556 0.001095106 0.57904539
## HubDist -0.15219624 -0.119165302 -0.11781342 -0.263770024 -0.09757715
##
## geolrngaggn elevationm wr_unrn solpawhcn slopern
## trngcv 0.23224359 0.32827534 -0.062341490 -0.14671433 -0.18198480
## bio14 0.16859387 0.45904523 -0.039418673 0.23912653 -0.02970640
## slope 0.17634243 0.28126622 0.056205249 -0.17955216 -0.15398297
## rugg500s 0.13202620 0.03578479 0.073818209 -0.20214947 -0.15782983
## pil_twim -0.43193057 -0.29768807 -0.206746105 0.17860252 0.26432828
## pil_twicv -0.18510842 -0.30821673 -0.106794102 -0.01207811 0.03160380
## pil_topocv 0.05077707 -0.06616776 -0.074939411 0.05138687 -0.13003649
## pil_slps 0.36090860 0.21818619 0.113523411 -0.22465768 -0.16139194
## pil_slpcv -0.14068400 -0.11781350 0.002507419 -0.04750079 0.10439130
## pil_elr3cv -0.21925198 -0.22561452 -0.023303594 -0.02096902 0.01645636
## mrvbf -0.21208074 -0.11948245 -0.096642533 0.13166310 0.33641174
## mrrtf 0.02960894 -0.01027911 0.010595057 0.15954042 -0.03485441
## minfertf -0.23450476 -0.24459358 -0.105836620 0.13802186 0.17182369
## lf7rup -0.19710387 -0.20014064 -0.022543060 0.10119036 0.22857791
## hstructn -0.03969216 0.01344205 -0.074267122 0.14136903 0.39431854
## geolrngaggn 1.00000000 0.23599976 0.100423279 -0.10953559 -0.10709799
## elevationm 0.23599976 1.00000000 -0.054646005 0.08226795 -0.05950478
## wr_unrn 0.10042328 -0.05464600 1.000000000 -0.64879855 -0.09753958
## solpawhcn -0.10953559 0.08226795 -0.648798550 1.00000000 0.16282807
## slopern -0.10709799 -0.05950478 -0.097539581 0.16282807 1.00000000
## MIN_AGE_MA -0.07323784 -0.14465058 0.055594693 0.18305242 0.08046612
## HubDist -0.01398595 0.23769455 0.062436255 -0.16628773 -0.09694540
##
## MIN_AGE_MA HubDist
## trngcv -0.468541755 0.25900610
## bio14 0.419619739 -0.18161858
## slope -0.180981491 0.19846532
## rugg500s -0.332963519 0.15405198
## pil_twim -0.133647611 0.03071807
## pil_twicv -0.195963275 0.03937561
## pil_topocv -0.124023011 0.10601700
## pil_slps -0.145357804 0.17613904
## pil_slpcv -0.114125936 0.26684264
## pil_elr3cv -0.252505000 0.12772916
## mrvbf 0.007300140 -0.15219624
## mrrtf 0.323227324 -0.11916530
## minfertf -0.091945560 -0.11781342
## lf7rup 0.001095106 -0.26377002
## hstructn 0.579045392 -0.09757715
## geolrngaggn -0.073237842 -0.01398595
## elevationm -0.144650575 0.23769455
## wr_unrn 0.055594693 0.06243625
## solpawhcn 0.183052422 -0.16628773
## slopern 0.080466116 -0.09694540
## MIN_AGE_MA 1.000000000 -0.20084125
## HubDist -0.200841251 1.00000000

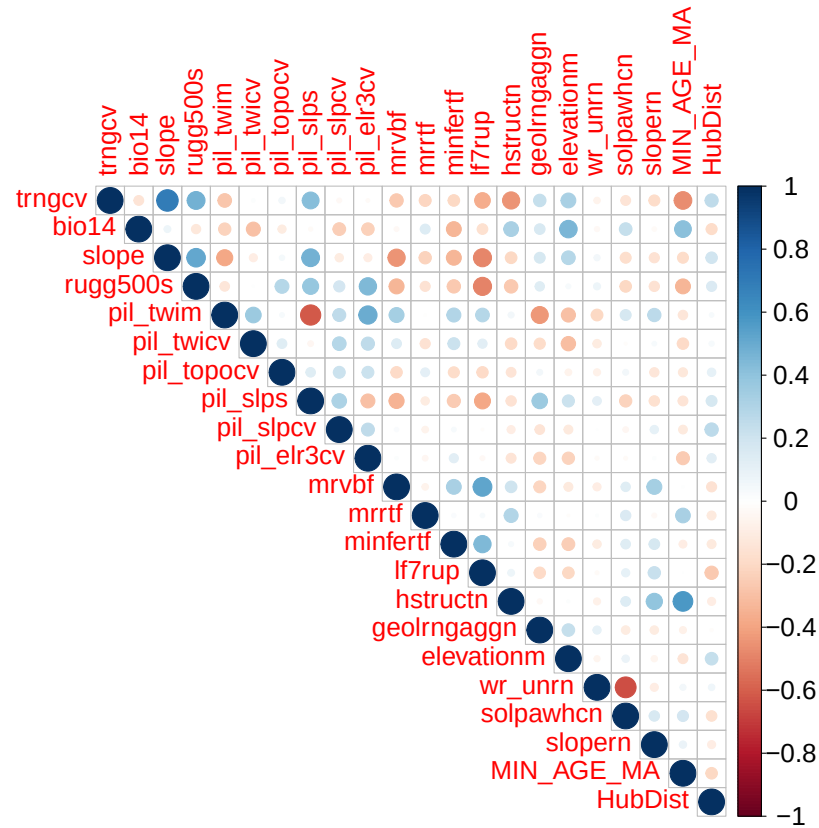
```

corrplot

```
corrplot(correlations, method = "circle", tl.cex = 1.5, cl.cex = 1.5)
```



```
corrplot(correlations, method = "circle", type = "upper", tl.cex = 1.5, cl.cex = 1.5)
```

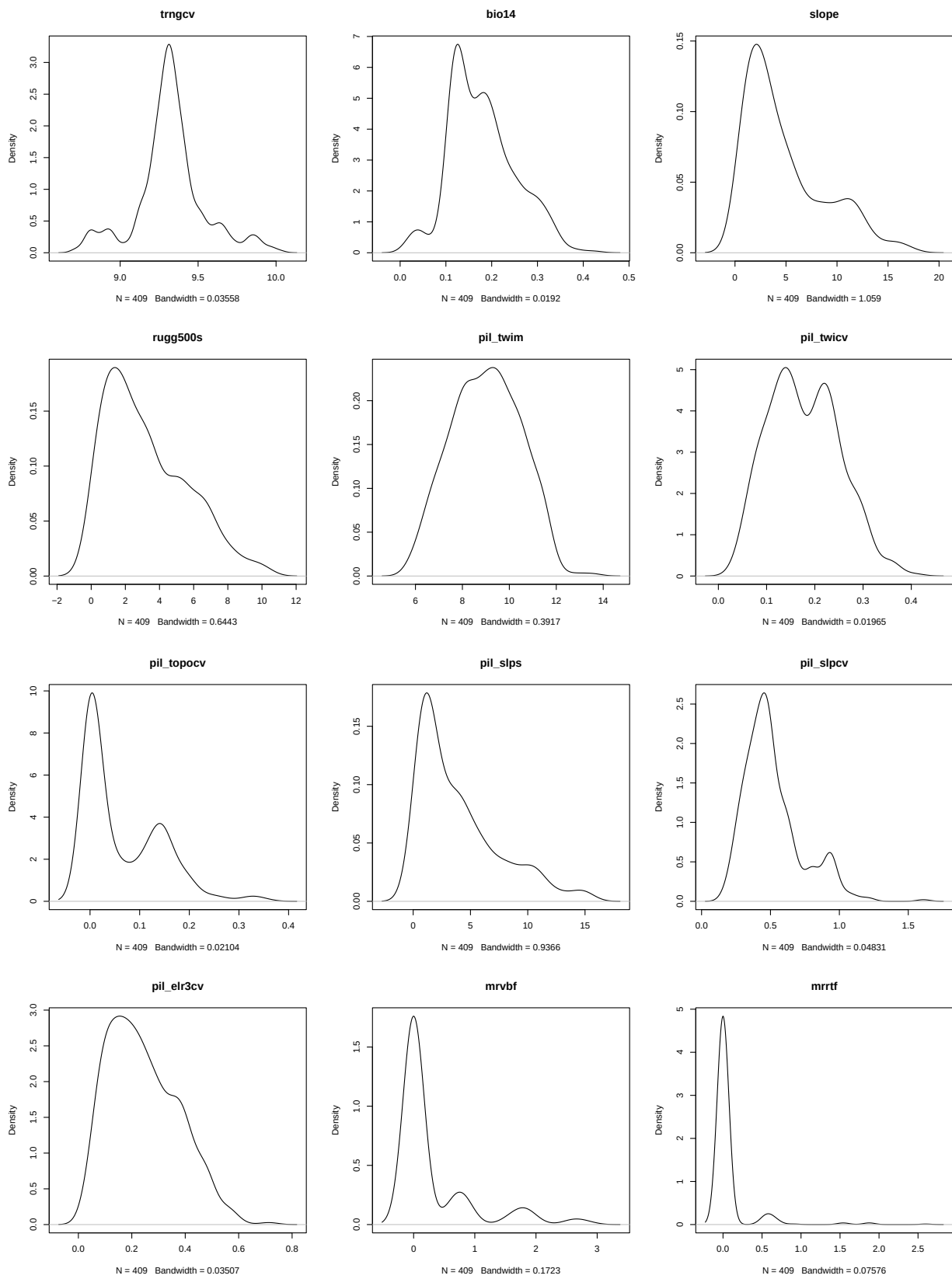


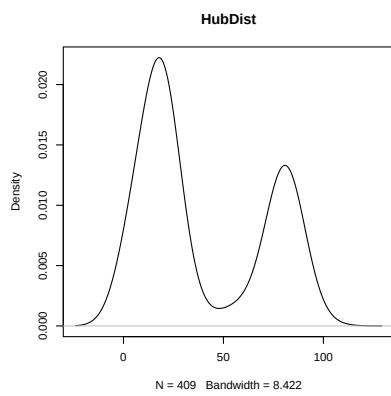
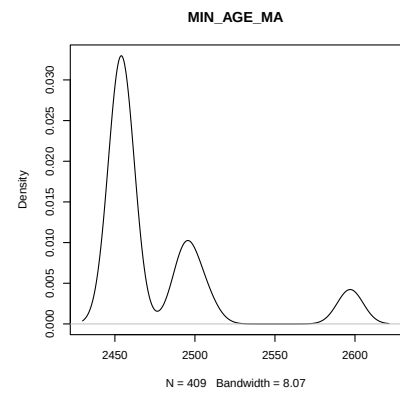
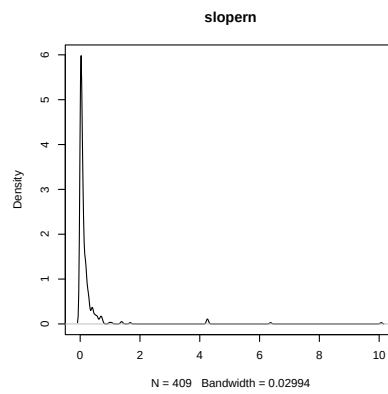
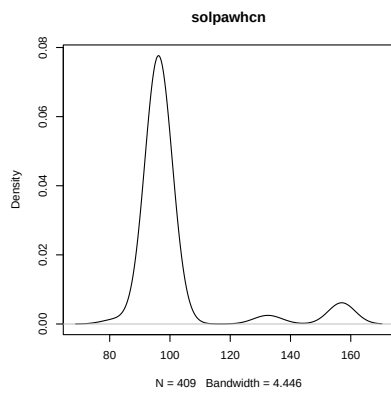
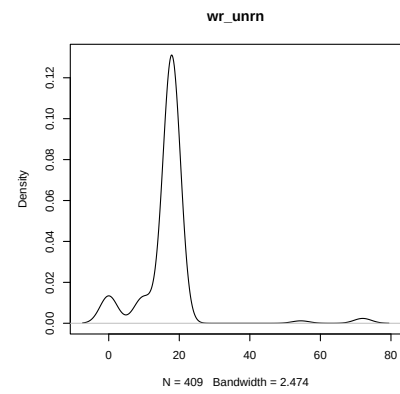
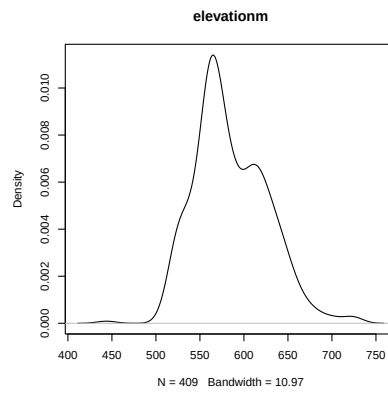
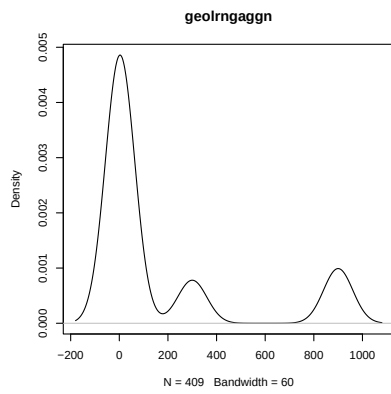
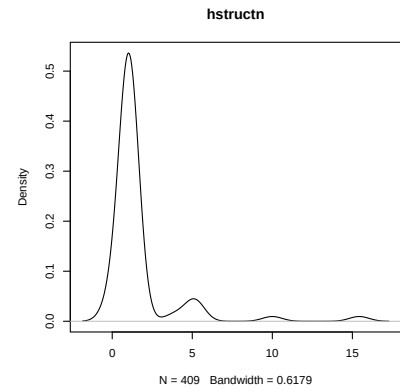
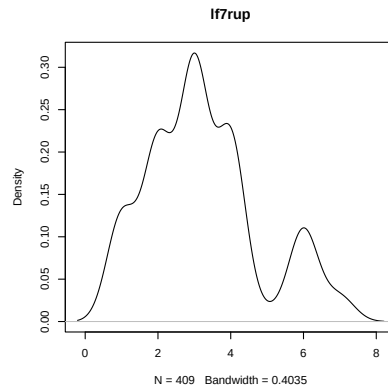
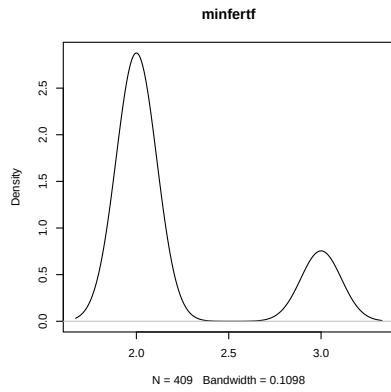
Covariates Density Plots

```
dim(data2)
```

```
## [1] 409 29
```

```
# Create a density plot for each column
par(mfrow = c(2, 3)) # Arrange plots in a 2x3 grid
for (col in seq_along(selectcol_Data2)) {
  plot(density(selectcol_Data2[, col]), main = colnames(selectcol_Data2)[col])
}
```





```

set.seed(78945)
Index1 <- createDataPartition(data2$true_troglofauna, p=0.5, list=FALSE)
data_train <- data2[ Index1,]
data_prov <- data2[-Index1,]
dim(data_train)

```

```
## [1] 205 29
```

```
summary(data_train)
```

```

##      trngcv      bio14      slope      rugg500s
##  Min.   :8.713   Min.   :0.02988   Min.   : 0.3154   Min.   : 0.02999
## 1st Qu.:9.222   1st Qu.:0.12881   1st Qu.: 1.7353   1st Qu.: 1.26590
## Median :9.320   Median :0.18033   Median : 3.7947   Median : 2.63082
## Mean   :9.319   Mean   :0.18687   Mean   : 4.9174   Mean   : 3.12256
## 3rd Qu.:9.411   3rd Qu.:0.23410   3rd Qu.: 7.2988   3rd Qu.: 4.78618
## Max.   :9.972   Max.   :0.38452   Max.   :17.2472   Max.   :10.10635
##
##      pil_twim      pil_twicv      pil_topocv      pil_slps
##  Min.   : 5.975   Min.   :0.03184   Min.   :0.00000   Min.   : 0.08751
## 1st Qu.: 8.095   1st Qu.:0.12767   1st Qu.:0.00000   1st Qu.: 1.02469
## Median : 9.164   Median :0.16995   Median :0.01947   Median : 2.37349
## Mean   : 9.078   Mean   :0.17922   Mean   :0.06160   Mean   : 3.65220
## 3rd Qu.:10.071   3rd Qu.:0.22963   3rd Qu.:0.12689   3rd Qu.: 5.18772
## Max.   :13.544   Max.   :0.36039   Max.   :0.33982   Max.   :14.88655
##
##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
##  Min.   :0.1908   Min.   :0.03157   Min.   :0.00000   Min.   :0.00000
## 1st Qu.:0.3766   1st Qu.:0.14646   1st Qu.:0.00000   1st Qu.:0.00000
## Median :0.4692   Median :0.23215   Median :0.00000   Median :0.00000
## Mean   :0.4998   Mean   :0.24334   Mean   :0.33910   Mean   :0.08205
## 3rd Qu.:0.5842   3rd Qu.:0.31863   3rd Qu.:0.00000   3rd Qu.:0.00000
## Max.   :1.2105   Max.   :0.59121   Max.   :2.85790   Max.   :2.59788
##
##      minfertf      lf7rup      hstructn      geolrngaggn
##  Min.   :2.00   Min.   :1.00   Min.   : 0.000   Min.   : 0.0115
## 1st Qu.:2.00   1st Qu.:2.00   1st Qu.: 1.043   1st Qu.: 2.5880
## Median :2.00   Median :3.00   Median : 1.043   Median : 2.5880
## Mean   :2.18   Mean   :3.21   Mean   : 1.877   Mean   :176.0964
## 3rd Qu.:2.00   3rd Qu.:4.00   3rd Qu.: 1.043   3rd Qu.:300.0000
## Max.   :3.00   Max.   :7.00   Max.   :15.429   Max.   :900.0000
##
##      elevationm      wr_unrn      solpawhcn      slopern
##  Min.   :443.9   Min.   : 0.00   Min.   : 82.0   Min.   : 0.00214
## 1st Qu.:558.7   1st Qu.:17.86   1st Qu.: 96.0   1st Qu.: 0.02306
## Median :582.6   Median :17.86   Median : 96.0   Median : 0.07592
## Mean   :587.3   Mean   :17.20   Mean   :101.6   Mean   : 0.19765
## 3rd Qu.:619.4   3rd Qu.:17.86   3rd Qu.: 96.0   3rd Qu.: 0.17402
## Max.   :723.7   Max.   :72.00   Max.   :157.0   Max.   :10.06944
##
##      MIN_AGE_MA      HubDist      month_collection
##  Min.   :2454   Min.   : 2.087   April   :31

```



```

## 1st Qu.:2454    1st Qu.: 16.862    March    :27
## Median :2454    Median : 21.842    May      :24
## Mean   :2479    Mean   : 38.612    January  :23
## 3rd Qu.:2494    3rd Qu.: 76.503    December :21
## Max.    :2597    Max.    :104.012    September:19
##                                     (Other)  :60
##                                     ROCKTYPE1
## sedimentary carbonate                : 21
## sedimentary other chemical or biochemical:150
## sedimentary siliciclastic            : 34
##
##
##
##                                     FORMATION
## Brockman Iron Formation                :132
## Marra Mamba Iron Formation             : 18
## Mount McRae Shale and Mount Sylvia Formation: 34
## Wittenoom Formation                   : 21
##
##
##
##                                     HubName
## anticline, exposed                    :122
## exposed                               : 10
## strike-slip, exposed, showing relative dextral displacement: 7
## syncline, exposed                     : 66
##
##
##
## true_troglofauna    LATITUDE    LONGITUDE
## 0: 74                Min.      :-22.61    Min.      :117.0
## 1:131                1st Qu.   :-22.25    1st Qu.   :117.6
##                     Median    :-22.16    Median    :117.8
##                     Mean      :-22.22    Mean      :117.7
##                     3rd Qu.   :-22.12    3rd Qu.   :117.9
##                     Max.      :-22.04    Max.      :118.0
##
##

```

```

data_train <- data_train[, -which(names(data_train) %in% c("LATITUDE", "LONGITUDE"))]
dim(data_train)

```

```

## [1] 205 27

```

```

head(data_train)

```

```

##      trngcv  bio14    slope rugg500s pil_twim pil_twicv pil_topocv pil_slps
## 173 9.22221 0.11984 2.62358 3.22619 8.90416 0.15090 0.15976 0.80723
## 50  9.83993 0.22232 15.87183 10.08672 7.72970 0.14045 0.02834 7.59460
## 458 9.28573 0.18881 10.57231 5.03119 8.63300 0.08212 0.00000 1.58531
## 383 9.32038 0.13525 1.22187 2.82780 9.56502 0.15810 0.00000 0.72478
## 24  9.40258 0.11255 6.00854 3.17529 8.35731 0.35754 0.00000 4.45767
## 246 9.41757 0.16629 8.93139 5.11192 8.63891 0.23263 0.05131 10.49186

```

```

##      pil_slpcv pil_elr3cv   mrvbf mrrtf minfertf lf7rup hstructn geolrngagn
## 173  0.25382   0.30016 0.00000    0      2      4  1.04286   2.5880
## 50   0.40627   0.25831 0.00000    0      2      1  1.04286   2.5880
## 458  0.33915   0.42026 0.00000    0      2      4  1.04286   2.5880
## 383  0.41914   0.56449 0.85295    0      3      2  1.04286   0.0115
## 24   0.57378   0.26367 0.00000    0      2      2  1.04286   2.5880
## 246  0.90002   0.24122 0.00000    0      2      3  1.04286   2.5880
##      elevationm wr_unrn solpawhcn slopern MIN_AGE_MA  HubDist month_collection
## 173   530.6564 17.85714      96 0.03171      2454 16.25063      September
## 50    603.8304 17.85714      96 0.01338      2454 23.47565      March
## 458   592.9053 17.85714      96 0.01833      2454 14.07495      April
## 383   561.8668 17.85714      96 0.12433      2454 15.77570      January
## 24    558.6679 17.85714      96 0.06396      2494 22.50256      December
## 246   601.5659 17.85714      96 0.03762      2454 86.82374      March
##
##                                ROCKTYPE1
## 173 sedimentary other chemical or biochemical
## 50  sedimentary other chemical or biochemical
## 458 sedimentary other chemical or biochemical
## 383 sedimentary other chemical or biochemical
## 24      sedimentary siliciclastic
## 246 sedimentary other chemical or biochemical
##
##                                FORMATION                                HubName
## 173                                Brockman Iron Formation anticline, exposed
## 50                                Brockman Iron Formation anticline, exposed
## 458                                Brockman Iron Formation anticline, exposed
## 383                                Brockman Iron Formation anticline, exposed
## 24  Mount McRae Shale and Mount Sylvia Formation anticline, exposed
## 246                                Brockman Iron Formation syncline, exposed
##
##      true_troglofauna
## 173                    0
## 50                    1
## 458                    0
## 383                    1
## 24                    0
## 246                    1

```

```
summary(data_train)
```

```

##      trngcv      bio14      slope      rugg500s
## Min.   :8.713   Min.   :0.02988   Min.   : 0.3154   Min.   : 0.02999
## 1st Qu.:9.222   1st Qu.:0.12881   1st Qu.: 1.7353   1st Qu.: 1.26590
## Median :9.320   Median :0.18033   Median : 3.7947   Median : 2.63082
## Mean   :9.319   Mean   :0.18687   Mean   : 4.9174   Mean   : 3.12256
## 3rd Qu.:9.411   3rd Qu.:0.23410   3rd Qu.: 7.2988   3rd Qu.: 4.78618
## Max.   :9.972   Max.   :0.38452   Max.   :17.2472   Max.   :10.10635
##
##      pil_twim      pil_twicv      pil_topocv      pil_slps
## Min.   : 5.975   Min.   :0.03184   Min.   :0.00000   Min.   : 0.08751
## 1st Qu.: 8.095   1st Qu.:0.12767   1st Qu.:0.00000   1st Qu.: 1.02469
## Median : 9.164   Median :0.16995   Median :0.01947   Median : 2.37349
## Mean   : 9.078   Mean   :0.17922   Mean   :0.06160   Mean   : 3.65220
## 3rd Qu.:10.071   3rd Qu.:0.22963   3rd Qu.:0.12689   3rd Qu.: 5.18772
## Max.   :13.544   Max.   :0.36039   Max.   :0.33982   Max.   :14.88655
##

```

```

##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
## Min.      :0.1908    Min.      :0.03157    Min.      :0.0000    Min.      :0.00000
## 1st Qu.:0.3766    1st Qu.:0.14646    1st Qu.:0.0000    1st Qu.:0.00000
## Median :0.4692    Median :0.23215    Median :0.0000    Median :0.00000
## Mean   :0.4998    Mean   :0.24334    Mean   :0.3391    Mean   :0.08205
## 3rd Qu.:0.5842    3rd Qu.:0.31863    3rd Qu.:0.0000    3rd Qu.:0.00000
## Max.   :1.2105    Max.   :0.59121    Max.   :2.8579    Max.   :2.59788
##
##      minfertf      lf7rup      hstructn      geolrngagn
## Min.      :2.00    Min.      :1.00    Min.      : 0.000    Min.      : 0.0115
## 1st Qu.:2.00    1st Qu.:2.00    1st Qu.: 1.043    1st Qu.: 2.5880
## Median :2.00    Median :3.00    Median : 1.043    Median : 2.5880
## Mean   :2.18    Mean   :3.21    Mean   : 1.877    Mean   :176.0964
## 3rd Qu.:2.00    3rd Qu.:4.00    3rd Qu.: 1.043    3rd Qu.:300.0000
## Max.   :3.00    Max.   :7.00    Max.   :15.429    Max.   :900.0000
##
##      elevationm      wr_unrn      solpawhcn      slopern
## Min.      :443.9    Min.      : 0.00    Min.      : 82.0    Min.      : 0.00214
## 1st Qu.:558.7    1st Qu.:17.86    1st Qu.: 96.0    1st Qu.: 0.02306
## Median :582.6    Median :17.86    Median : 96.0    Median : 0.07592
## Mean   :587.3    Mean   :17.20    Mean   :101.6    Mean   : 0.19765
## 3rd Qu.:619.4    3rd Qu.:17.86    3rd Qu.: 96.0    3rd Qu.: 0.17402
## Max.   :723.7    Max.   :72.00    Max.   :157.0    Max.   :10.06944
##
##      MIN_AGE_MA      HubDist      month_collection
## Min.      :2454    Min.      : 2.087    April      :31
## 1st Qu.:2454    1st Qu.: 16.862    March      :27
## Median :2454    Median : 21.842    May        :24
## Mean   :2479    Mean   : 38.612    January    :23
## 3rd Qu.:2494    3rd Qu.: 76.503    December   :21
## Max.   :2597    Max.   :104.012    September  :19
##                                     (Other)   :60
##                                     ROCKTYPE1
## sedimentary carbonate                  : 21
## sedimentary other chemical or biochemical:150
## sedimentary siliciclastic              : 34
##
##
##                                     FORMATION
## Brockman Iron Formation                  :132
## Marra Mamba Iron Formation              : 18
## Mount McRae Shale and Mount Sylvia Formation: 34
## Wittenoom Formation                    : 21
##
##
##                                     HubName
## anticline, exposed                      :122
## exposed                                : 10
## strike-slip, exposed, showing relative dextral displacement: 7
## syncline, exposed                      : 66
##

```

```
##
##
## true_troglofauna
## 0: 74
## 1:131
##
##
##
##
##
```

```
set.seed(78945)
```

```
trainIndex1 <- createDataPartition(data_prov$true_troglofauna, p=0.6, list=FALSE)
data_test <- data_prov[ trainIndex1,]
finalTest1 <- data_prov[-trainIndex1,]
```

```
summary(data_test)
```

```
##      trngcv      bio14      slope      rugg500s
## Min.   : 8.779   Min.   :0.02054   Min.   : 0.3373   Min.   : 0.1128
## 1st Qu.: 9.241   1st Qu.:0.11885   1st Qu.: 2.0234   1st Qu.: 1.4004
## Median : 9.300   Median :0.16241   Median : 3.1520   Median : 2.9601
## Mean   : 9.326   Mean   :0.16978   Mean   : 4.6339   Mean   : 3.3532
## 3rd Qu.: 9.402   3rd Qu.:0.20572   3rd Qu.: 6.2128   3rd Qu.: 5.0275
## Max.   :10.026   Max.   :0.33206   Max.   :16.3552   Max.   :10.1064
##
##      pil_twim      pil_twicv      pil_topocv      pil_slps
## Min.   : 5.944   Min.   :0.03919   Min.   :0.00000   Min.   : 0.1607
## 1st Qu.: 8.005   1st Qu.:0.12469   1st Qu.:0.00000   1st Qu.: 1.2269
## Median : 8.975   Median :0.16440   Median :0.01535   Median : 2.7404
## Mean   : 9.050   Mean   :0.17676   Mean   :0.06513   Mean   : 4.0601
## 3rd Qu.:10.169   3rd Qu.:0.21979   3rd Qu.:0.13731   3rd Qu.: 5.5650
## Max.   :11.582   Max.   :0.40758   Max.   :0.35341   Max.   :15.2506
##
##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
## Min.   :0.1719   Min.   :0.05163   Min.   :0.00000   Min.   :0.00000
## 1st Qu.:0.3834   1st Qu.:0.12926   1st Qu.:0.00000   1st Qu.:0.00000
## Median :0.4809   Median :0.22894   Median :0.00000   Median :0.00000
## Mean   :0.5469   Mean   :0.25332   Mean   :0.3064   Mean   :0.03911
## 3rd Qu.:0.6665   3rd Qu.:0.37929   3rd Qu.:0.5784   3rd Qu.:0.00000
## Max.   :1.2105   Max.   :0.71240   Max.   :1.9928   Max.   :1.56825
##
##      minfertf      lf7rup      hstructn      geolrngaggn
## Min.   :2.000   Min.   :1.000   Min.   : 0.000   Min.   : 0.0115
## 1st Qu.:2.000   1st Qu.:2.000   1st Qu.: 1.043   1st Qu.: 2.5880
## Median :2.000   Median :3.000   Median : 1.043   Median : 2.5880
## Mean   :2.236   Mean   :3.341   Mean   : 1.340   Mean   :168.6057
## 3rd Qu.:2.000   3rd Qu.:4.000   3rd Qu.: 1.043   3rd Qu.:21.9000
## Max.   :3.000   Max.   :7.000   Max.   :15.429   Max.   :900.0000
##
##      elevationm      wr_unrn      solpawhcn      slopern
## Min.   :518.3   Min.   : 0.00   Min.   : 82.0   Min.   :0.00162
```

```

## 1st Qu.:560.5 1st Qu.:17.86 1st Qu.: 96.0 1st Qu.:0.02080
## Median :576.1 Median :17.86 Median : 96.0 Median :0.06480
## Mean :583.6 Mean :16.97 Mean :100.3 Mean :0.21548
## 3rd Qu.:607.7 3rd Qu.:17.86 3rd Qu.: 96.0 3rd Qu.:0.18920
## Max. :721.9 Max. :72.00 Max. :157.0 Max. :6.36852
##
## MIN_AGE_MA HubDist month_collection
## Min. :2454 Min. : 1.294 September:18
## 1st Qu.:2454 1st Qu.:15.756 November :18
## Median :2454 Median :21.772 May :17
## Mean :2471 Mean :39.947 January :16
## 3rd Qu.:2494 3rd Qu.:79.998 March :14
## Max. :2597 Max. :91.703 April :13
## (Other) :27
## ROCKTYPE1
## sedimentary carbonate : 4
## sedimentary other chemical or biochemical:97
## sedimentary siliciclastic :22
##
##
## FORMATION
## Brockman Iron Formation :90
## Marra Mamba Iron Formation : 7
## Mount McRae Shale and Mount Sylvia Formation:22
## Wittenoom Formation : 4
##
##
## HubName
## anticline, exposed :70
## exposed : 5
## strike-slip, exposed, showing relative dextral displacement: 1
## syncline, exposed :47
##
##
## true_troglofauna LATITUDE LONGITUDE
## 0:44 Min. :-22.61 Min. :117.1
## 1:79 1st Qu.: -22.21 1st Qu.:117.7
## Median :-22.16 Median :117.8
## Mean :-22.20 Mean :117.8
## 3rd Qu.: -22.13 3rd Qu.:117.9
## Max. :-22.09 Max. :118.0
##

```

```
dim(data_test)
```

```
## [1] 123 29
```

```
summary(finalTest1)
```

```

##      trngcv          bio14          slope          rugg500s
## Min.    :8.808   Min.    :0.01801   Min.    : 0.2666   Min.    :0.0407
## 1st Qu.:9.244   1st Qu.:0.11957   1st Qu.: 2.0234   1st Qu.:1.4870
## Median :9.320   Median :0.17200   Median : 3.3626   Median :2.7764
## Mean    :9.335   Mean    :0.17983   Mean    : 5.0533   Mean    :3.3959
## 3rd Qu.:9.410   3rd Qu.:0.22522   3rd Qu.: 7.4120   3rd Qu.:5.2247
## Max.    :9.889   Max.    :0.42288   Max.    :15.8156   Max.    :9.9756
##
##      pil_twim      pil_twicv      pil_topocv      pil_slps
## Min.    : 5.749   Min.    :0.04324   Min.    :0.00000   Min.    : 0.1344
## 1st Qu.: 7.738   1st Qu.:0.11417   1st Qu.:0.00000   1st Qu.: 1.2658
## Median : 8.587   Median :0.16848   Median :0.02168   Median : 3.9729
## Mean    : 8.762   Mean    :0.17586   Mean    :0.07001   Mean    : 4.5940
## 3rd Qu.: 9.944   3rd Qu.:0.23585   3rd Qu.:0.13900   3rd Qu.: 6.9364
## Max.    :11.582   Max.    :0.36698   Max.    :0.32611   Max.    :12.7276
##
##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
## Min.    :0.1797   Min.    :0.0359   Min.    :0.0000   Min.    :0.00000
## 1st Qu.:0.3642   1st Qu.:0.1489   1st Qu.:0.0000   1st Qu.:0.00000
## Median :0.4696   Median :0.2056   Median :0.0000   Median :0.00000
## Mean    :0.5129   Mean    :0.2297   Mean    :0.2218   Mean    :0.08481
## 3rd Qu.:0.6206   3rd Qu.:0.3127   3rd Qu.:0.0000   3rd Qu.:0.00000
## Max.    :1.6098   Max.    :0.5645   Max.    :2.8219   Max.    :1.88966
##
##      minfertf      lf7rup      hstructn      geolrngaggn
## Min.    :2.000   Min.    :1.000   Min.    : 0.000   Min.    : 0.0115
## 1st Qu.:2.000   1st Qu.:2.000   1st Qu.: 1.043   1st Qu.: 2.5880
## Median :2.000   Median :3.000   Median : 1.043   Median : 2.5880
## Mean    :2.235   Mean    :3.099   Mean    : 1.535   Mean    :165.4988
## 3rd Qu.:2.000   3rd Qu.:4.000   3rd Qu.: 1.043   3rd Qu.:300.0000
## Max.    :3.000   Max.    :7.000   Max.    :15.429   Max.    :900.0000
##
##      elevationm      wr_unrn      solpawhcn      slopern
## Min.    :515.2   Min.    : 0.00   Min.    : 96.0   Min.    :0.00179
## 1st Qu.:559.7   1st Qu.:17.86   1st Qu.: 96.0   1st Qu.:0.01995
## Median :570.5   Median :17.86   Median : 96.0   Median :0.04257
## Mean    :582.6   Mean    :15.42   Mean    :102.1   Mean    :0.24617
## 3rd Qu.:606.6   3rd Qu.:17.86   3rd Qu.: 96.0   3rd Qu.:0.10689
## Max.    :726.1   Max.    :17.86   Max.    :157.0   Max.    :4.28009
##
##      MIN_AGE_MA      HubDist      month_collection
## Min.    :2454   Min.    : 1.931   September:13
## 1st Qu.:2454   1st Qu.:16.501   December :12
## Median :2454   Median :21.793   January  :11
## Mean    :2482   Mean    :40.231   March    : 9
## 3rd Qu.:2494   3rd Qu.:76.311   May      : 9
## Max.    :2597   Max.    :88.221   June     : 7
##                      (Other) :20
##                      ROCKTYPE1
## sedimentary carbonate      : 1
## sedimentary other chemical or biochemical:61
## sedimentary siliciclastic   :19
##
##

```

```
##
##
## FORMATION
## Brockman Iron Formation :51
## Marra Mamba Iron Formation :10
## Mount McRae Shale and Mount Sylvia Formation:19
## Wittenoom Formation : 1
##
##
##
## HubName
## anticline, exposed :43
## exposed : 6
## strike-slip, exposed, showing relative dextral displacement: 2
## syncline, exposed :30
##
##
##
## true_troglofauna LATITUDE LONGITUDE
## 0:29 Min. :-22.64 Min. :117.1
## 1:52 1st Qu.:-22.22 1st Qu.:117.6
## Median :-22.16 Median :117.8
## Mean :-22.21 Mean :117.7
## 3rd Qu.:-22.12 3rd Qu.:117.9
## Max. :-22.09 Max. :118.0
##
```

```
dim(finalTest1)
```

```
## [1] 81 29
```

```
data_test <- data_test[, -which(names(data2) %in% c("LATITUDE", "LONGITUDE"))]
finalTest <- finalTest1[, -which(names(data2) %in% c("LATITUDE", "LONGITUDE"))]
sapply(data_test,class);sapply(finalTest1,class)
```

```
##      trngcv      bio14      slope      rugg500s
##      "numeric"      "numeric"      "numeric"      "numeric"
##      pil_twim      pil_twicv      pil_topocv      pil_slps
##      "numeric"      "numeric"      "numeric"      "numeric"
##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
##      "numeric"      "numeric"      "numeric"      "numeric"
##      minfertf      lf7rup      hstructn      geolrngaggn
##      "numeric"      "numeric"      "numeric"      "numeric"
##      elevationm      wr_unrn      solpawhcn      slopern
##      "numeric"      "numeric"      "numeric"      "numeric"
##      MIN_AGE_MA      HubDist month_collection      ROCKTYPE1
##      "numeric"      "numeric"      "factor"      "factor"
##      FORMATION      HubName true_troglofauna
##      "factor"      "factor"      "factor"

##      trngcv      bio14      slope      rugg500s
```

```
##      "numeric"      "numeric"      "numeric"      "numeric"
##      pil_twim      pil_twicv      pil_topocv      pil_slps
##      "numeric"      "numeric"      "numeric"      "numeric"
##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
##      "numeric"      "numeric"      "numeric"      "numeric"
##      minfertif      lf7rup      hstructn      geolrngaggn
##      "numeric"      "numeric"      "numeric"      "numeric"
##      elevationm      wr_unrn      solpawhcn      slopern
##      "numeric"      "numeric"      "numeric"      "numeric"
##      MIN_AGE_MA      HubDist month_collection      ROCKTYPE1
##      "numeric"      "numeric"      "factor"      "factor"
##      FORMATION      HubName true_troglofauna      LATITUDE
##      "factor"      "factor"      "factor"      "numeric"
##      LONGITUDE
##      "numeric"
```

```
head(data_test)
```

```
##      trngcv      bio14      slope      rugg500s      pil_twim      pil_twicv      pil_topocv      pil_slps
## 538 8.94549 0.20508 0.33730 0.38791 10.79223 0.13333 0.00000 0.16074
## 580 9.30815 0.15320 3.35630 6.98694 8.78567 0.28006 0.00000 1.81942
## 449 9.61349 0.28381 16.35521 2.20207 5.94411 0.04918 0.00000 6.81287
## 149 9.41111 0.11746 4.08951 0.66187 7.66015 0.20674 0.00000 2.03130
## 304 9.29545 0.11761 2.02341 3.37463 11.58232 0.19793 0.05766 3.97289
## 15 8.77937 0.29834 1.52777 1.09473 7.32769 0.07640 0.00000 4.35278
##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf      minfertif      lf7rup      hstructn      geolrngaggn
## 538 0.61696 0.26306 1.5954 0.00000 3 4 15.42857 0.0115
## 580 0.50594 0.53529 0.0000 0.00000 2 3 1.04286 2.5880
## 449 0.25979 0.12502 0.0000 0.00000 2 4 1.04286 900.0000
## 149 0.42033 0.09569 0.0000 0.00000 2 3 1.04286 2.5880
## 304 0.94816 0.42085 0.0000 0.00000 2 4 1.04286 2.5880
## 15 0.60188 0.08611 0.0000 0.91144 2 4 3.77143 300.0000
##      elevationm      wr_unrn      solpawhcn      slopern      MIN_AGE_MA      HubDist      month_collection
## 538 593.5888 8.33333 131 6.36852 2506 17.36909 June
## 580 606.9091 17.85714 96 0.62784 2454 82.98079 February
## 449 721.9289 17.85714 96 0.00249 2454 88.50439 April
## 149 561.2512 17.85714 96 0.16245 2494 21.98282 April
## 304 560.4824 17.85714 96 0.27924 2454 81.98112 May
## 15 548.1772 72.00000 82 0.05721 2597 56.67423 November
##      ROCKTYPE1
## 538      sedimentary carbonate
## 580 sedimentary other chemical or biochemical
## 449 sedimentary other chemical or biochemical
## 149      sedimentary siliciclastic
## 304 sedimentary other chemical or biochemical
## 15  sedimentary other chemical or biochemical
##      FORMATION      HubName
## 538      Wittenoom Formation anticline, exposed
## 580      Brockman Iron Formation syncline, exposed
## 449      Brockman Iron Formation syncline, exposed
## 149 Mount McRae Shale and Mount Sylvia Formation anticline, exposed
## 304      Brockman Iron Formation syncline, exposed
## 15      Marra Mamba Iron Formation anticline, exposed
##      true_troglofauna
```



```
## 538      0
## 580      1
## 449      1
## 149      1
## 304      1
## 15       1
```

```
dim(data_test)
```

```
## [1] 123 27
```

```
summary(data_test)
```

```
##      trngcv      bio14      slope      rugg500s
## Min.   : 8.779   Min.   :0.02054   Min.   : 0.3373   Min.   : 0.1128
## 1st Qu.: 9.241   1st Qu.:0.11885   1st Qu.: 2.0234   1st Qu.: 1.4004
## Median : 9.300   Median :0.16241   Median : 3.1520   Median : 2.9601
## Mean   : 9.326   Mean   :0.16978   Mean   : 4.6339   Mean   : 3.3532
## 3rd Qu.: 9.402   3rd Qu.:0.20572   3rd Qu.: 6.2128   3rd Qu.: 5.0275
## Max.   :10.026   Max.   :0.33206   Max.   :16.3552   Max.   :10.1064
##
##      pil_twim      pil_twicv      pil_topocv      pil_slps
## Min.   : 5.944   Min.   :0.03919   Min.   :0.00000   Min.   : 0.1607
## 1st Qu.: 8.005   1st Qu.:0.12469   1st Qu.:0.00000   1st Qu.: 1.2269
## Median : 8.975   Median :0.16440   Median :0.01535   Median : 2.7404
## Mean   : 9.050   Mean   :0.17676   Mean   :0.06513   Mean   : 4.0601
## 3rd Qu.:10.169   3rd Qu.:0.21979   3rd Qu.:0.13731   3rd Qu.: 5.5650
## Max.   :11.582   Max.   :0.40758   Max.   :0.35341   Max.   :15.2506
##
##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
## Min.   :0.1719   Min.   :0.05163   Min.   :0.0000   Min.   :0.00000
## 1st Qu.:0.3834   1st Qu.:0.12926   1st Qu.:0.0000   1st Qu.:0.00000
## Median :0.4809   Median :0.22894   Median :0.0000   Median :0.00000
## Mean   :0.5469   Mean   :0.25332   Mean   :0.3064   Mean   :0.03911
## 3rd Qu.:0.6665   3rd Qu.:0.37929   3rd Qu.:0.5784   3rd Qu.:0.00000
## Max.   :1.2105   Max.   :0.71240   Max.   :1.9928   Max.   :1.56825
##
##      minfertf      lf7rup      hstructn      geolrngaggn
## Min.   :2.000   Min.   :1.000   Min.   : 0.000   Min.   : 0.0115
## 1st Qu.:2.000   1st Qu.:2.000   1st Qu.: 1.043   1st Qu.: 2.5880
## Median :2.000   Median :3.000   Median : 1.043   Median : 2.5880
## Mean   :2.236   Mean   :3.341   Mean   : 1.340   Mean   :168.6057
## 3rd Qu.:2.000   3rd Qu.:4.000   3rd Qu.: 1.043   3rd Qu.: 21.9000
## Max.   :3.000   Max.   :7.000   Max.   :15.429   Max.   :900.0000
##
##      elevationm      wr_unrn      solpawhcn      slopern
## Min.   :518.3   Min.   : 0.00   Min.   : 82.0   Min.   :0.00162
## 1st Qu.:560.5   1st Qu.:17.86   1st Qu.: 96.0   1st Qu.:0.02080
## Median :576.1   Median :17.86   Median : 96.0   Median :0.06480
## Mean   :583.6   Mean   :16.97   Mean   :100.3   Mean   :0.21548
## 3rd Qu.:607.7   3rd Qu.:17.86   3rd Qu.: 96.0   3rd Qu.:0.18920
## Max.   :721.9   Max.   :72.00   Max.   :157.0   Max.   :6.36852
##
```

```

##      MIN_AGE_MA      HubDist      month_collection
## Min.      :2454      Min.      : 1.294      September:18
## 1st Qu.:2454      1st Qu.:15.756      November :18
## Median :2454      Median :21.772      May       :17
## Mean    :2471      Mean    :39.947      January   :16
## 3rd Qu.:2494      3rd Qu.:79.998      March     :14
## Max.    :2597      Max.    :91.703      April     :13
##                                     (Other)  :27
##                                     ROCKTYPE1
## sedimentary carbonate                : 4
## sedimentary other chemical or biochemical:97
## sedimentary siliciclastic            :22
##
##
##
##                                     FORMATION
## Brockman Iron Formation                :90
## Marra Mamba Iron Formation             : 7
## Mount McRae Shale and Mount Sylvia Formation:22
## Wittenoom Formation                   : 4
##
##
##
##                                     HubName
## anticline, exposed                    :70
## exposed                               : 5
## strike-slip, exposed, showing relative dextral displacement: 1
## syncline, exposed                     :47
##
##
##
## true_troglofauna
## 0:44
## 1:79
##
##
##
##

```

```
names(data_test)
```

```

## [1] "trngcv"      "bio14"      "slope"      "rugg500s"
## [5] "pil_twim"    "pil_twicv"  "pil_topocv" "pil_slps"
## [9] "pil_slpcv"    "pil_elr3cv" "mrvbf"      "mrrtf"
## [13] "minfertf"    "lf7rup"     "hstructn"   "geolrngaggn"
## [17] "elevationm"  "wr_unrn"    "solpawhcn"  "slopern"
## [21] "MIN_AGE_MA"  "HubDist"    "month_collection" "ROCKTYPE1"
## [25] "FORMATION"   "HubName"    "true_troglofauna"

```

```
head(finalTest)
```

```
##      trngcv  bio14  slope rugg500s pil_twim pil_twicv pil_topocv pil_slps
```

```

## 95 9.88856 0.25342 10.56614 4.74161 8.49100 0.23228 0.00000 6.65473
## 626 8.80830 0.28342 2.23671 1.25996 7.13560 0.05745 0.00000 2.40606
## 607 9.32483 0.28510 2.18706 0.46594 9.42484 0.23025 0.17897 1.49514
## 623 8.89422 0.31750 3.19405 2.99698 7.67751 0.12228 0.00000 1.90480
## 18 9.33742 0.09946 6.24483 3.65277 10.97026 0.25139 0.00000 4.37832
## 470 9.14680 0.22522 2.94459 7.08095 7.82185 0.23485 0.19927 10.64527
## pil_slpcv pil_elr3cv mrvbf mrrtf minfertf lf7rup hstructn geolrngagn
## 95 0.37579 0.12946 0 0.00000 2 2 1.04286 900.000
## 626 0.47131 0.28907 0 0.00000 2 4 5.14286 300.000
## 607 0.50164 0.16386 0 0.70467 2 3 0.00000 300.000
## 623 0.27703 0.07601 0 0.00000 2 4 5.14286 300.000
## 18 0.79251 0.24706 0 0.00000 2 3 1.04286 2.588
## 470 0.65664 0.24246 0 0.00000 2 4 1.04286 300.000
## elevationm wr_unrn solpawhcn slopern MIN_AGE_MA HubDist month_collection
## 95 627.5781 17.85714 96 0.01311 2454 77.101771 April
## 626 542.7686 10.00000 100 0.04463 2597 7.811622 June
## 607 570.4954 0.00000 157 0.09306 2597 16.627614 September
## 623 566.4947 10.00000 100 0.00955 2597 7.772825 November
## 18 525.2358 17.85714 96 4.24974 2494 21.317005 December
## 470 646.1981 17.85714 96 0.20195 2494 15.040323 June
## ROCKTYPE1
## 95 sedimentary other chemical or biochemical
## 626 sedimentary other chemical or biochemical
## 607 sedimentary other chemical or biochemical
## 623 sedimentary other chemical or biochemical
## 18 sedimentary siliciclastic
## 470 sedimentary siliciclastic
## FORMATION HubName
## 95 Brockman Iron Formation syncline, exposed
## 626 Marra Mamba Iron Formation exposed
## 607 Marra Mamba Iron Formation anticline, exposed
## 623 Marra Mamba Iron Formation exposed
## 18 Mount McRae Shale and Mount Sylvia Formation anticline, exposed
## 470 Mount McRae Shale and Mount Sylvia Formation anticline, exposed
## true_troglofauna
## 95 1
## 626 1
## 607 1
## 623 1
## 18 0
## 470 1

```

```
dim(finalTest)
```

```
## [1] 81 27
```

```
summary(finalTest)
```

```

##      trngcv      bio14      slope      rugg500s
## Min. :8.808 Min. :0.01801 Min. : 0.2666 Min. :0.0407
## 1st Qu.:9.244 1st Qu.:0.11957 1st Qu.: 2.0234 1st Qu.:1.4870
## Median :9.320 Median :0.17200 Median : 3.3626 Median :2.7764
## Mean :9.335 Mean :0.17983 Mean : 5.0533 Mean :3.3959

```

```

## 3rd Qu.:9.410 3rd Qu.:0.22522 3rd Qu.: 7.4120 3rd Qu.:5.2247
## Max. :9.889 Max. :0.42288 Max. :15.8156 Max. :9.9756
##
## pil_twim pil_twicv pil_topocv pil_slps
## Min. : 5.749 Min. :0.04324 Min. :0.00000 Min. : 0.1344
## 1st Qu.: 7.738 1st Qu.:0.11417 1st Qu.:0.00000 1st Qu.: 1.2658
## Median : 8.587 Median :0.16848 Median :0.02168 Median : 3.9729
## Mean : 8.762 Mean :0.17586 Mean :0.07001 Mean : 4.5940
## 3rd Qu.: 9.944 3rd Qu.:0.23585 3rd Qu.:0.13900 3rd Qu.: 6.9364
## Max. :11.582 Max. :0.36698 Max. :0.32611 Max. :12.7276
##
## pil_slpcv pil_elr3cv mrvbf mrrtf
## Min. :0.1797 Min. :0.0359 Min. :0.0000 Min. :0.00000
## 1st Qu.:0.3642 1st Qu.:0.1489 1st Qu.:0.0000 1st Qu.:0.00000
## Median :0.4696 Median :0.2056 Median :0.0000 Median :0.00000
## Mean :0.5129 Mean :0.2297 Mean :0.2218 Mean :0.08481
## 3rd Qu.:0.6206 3rd Qu.:0.3127 3rd Qu.:0.0000 3rd Qu.:0.00000
## Max. :1.6098 Max. :0.5645 Max. :2.8219 Max. :1.88966
##
## minfertf lf7rup hstructn geolrngaggn
## Min. :2.000 Min. :1.000 Min. : 0.000 Min. : 0.0115
## 1st Qu.:2.000 1st Qu.:2.000 1st Qu.: 1.043 1st Qu.: 2.5880
## Median :2.000 Median :3.000 Median : 1.043 Median : 2.5880
## Mean :2.235 Mean :3.099 Mean : 1.535 Mean :165.4988
## 3rd Qu.:2.000 3rd Qu.:4.000 3rd Qu.: 1.043 3rd Qu.:300.0000
## Max. :3.000 Max. :7.000 Max. :15.429 Max. :900.0000
##
## elevationm wr_unrn solpawhcn slopern
## Min. :515.2 Min. : 0.00 Min. : 96.0 Min. :0.00179
## 1st Qu.:559.7 1st Qu.:17.86 1st Qu.: 96.0 1st Qu.:0.01995
## Median :570.5 Median :17.86 Median : 96.0 Median :0.04257
## Mean :582.6 Mean :15.42 Mean :102.1 Mean :0.24617
## 3rd Qu.:606.6 3rd Qu.:17.86 3rd Qu.: 96.0 3rd Qu.:0.10689
## Max. :726.1 Max. :17.86 Max. :157.0 Max. :4.28009
##
## MIN_AGE_MA HubDist month_collection
## Min. :2454 Min. : 1.931 September:13
## 1st Qu.:2454 1st Qu.:16.501 December :12
## Median :2454 Median :21.793 January :11
## Mean :2482 Mean :40.231 March : 9
## 3rd Qu.:2494 3rd Qu.:76.311 May : 9
## Max. :2597 Max. :88.221 June : 7
## (Other) :20
## ROCKTYPE1
## sedimentary carbonate : 1
## sedimentary other chemical or biochemical:61
## sedimentary siliciclastic :19
##
##
## FORMATION
## Brockman Iron Formation :51
## Marra Mamba Iron Formation :10

```

```
## Mount McRae Shale and Mount Sylvia Formation:19
## Wittenoom Formation : 1
##
##
##
## HubName
## anticline, exposed :43
## exposed : 6
## strike-slip, exposed, showing relative dextral displacement: 2
## syncline, exposed :30
##
##
##
## true_troglofauna
## 0:29
## 1:52
##
##
##
##
```

```
names(finalTest)
```

```
## [1] "trngcv" "bio14" "slope" "rugg500s"
## [5] "pil_twim" "pil_twicv" "pil_topocv" "pil_slps"
## [9] "pil_slpcv" "pil_elr3cv" "mrvbf" "mrrtf"
## [13] "minfertf" "lf7rup" "hstructn" "geolrngaggn"
## [17] "elevationm" "wr_unrn" "solpawhcn" "slopern"
## [21] "MIN_AGE_MA" "HubDist" "month_collection" "ROCKTYPE1"
## [25] "FORMATION" "HubName" "true_troglofauna"
```

```
ContraProva<-data_test$true_troglofauna
summary(ContraProva)
```

```
## 0 1
## 44 79
```

confusion__matrix

```
get_confusion_elements <- function(caret_confusion_matrix) {
  tp <- as.numeric(caret_confusion_matrix$table[4]) # true positives
  fn <- as.numeric(caret_confusion_matrix$table[3]) # false negatives
  fp <- as.numeric(caret_confusion_matrix$table[2]) # false positives
  tn <- as.numeric(caret_confusion_matrix$table[1]) # true negatives
}
```

```

    return( c(tp, fp, tn, fn) )
}

calculate_mcc <- function(tp, fp, tn, fn) {
  # calculates Matthews correlation coefficient
  # tp - true positives
  # fp - false positives
  # tn - true negatives
  # fn - false negatives
  mcc <- ((tp * tn) - (fp * fn)) /
    (sqrt( (tp + fp) * (tp + fn)) * sqrt((tn + fp) * (tn + fn)) )
  return(mcc)
}

calculate_mcc1 <- function(caret_confusion_matrix) {
  # calculates Matthews correlation coefficient
  # tp - true positives
  # fp - false positives
  # tn - true negatives
  # fn - false negatives
  mcc <- ((caret_confusion_matrix$table[4] * caret_confusion_matrix$table[1]) - (caret_confusion_mat.
    (sqrt( (caret_confusion_matrix$table[4] + caret_confusion_matrix$table[2]) * (caret_confusi
  return(mcc)
}

calculate_F2 <- function(CM_predictions) {
dbF2_11<-((1+2^2)*CM_predictions$byClass["Precision"]*CM_predictions$byClass["Sensitivity"])/(2^2*CM_pr
dbF2_11<-as.numeric(dbF2_11)
return(dbF2_11)
}

```

Train

```

set.seed(49897)

grid <- expand.grid(.mtry=seq(from = 2, to = 26, by = 2))

trControl <- trainControl(
  method = "repeatedcv", # Resampling method
  repeats = 10,          # Number of repetitions for repeated cross-validation
  number = 5,            # Number of folds in each iteration of cross-validation
  classProbs = TRUE,     # Calculate class probabilities
  savePredictions = "final", # Save final predictions
  summaryFunction = twoClassSummary # Function for summarizing results (assumed to be defined elsewhere)
)

rf_mtry <- train(

```

```

make.names(true_troglofauna) ~ .,          # Formula for the model, predicting 'true_troglofauna'
data = data_train,                        # Training data
method = "rf",                            # Random Forest method
strata = data_train$true_troglofauna,      # Stratification based on the target variable
sampsize = c(min(sum(data_train$true_troglofauna == 0), sum(data_train$true_troglofauna == 1)),
              min(sum(data_train$true_troglofauna == 1), sum(data_train$true_troglofauna == 1))),
metric = "ROC",                           # Evaluation metric (Receiver Operating Characteristic)
tuneGrid = grid,
trControl = trControl,                    # Control parameters for the training process
importance = TRUE,                         # Calculate variable importance
ntree = 500                               # Number of trees in the Random Forest
)

```

#In the above code, sampsize = rep(sum(training\$class == 1), 2) means both the classes will have same frequency

```
rf_mtry
```

```

## Random Forest
##
## 205 samples
## 26 predictor
## 2 classes: 'X0', 'X1'
##
## No pre-processing
## Resampling: Cross-Validated (5 fold, repeated 10 times)
## Summary of sample sizes: 164, 164, 164, 165, 163, 164, ...
## Resampling results across tuning parameters:
##
##  mtry  ROC      Sens      Spec
##  2     0.6938863 0.3445714 0.8992308
##  4     0.6779315 0.3856190 0.8395442
##  6     0.6717848 0.3813333 0.8236467
##  8     0.6691434 0.3911429 0.8190598
## 10     0.6684148 0.3853333 0.8136752
## 12     0.6639166 0.3976190 0.8144729
## 14     0.6653675 0.3920000 0.8076353
## 16     0.6675443 0.4027619 0.8022222
## 18     0.6633946 0.3946667 0.8060399
## 20     0.6678145 0.4013333 0.8060684
## 22     0.6698575 0.4018095 0.8045014
## 24     0.6701758 0.4149524 0.8060684
## 26     0.6681216 0.4081905 0.8037322
##
## ROC was used to select the optimal model using the largest value.
## The final value used for the model was mtry = 2.

```

```
getTrainPerf(rf_mtry)
```

```

##      TrainROC TrainSens TrainSpec method
## 1 0.6938863 0.3445714 0.8992308      rf

```

```
summary(rf_mtry)
```

##	Length	Class	Mode
## call	8	-none-	call
## type	1	-none-	character
## predicted	205	factor	numeric
## err.rate	1500	-none-	numeric
## confusion	6	-none-	numeric
## votes	410	matrix	numeric
## oob.times	205	-none-	numeric
## classes	2	-none-	character
## importance	164	-none-	numeric
## importanceSD	123	-none-	numeric
## localImportance	0	-none-	NULL
## proximity	0	-none-	NULL
## ntree	1	-none-	numeric
## mtry	1	-none-	numeric
## forest	14	-none-	list
## y	205	factor	numeric
## test	0	-none-	NULL
## inbag	0	-none-	NULL
## xNames	41	-none-	character
## problemType	1	-none-	character
## tuneValue	1	data.frame	list
## obsLevels	2	-none-	character
## param	4	-none-	list

```
rf_mtry$bestTune$mtry
```

```
## [1] 2
```

```
rf_mtry$finalModel
```

```
##
## Call:
## randomForest(x = x, y = y, ntree = 500, mtry = param$mtry, strata = ..1,      sampsize = ..2, impor
##           Type of random forest: classification
##           Number of trees: 500
## No. of variables tried at each split: 2
##
##           OOB estimate of  error rate: 28.78%
## Confusion matrix:
##      X0  X1 class.error
## X0 27  47  0.63513514
## X1 12 119  0.09160305
```

```
rf_mtry$finalModel$confusion
```

```
##      X0  X1 class.error
## X0 27  47  0.63513514
## X1 12 119  0.09160305
```



```
summary(rf_mtry$pred)
```

```
##      mtry  pred    obs      X0      X1
## Min.   :2    X0: 387    X0: 740  Min.   :0.0140  Min.   :0.0360
## 1st Qu.:2    X1:1663    X1:1310  1st Qu.:0.2100  1st Qu.:0.5500
## Median :2                                Median :0.3230  Median :0.6770
## Mean   :2                                Mean   :0.3474  Mean   :0.6526
## 3rd Qu.:2                                3rd Qu.:0.4500  3rd Qu.:0.7900
## Max.   :2                                Max.   :0.9640  Max.   :0.9860
##      rowIndex  Resample
## Min.   : 1    Length:2050
## 1st Qu.: 52    Class :character
## Median :103    Mode  :character
## Mean   :103
## 3rd Qu.:154
## Max.   :205
```

```
summary(rf_mtry$pred$pred)
```

```
##      X0      X1
## 387 1663
```

```
summary(rf_mtry$pred$obs)
```

```
##      X0      X1
## 740 1310
```

```
head(rf_mtry$pred$X0,20)
```

```
## [1] 0.426 0.226 0.378 0.722 0.438 0.284 0.166 0.448 0.164 0.308 0.244 0.284
## [13] 0.290 0.118 0.158 0.154 0.396 0.360 0.304 0.220
```

```
head(rf_mtry$pred$X1,20)
```

```
## [1] 0.574 0.774 0.622 0.278 0.562 0.716 0.834 0.552 0.836 0.692 0.756 0.716
## [13] 0.710 0.882 0.842 0.846 0.604 0.640 0.696 0.780
```

```
summary(data_train$true_troglofauna)
```

```
##      0      1
## 74 131
```

```
head(rf_mtry$pred,20)
```

```
##      mtry pred obs      X0      X1 rowIndex  Resample
## 1      2   X1  X1 0.426 0.574     128 Fold1.Rep03
## 2      2   X1  X0 0.226 0.774     154 Fold2.Rep03
## 3      2   X1  X1 0.378 0.622     187 Fold1.Rep03
```

```
## 4      2    X0  X0 0.722 0.278      144 Fold2.Rep03
## 5      2    X1  X0 0.438 0.562       76 Fold3.Rep01
## 6      2    X1  X0 0.284 0.716      124 Fold1.Rep03
## 7      2    X1  X1 0.166 0.834      151 Fold1.Rep03
## 8      2    X1  X1 0.448 0.552      204 Fold4.Rep03
## 9      2    X1  X1 0.164 0.836       74 Fold3.Rep01
## 10     2    X1  X0 0.308 0.692      197 Fold1.Rep03
## 11     2    X1  X0 0.244 0.756      197 Fold1.Rep02
## 12     2    X1  X0 0.284 0.716      141 Fold2.Rep03
## 13     2    X1  X1 0.290 0.710      104 Fold5.Rep02
## 14     2    X1  X1 0.118 0.882      200 Fold1.Rep03
## 15     2    X1  X1 0.158 0.842      173 Fold1.Rep03
## 16     2    X1  X1 0.154 0.846      109 Fold2.Rep03
## 17     2    X1  X0 0.396 0.604      195 Fold3.Rep03
## 18     2    X1  X1 0.360 0.640      123 Fold1.Rep03
## 19     2    X1  X1 0.304 0.696      196 Fold1.Rep02
## 20     2    X1  X1 0.220 0.780      171 Fold3.Rep03
```

```
sapply(rf_mtry$pred, class)
```

```
##      mtry      pred      obs      X0      X1      rowIndex
## "numeric" "factor" "factor" "numeric" "numeric" "integer"
##      Resample
## "character"
```

```
rf_mtry$results$Sens
```

```
## [1] 0.3445714 0.3856190 0.3813333 0.3911429 0.3853333 0.3976190 0.3920000
## [8] 0.4027619 0.3946667 0.4013333 0.4018095 0.4149524 0.4081905
```

```
rf_mtry$results$SensSD
```

```
## [1] 0.1129201 0.1187946 0.1205017 0.1220932 0.1235770 0.1291989 0.1322301
## [8] 0.1250742 0.1253276 0.1232890 0.1253359 0.1250073 0.1385888
```

```
rf_mtry$results$Spec
```

```
## [1] 0.8992308 0.8395442 0.8236467 0.8190598 0.8136752 0.8144729 0.8076353
## [8] 0.8022222 0.8060399 0.8060684 0.8045014 0.8060684 0.8037322
```

```
rf_mtry$results$SpecSD
```

```
## [1] 0.05710551 0.07833820 0.08376055 0.07609411 0.08138075 0.07717975
## [7] 0.08210620 0.08032530 0.07543814 0.08250874 0.07885393 0.08171997
## [13] 0.08563722
```

```
rf_mtry$results$ROC
```

```
## [1] 0.6938863 0.6779315 0.6717848 0.6691434 0.6684148 0.6639166 0.6653675
## [8] 0.6675443 0.6633946 0.6678145 0.6698575 0.6701758 0.6681216
```

```
rf_mtry$results$ROCSD
```

```
## [1] 0.08073801 0.08315197 0.08505518 0.08669530 0.08547976 0.08745346
## [7] 0.08757180 0.09130348 0.09053936 0.08594373 0.08741470 0.08847475
## [13] 0.09059512
```

```
rf_mtry$finalModel$confusion
```

```
##      X0  X1 class.error
## X0 27  47  0.63513514
## X1 12 119  0.09160305
```

```
rf_mtry$finalModel$forest$cutoff
```

```
## [1] 0.5 0.5
```

```
result<-data.frame(Sens=rf_mtry$results$Sens,SensSD= rf_mtry$results$SensSD,Spec=rf_mtry$results$Spec,S
result
```

```
##      Sens      SensSD      Spec      SpecSD      ROC      ROCSD
## 1  0.3445714 0.1129201 0.8992308 0.05710551 0.6938863 0.08073801
## 2  0.3856190 0.1187946 0.8395442 0.07833820 0.6779315 0.08315197
## 3  0.3813333 0.1205017 0.8236467 0.08376055 0.6717848 0.08505518
## 4  0.3911429 0.1220932 0.8190598 0.07609411 0.6691434 0.08669530
## 5  0.3853333 0.1235770 0.8136752 0.08138075 0.6684148 0.08547976
## 6  0.3976190 0.1291989 0.8144729 0.07717975 0.6639166 0.08745346
## 7  0.3920000 0.1322301 0.8076353 0.08210620 0.6653675 0.08757180
## 8  0.4027619 0.1250742 0.8022222 0.08032530 0.6675443 0.09130348
## 9  0.3946667 0.1253276 0.8060399 0.07543814 0.6633946 0.09053936
## 10 0.4013333 0.1232890 0.8060684 0.08250874 0.6678145 0.08594373
## 11 0.4018095 0.1253359 0.8045014 0.07885393 0.6698575 0.08741470
## 12 0.4149524 0.1250073 0.8060684 0.08171997 0.6701758 0.08847475
## 13 0.4081905 0.1385888 0.8037322 0.08563722 0.6681216 0.09059512
```

```
rf_mtry$finalModel$forest$cutoff
```

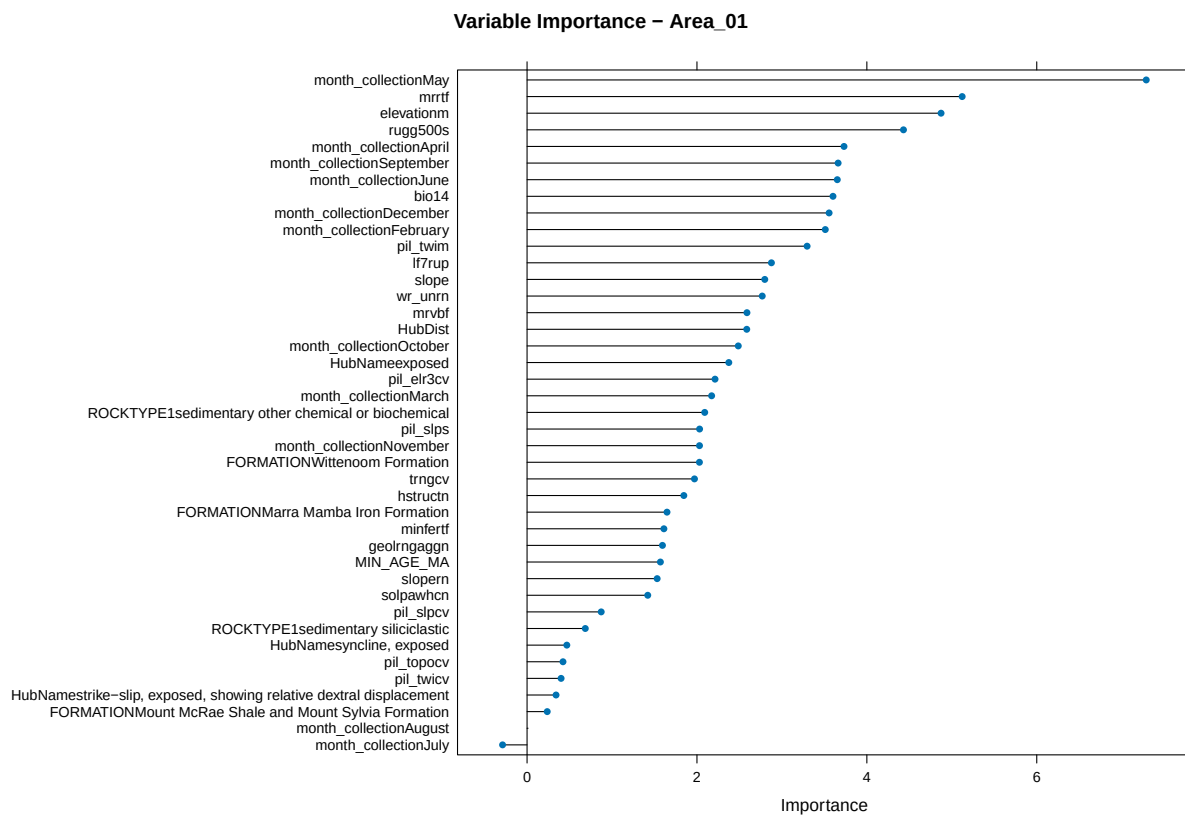
```
## [1] 0.5 0.5
```

```
V_Importance_rf_mtry <- varImp(rf_mtry, scale = FALSE)
V_Importance_rf_mtry
```

```
## rf variable importance
##
##      only 20 most important variables shown (out of 41)
##
##      Importance
## month_collectionMay      7.289
## mrrtf                    5.122
## elevationm               4.873
```

```
## rugg500s 4.431
## month_collectionApril 3.731
## month_collectionSeptember 3.662
## month_collectionJune 3.652
## bio14 3.601
## month_collectionDecember 3.556
## month_collectionFebruary 3.511
## pil_twim 3.297
## lf7rup 2.876
## slope 2.798
## wr_unrn 2.768
## mrvbf 2.588
## HubDist 2.586
## month_collectionOctober 2.486
## HubNameexposed 2.375
## pil_elr3cv 2.213
## month_collectionMarch 2.173
```

```
plot(V_Imp_rf_mtry, main="Variable Importance - Area_01")
```

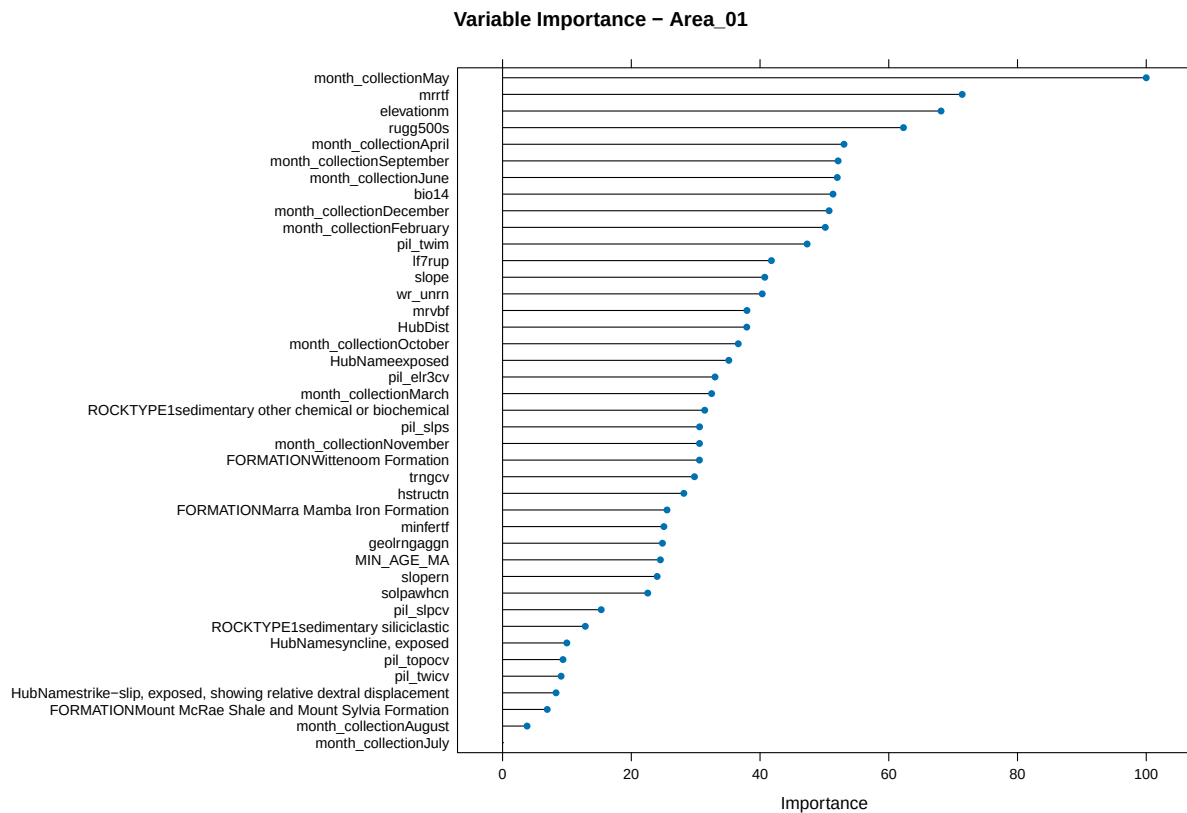


```
V_Imp_rf_mtry <- varImp(rf_mtry, scale = TRUE)
V_Imp_rf_mtry
```

```
## rf variable importance
##
## only 20 most important variables shown (out of 41)
```

```
##
## Importance
## month_collectionMay      100.00
## mrrtf                    71.40
## elevationm               68.12
## rugg500s                 62.28
## month_collectionApril    53.04
## month_collectionSeptember 52.13
## month_collectionJune     52.00
## bio14                   51.33
## month_collectionDecember 50.73
## month_collectionFebruary 50.14
## pil_twim                 47.31
## lf7rup                   41.77
## slope                    40.74
## wr_unrn                  40.34
## mrvbf                    37.97
## HubDist                  37.93
## month_collectionOctober  36.62
## HubNameexposed           35.15
## pil_elr3cv               33.01
## month_collectionMarch    32.48
```

```
plot(V_Imp_rf_mtry, main="Variable Importance - Area_01")
```



PDP

```
library(pdp)
```

```
## Warning: package 'pdp' was built under R version 4.3.3
```

```
# Set file path for saving the plot
```

```
file_path <- "C:/Users/23276776/OneDrive - UWA/DARE/9 - Paper 1/1_Figures/Arrumando_Figuras_LaTeXxX/Ar
```

```
# Open a PDF device
```

```
pdf(file = file_path, width = 9, height = 6) # Adjust width and height as needed
```

```
# Increase left margin (bottom, left, top, right)
```

```
par(mar = c(5, 7, 4, 2))
```

```
name2 <- c("1", "2", "3", "4", "5", "6", "7", "8", "9", "10", "11", "12")
```

```
pd_month_collection <- partial(rf_mtry, pred.var = "month_collection", quantiles = FALSE, prob = TRUE, v
```

```
# Create the plot
```

```
plot(pd_month_collection,
```

```
  main = "Area 1 - Month of collection",
```

```
  ylim = c(0.0, 1.0),
```

```
  names = name2,
```

```
  cex.lab = 1.4,      # Font size for axis titles
```

```
  cex.axis = 1.9,     # Font size for axis numbers
```

```
  cex.main = 1.9,     # Font size for the title
```

```
  cex = 1.6,          # Font size for any other text
```

```
  ylab = expression("Predicted outcome (" *  $\hat{y}$  * ")") # Set the y-axis title
```

```
)
```

```
# Add text annotations
```

```
text(1.5, 0.1,
```

```
  "1 - Jan\n2 - Feb\n3 - Mar\n4 - Apr\n5 - May\n6 - Jun",
```

```
  cex = 1.4,
```

```
  adj = c(0, 0))
```

```
text(3.5, 0.1,
```

```
  "7 - Jul\n8 - Aug\n9 - Sept\n10 - Oct\n11 - Nov\n12 - Dec",
```

```
  cex = 1.4,
```

```
  adj = c(0, 0))
```

```
# Close the PDF device
```

```
dev.off()
```

```
## pdf
```

```
## 2
```

```

#
# # Define names for the categories in UNITNAME
# unitname_labels <- c("Brockman Iron Formation",
#                      "Wittenoom Formation",
#                      "Mount McRae Shale and Mount Sylvia Formation",
#                      "Marra Mamba Iron Formation")
#
# # Compute partial dependence for UNITNAME
# pd_unitname <- partial(rf_mtry, pred.var = "UNITNAME", quantiles = FALSE, prob = TRUE, which.class =
# #
# # Create the plot
# plot(pd_unitname,
#      main = "Partial Dependence of UNITNAME",
#      ylim = c(0.0, 1.0),
#      names = unitname_labels,
#      cex.lab = 1,      # Font size for axis titles
#      cex.axis = 1,     # Font size for axis numbers
#      cex.main = 1.9,   # Font size for the title
#      cex = 1.6,        # Font size for any other text
#      ylab = expression("Predicted outcome (" * hat(y) * ")"), # Set the y-axis title
#      las = 2           # Rotate x-axis labels for better readability
# )
#
# # Add text annotations if needed (example for grouping units)
# text(1.5, 0.1,
#      "Brockman Iron Formation\nWittenoom Formation",
#      cex = 1.4,
#      adj = c(0, 0))
#
# text(3.5, 0.1,
#      "Mount McRae Shale and Mount Sylvia Formation\nMarra Mamba Iron Formation",
#      cex = 1.4,
#      adj = c(0, 0))

```

PDP HubDist

```

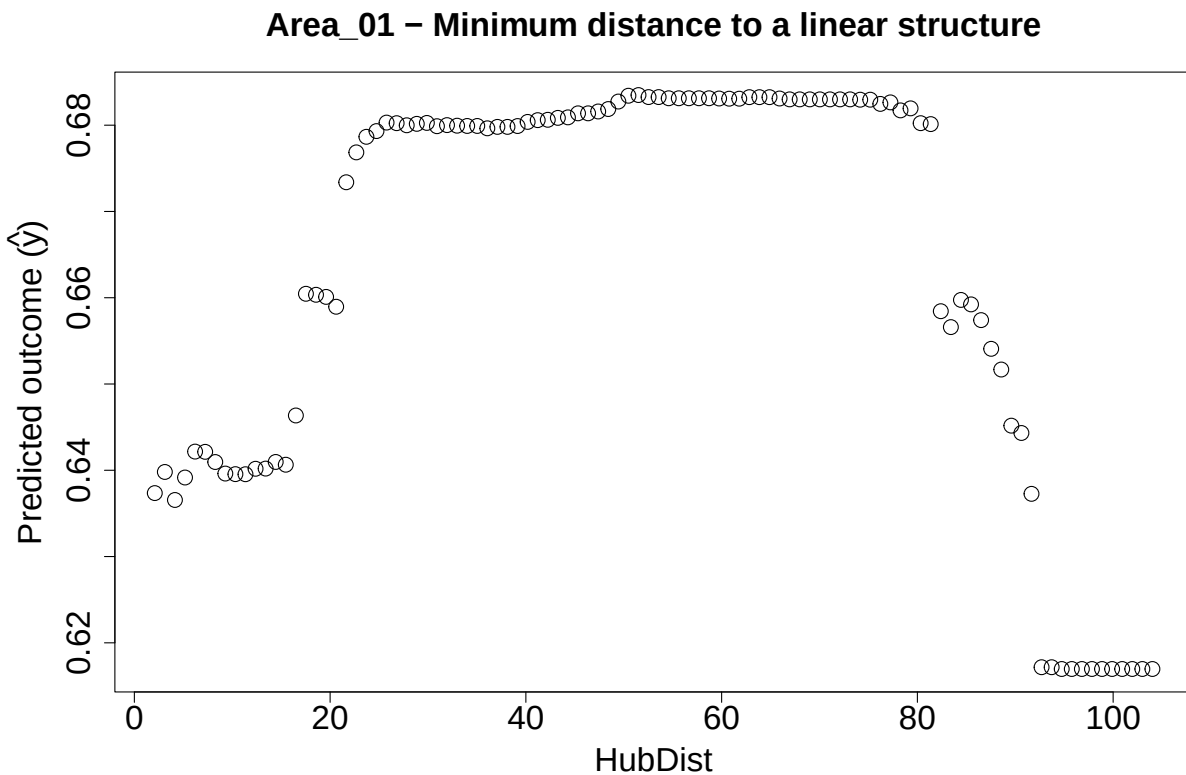
# Set larger margins: c(bottom, left, top, right)
par(mar = c(5.5, 6, 4.5, 2) + 0.1) # Adjust as needed

# Create partial dependence data
pd_HubDist <- partial(rf_mtry, pred.var = "HubDist",
                     type = "classification", prob = TRUE,
                     which.class = "X1", grid.resolution = 100)

# Plot the PDP
plot(pd_HubDist,
     main = "Area_01 - Minimum distance to a linear structure",
     ylab = expression("Predicted outcome (" * hat(y) * ")"),
     cex.lab = 1.9,      # Axis label font size
     cex.axis = 1.9,     # Axis number font size
     cex.main = 1.9,     # Title font size

```

```
)
    cex = 1.9          # General text size
```



```
# Open a PDF device with desired size
pdf("pd_area_1_HubDist.pdf", width = 8, height = 6)

# Set larger margins: c(bottom, left, top, right)
par(mar = c(5.5, 6, 4.5, 2) + 0.1) # Adjust as needed

# Create partial dependence data
pd_HubDist <- partial(rf_mtry, pred.var = "HubDist",
                     type = "classification", prob = TRUE,
                     which.class = "X1", grid.resolution = 100)

# Plot the PDP
plot(pd_HubDist,
     main = "Area_01 - Minimum distance to a linear structure",
     ylab = expression("Predicted outcome (" * hat(y) * ")"),
     cex.lab = 1.9,      # Axis label font size
     cex.axis = 1.9,     # Axis number font size
     cex.main = 1.9,     # Title font size
     cex = 1.9          # General text size
)

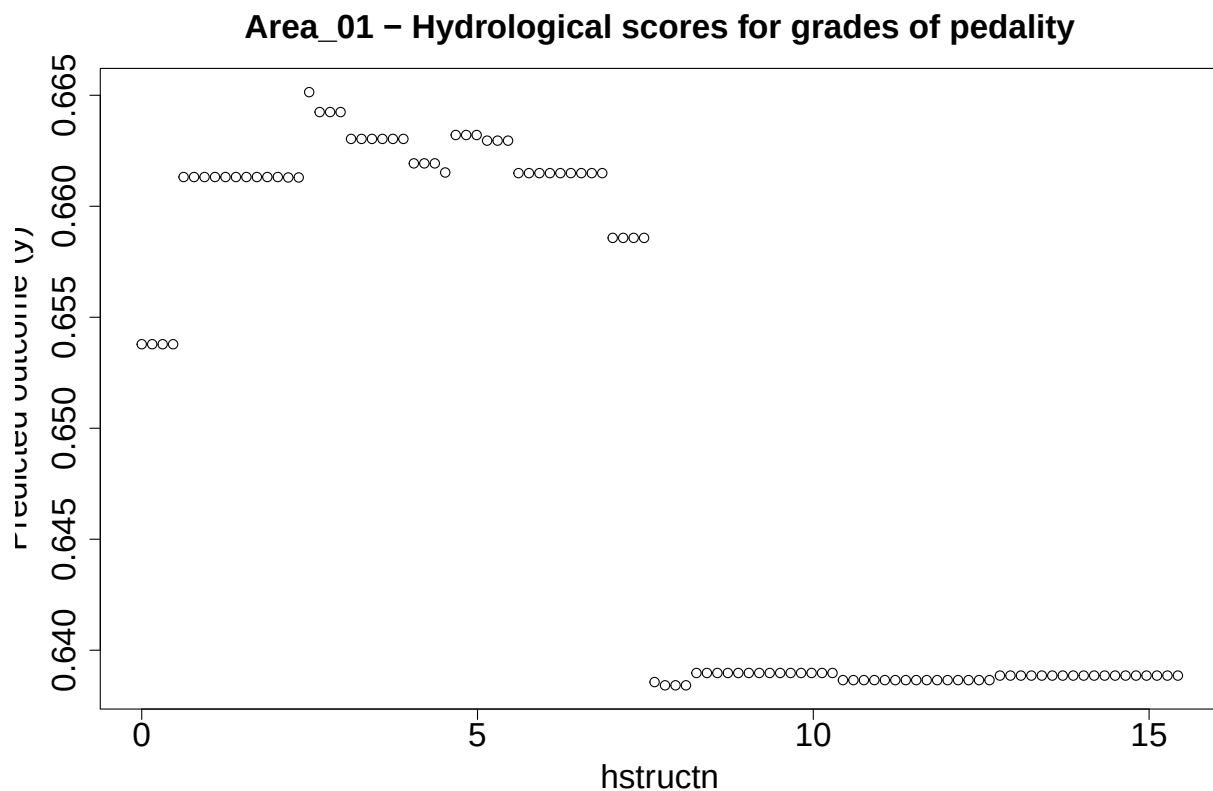
# Close the PDF device
dev.off()
```



```
## pdf
## 2
```

PDP hstructn

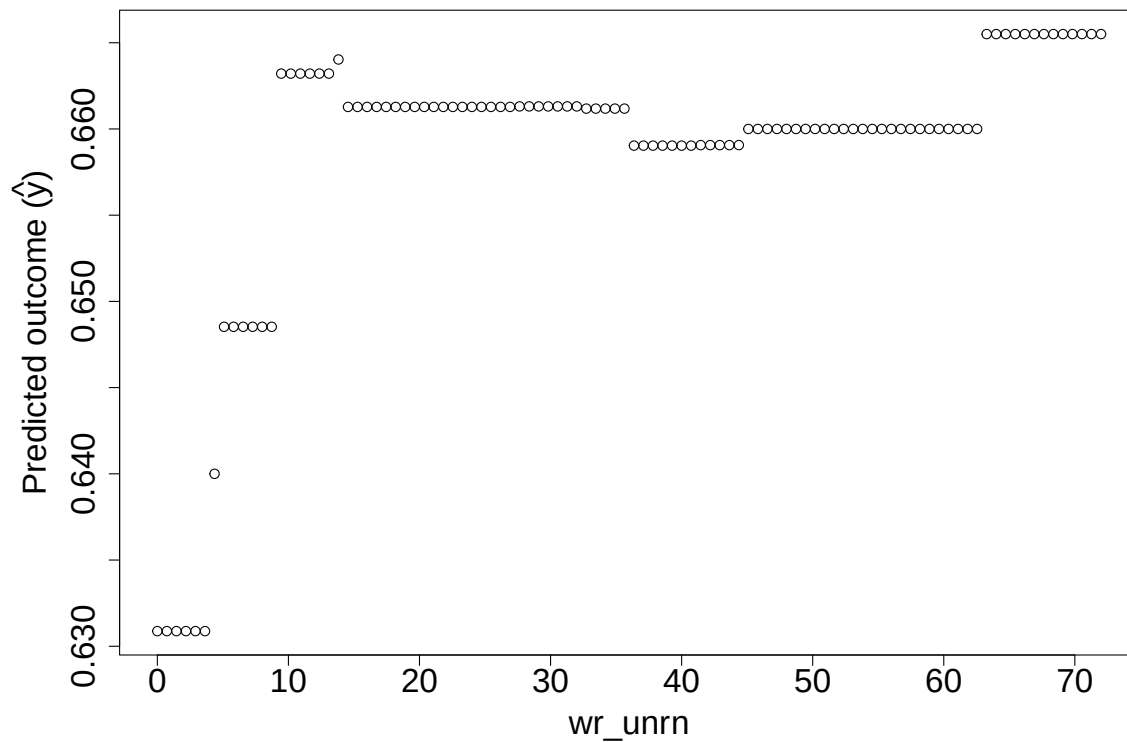
```
pd_hstructn <- partial(rf_mtry, pred.var = "hstructn", type = "classification", prob=TRUE, which.class="X")
plot(pd_hstructn, main = "Area_01 - Hydrological scores for grades of pedality", ylab = expression("Predicted outcome (" * hat(y) * ")"),
     cex.lab = 1.9,      # Font size for axis titles
     cex.axis = 1.9,     # Font size for axis numbers
     cex.main = 1.9,     # Font size for the title
     cex = 1.2)          # Font size for any other text
```



```
# Set larger left margin: mar = c(bottom, left, top, right)
par(mar = c(5, 7, 4, 4))

pd_wr_unrn <- partial(rf_mtry, pred.var = "wr_unrn", type = "classification", prob = TRUE, which.class = "X")
plot(pd_wr_unrn,
     main = "Area_01 - Proportion of soil with unreliable water retention properties",
     ylab = expression("Predicted outcome (" * hat(y) * ")"),
     cex.lab = 1.9,
     cex.axis = 1.9,
     cex.main = 1.9,
     cex = 1.2)
```

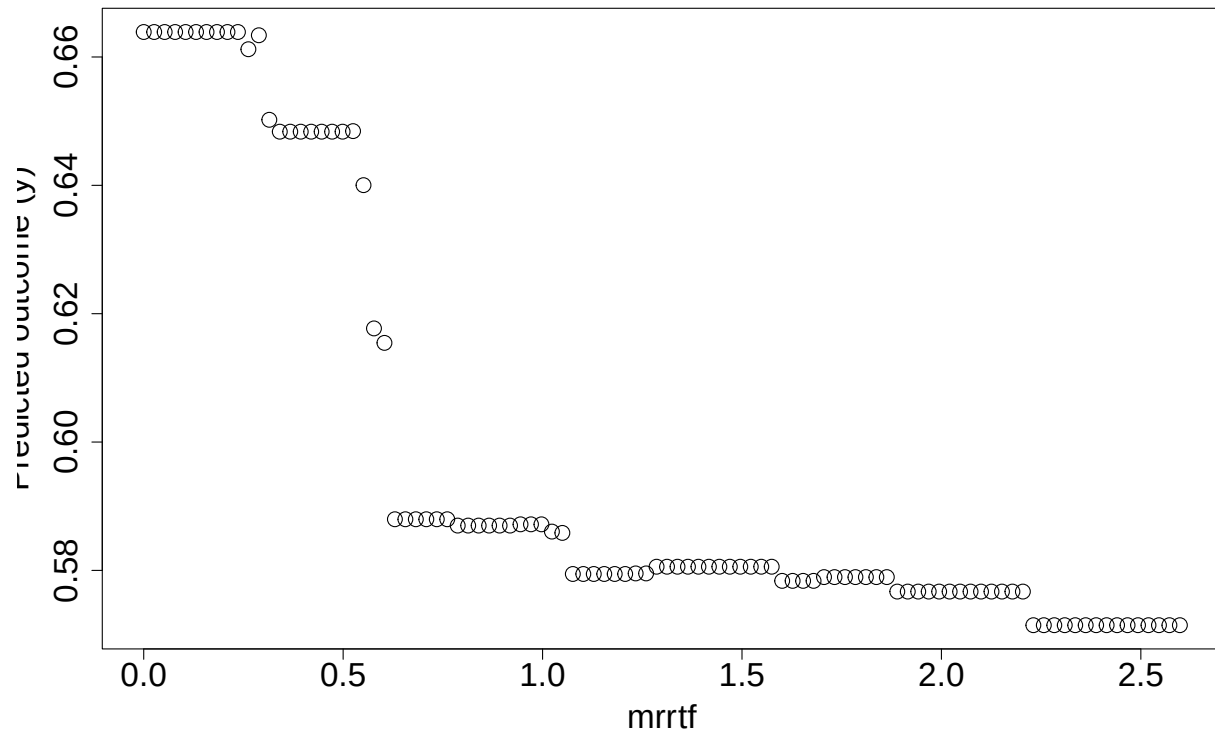
Area_01 – Proportion of soil with unreliable water retention properties



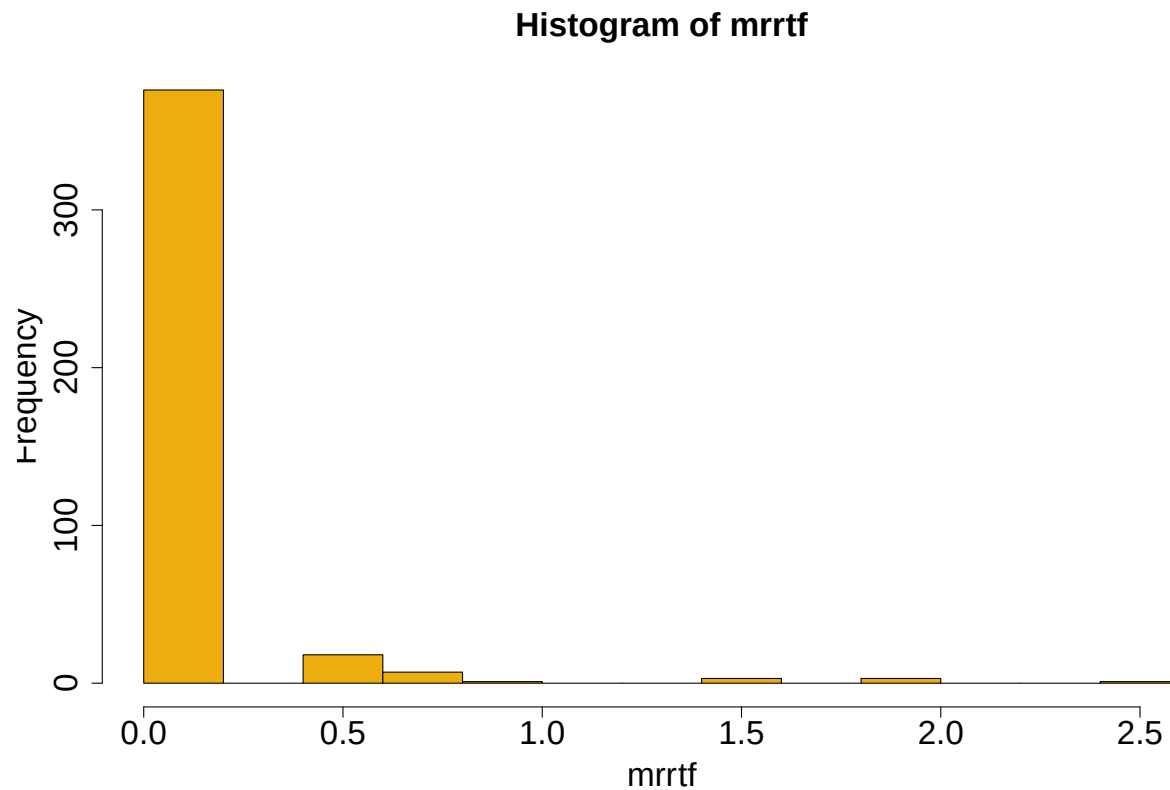
PDP mrrtf

```
pd_mrrtf <- partial(rf_mtry, pred.var = "mrrtf", type = "classification", prob=TRUE, which.class="X1", grid.resolution=10)
plot(pd_mrrtf, main = "Area_01 - Multi-resolution ridgetop flatness index", ylab = expression("Predicted outcome (Ŷ)"),
     cex.lab = 1.9,      # Font size for axis titles
     cex.axis = 1.9,    # Font size for axis numbers
     cex.main = 1.9,    # Font size for the title
     cex = 1.9)         # Font size for any other text
```

Area_01 – Multi-resolution ridgetop flatness index

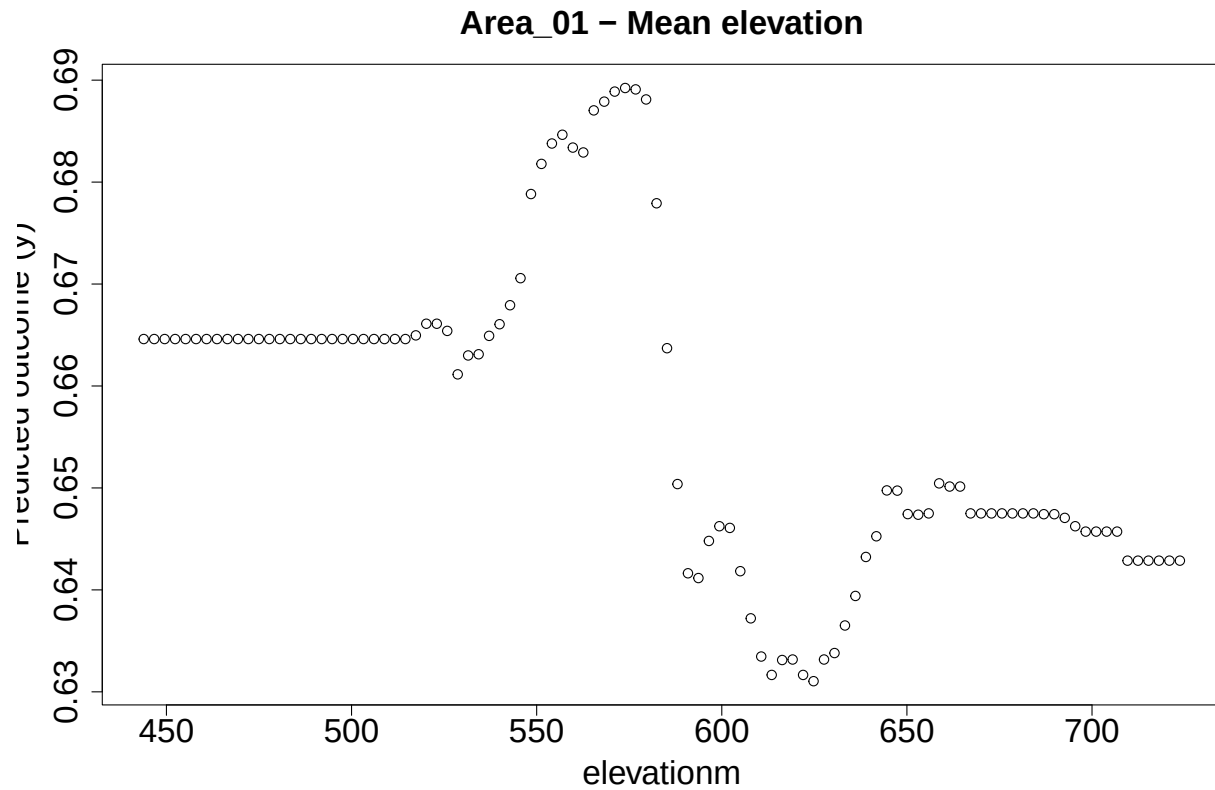


```
# Histogram of mrrtf
hist(data2$mrrtf,
      main = "Histogram of mrrtf",
      xlab = "mrrtf",
      border = "black",
      cex.lab = 1.9,      # Font size for axis titles
      cex.axis = 1.9,     # Font size for axis numbers
      cex.main = 1.9,     # Font size for the title
      cex = 1.9,          # Font size for any other text
      col = "darkgoldenrod2") # Set color for histogram bars
```

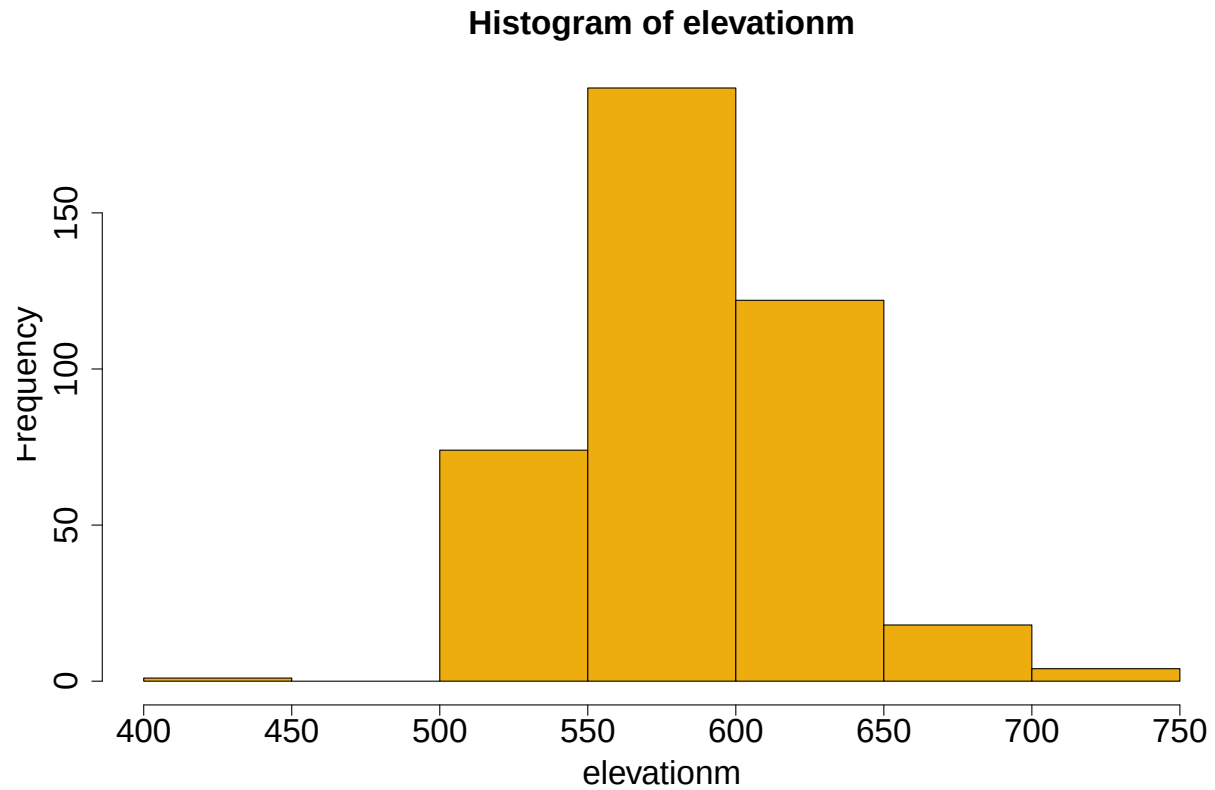


PDP elevationm

```
pd_elevationm <- partial(rf_mtry, pred.var = "elevationm", type = "classification", prob=TRUE, which.class = 1)
plot(pd_elevationm, main = "Area_01 - Mean elevation", ylab = expression("Predicted outcome (" * hat(y))),
     cex.lab = 1.9,      # Font size for axis titles
     cex.axis = 1.9,     # Font size for axis numbers
     cex.main = 1.9,     # Font size for the title
     cex = 1.2)          # Font size for any other text
```



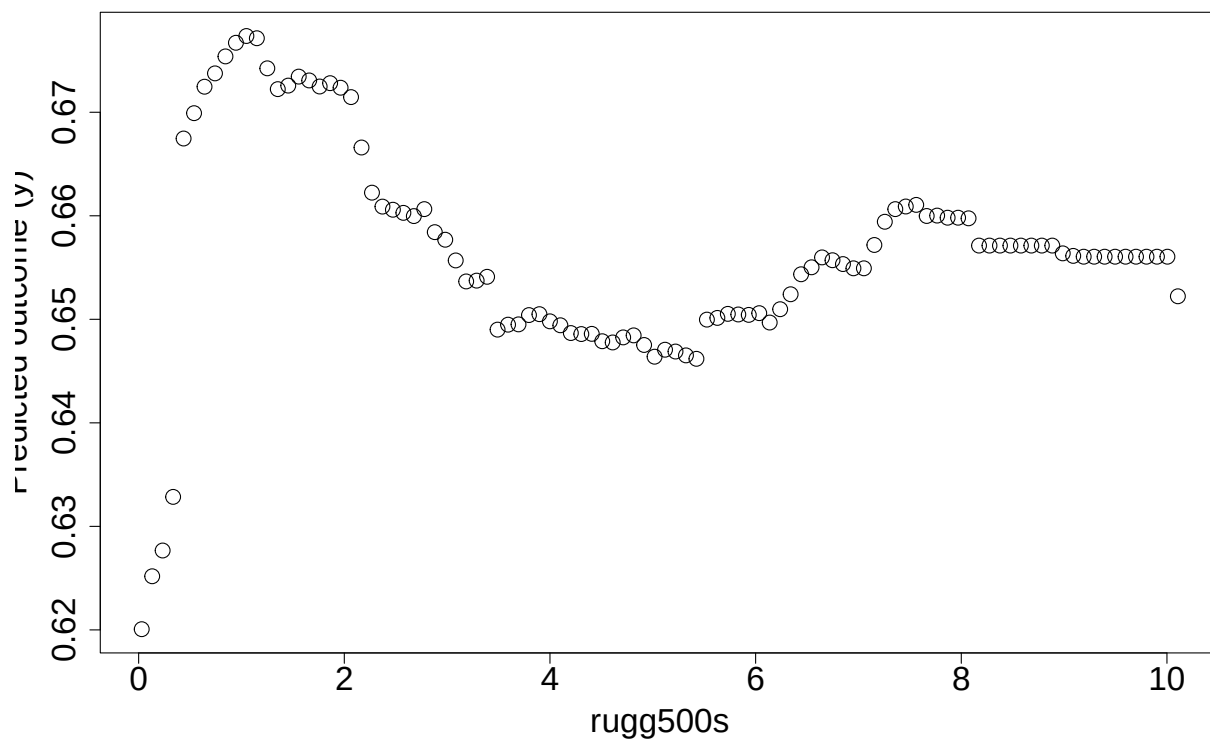
```
# Histogram of elevationm
hist(data2$elevationm,
      main = "Histogram of elevationm",
      xlab = "elevationm",
      border = "black",
      cex.lab = 1.9,      # Font size for axis titles
      cex.axis = 1.9,    # Font size for axis numbers
      cex.main = 1.9,    # Font size for the title
      cex = 1.9,         # Font size for any other text
      col = "darkgoldenrod2") # Set color for histogram bars
```



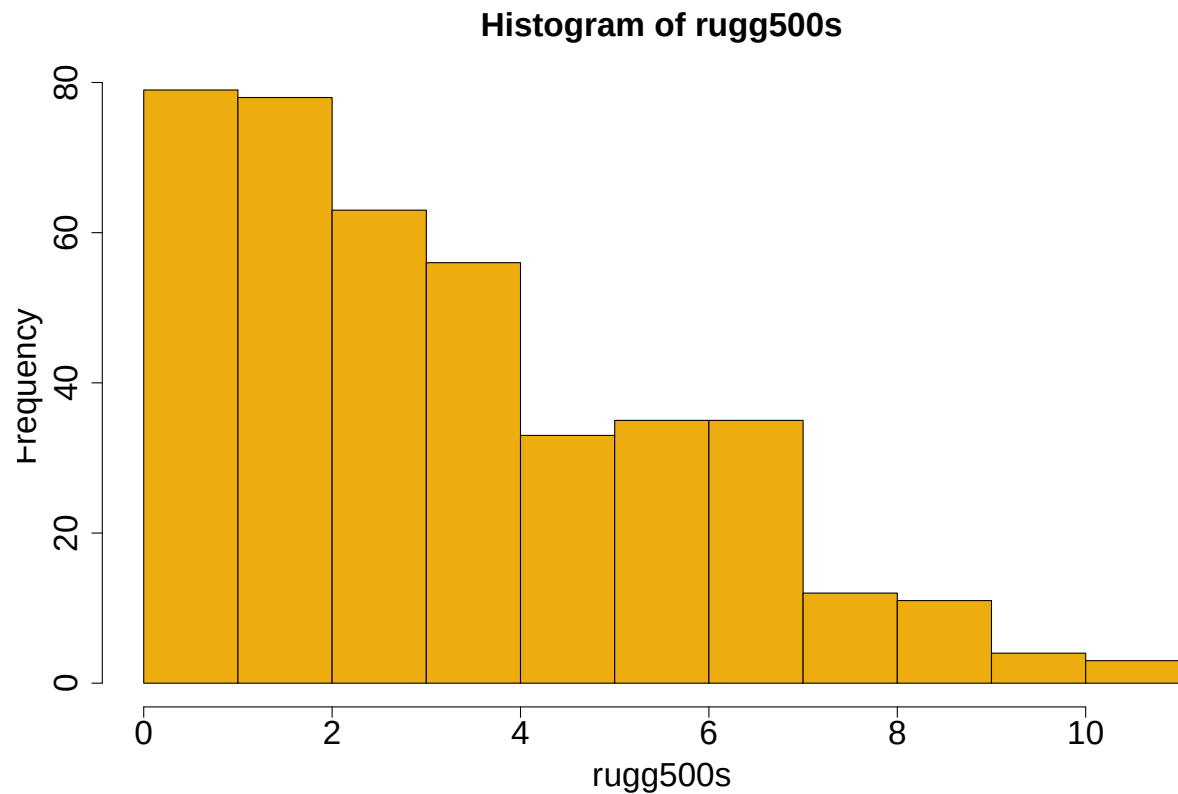
PDP rugg500s

```
pd_rugg500s <- partial(rf_mtry, pred.var = "rugg500s", type = "classification", prob=TRUE, which.class="X")
plot(pd_rugg500s, main = "Area_01 - Standard deviation of terrain ruggedness", ylab = expression("Predi
      cex.lab = 1.9,      # Font size for axis titles
      cex.axis = 1.9,    # Font size for axis numbers
      cex.main = 1.9,    # Font size for the title
      cex = 1.9)        # Font size for any other text
```

Area_01 – Standard deviation of terrain ruggedness

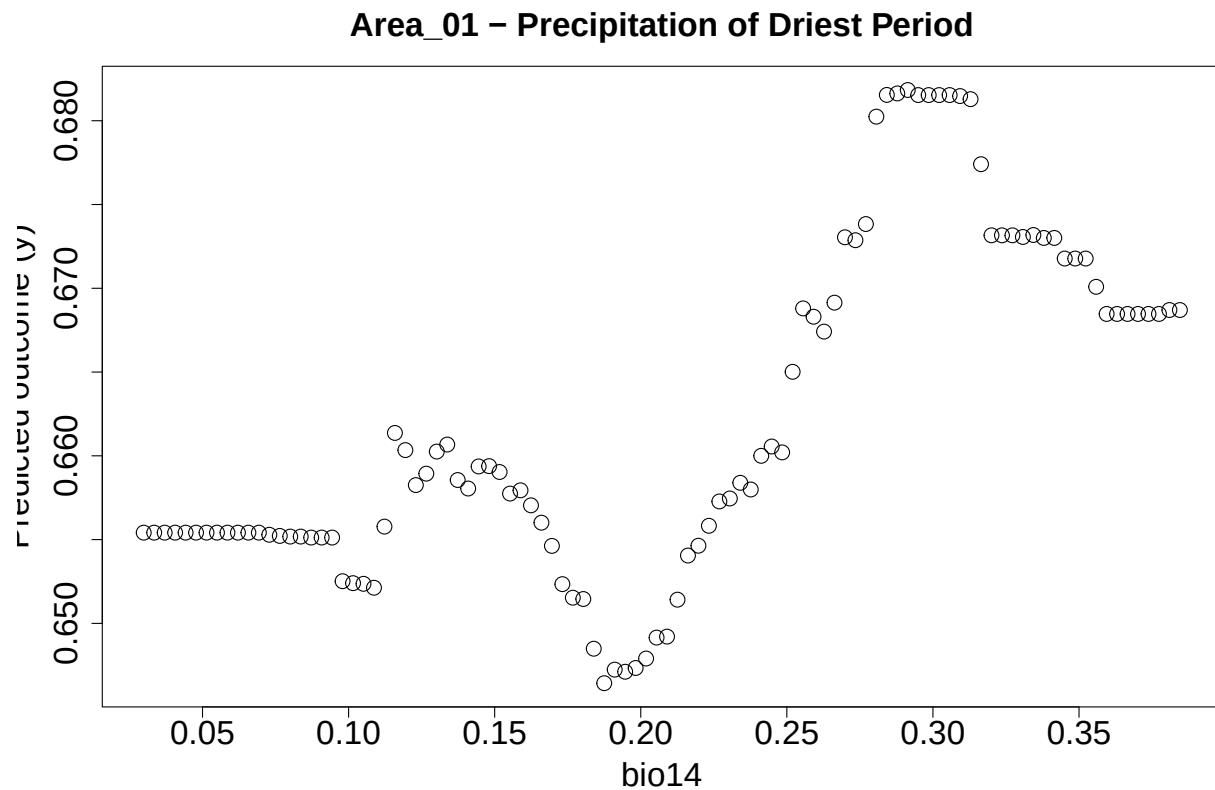


```
# Histogram of rugg500s
hist(data2$rugg500s,
      main = "Histogram of rugg500s",
      xlab = "rugg500s",
      border = "black",
      cex.lab = 1.9,      # Font size for axis titles
      cex.axis = 1.9,     # Font size for axis numbers
      cex.main = 1.9,     # Font size for the title
      cex = 1.9,          # Font size for any other text
      col = "darkgoldenrod2") # Set color for histogram bars
```

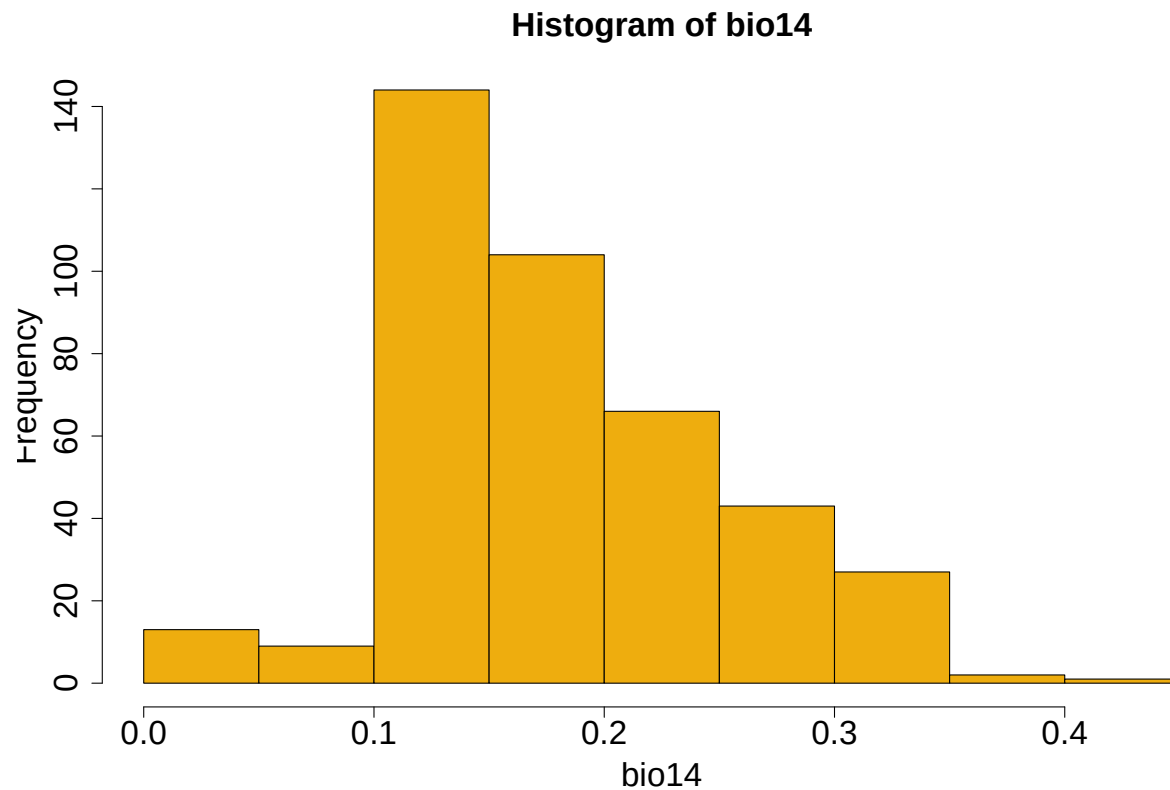


PDP bio14

```
pd_bio14 <- partial(rf_mtry, pred.var = "bio14", type = "classification", prob=TRUE, which.class="X1", grid=100)
plot(pd_bio14, main = "Area_01 - Precipitation of Driest Period", ylab = expression("Predicted outcome"),
     cex.lab = 1.9,      # Font size for axis titles
     cex.axis = 1.9,    # Font size for axis numbers
     cex.main = 1.9,    # Font size for the title
     cex = 1.9)         # Font size for any other text
```

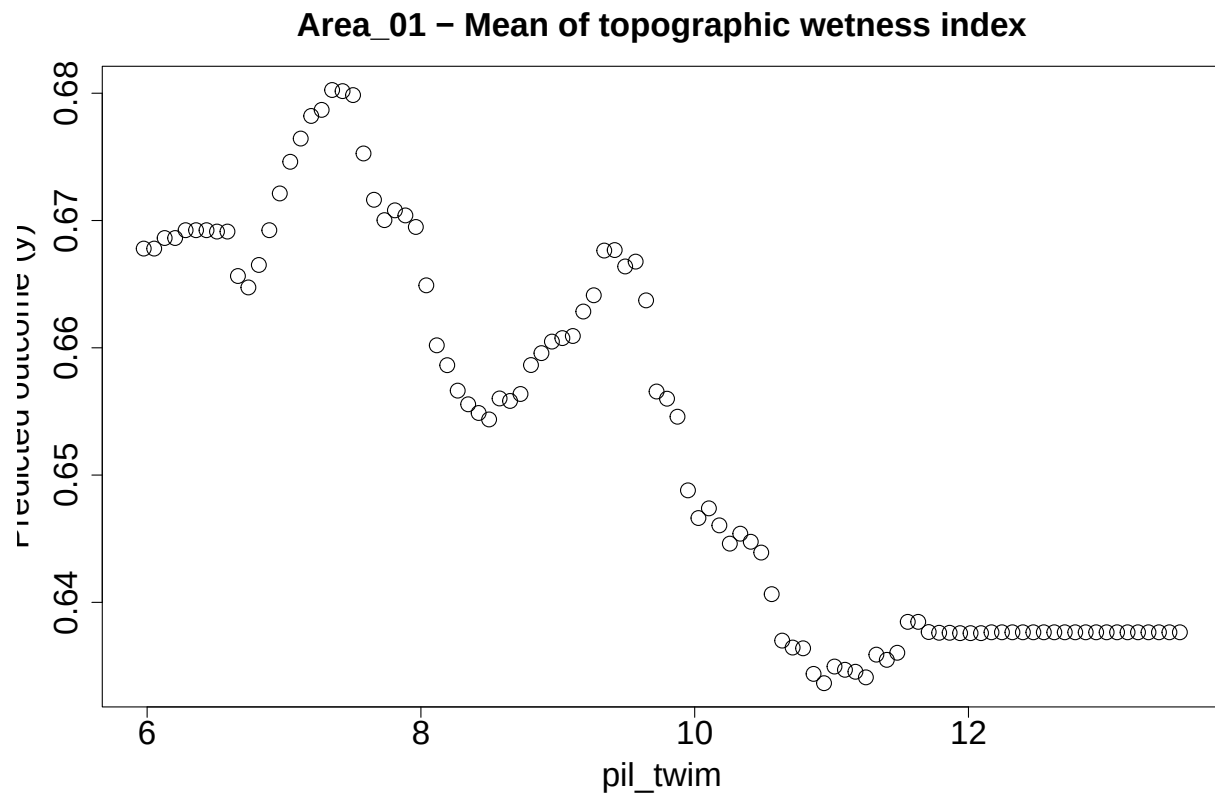



```
# Histogram of bio14
hist(data2$bio14,
      main = "Histogram of bio14",
      xlab = "bio14",
      border = "black",
      cex.lab = 1.9,      # Font size for axis titles
      cex.axis = 1.9,    # Font size for axis numbers
      cex.main = 1.9,    # Font size for the title
      cex = 1.9,         # Font size for any other text
      col = "darkgoldenrod2") # Set color for histogram bars
```

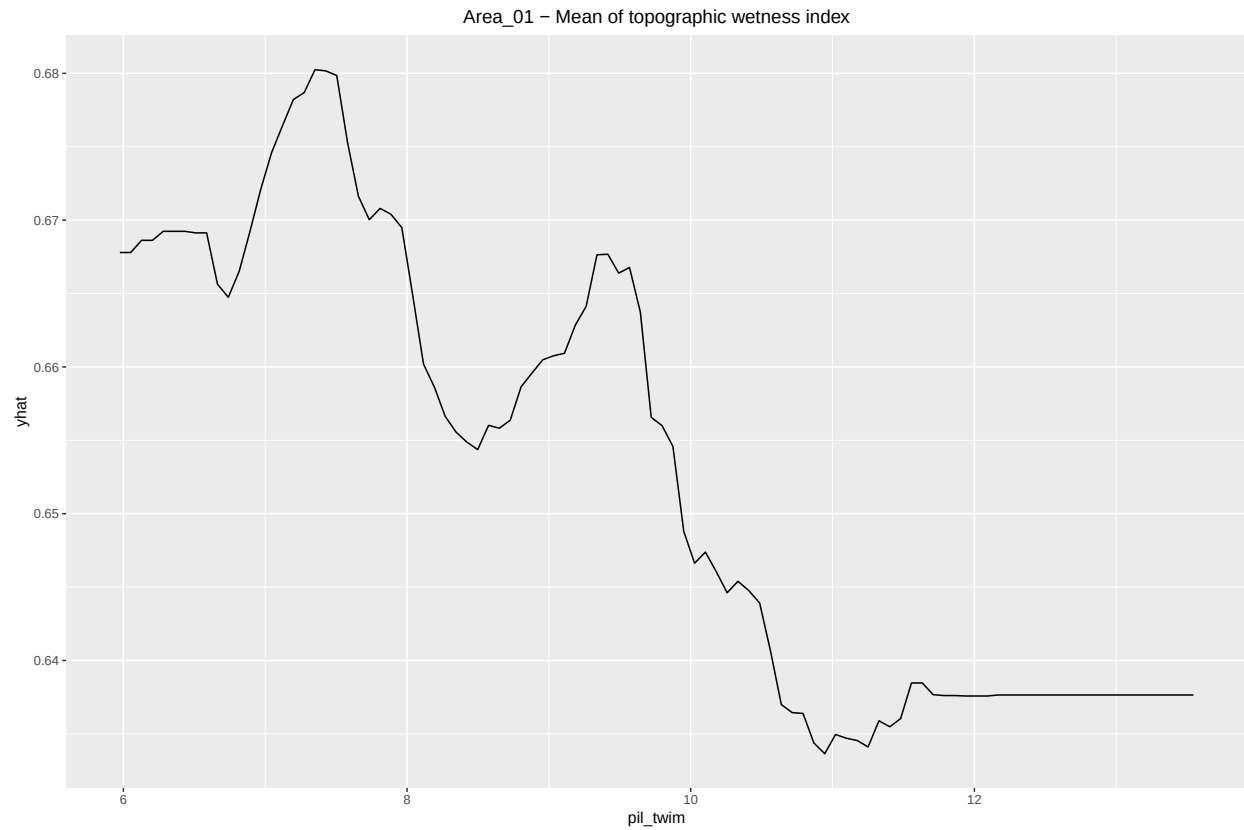


PDP pil_twim

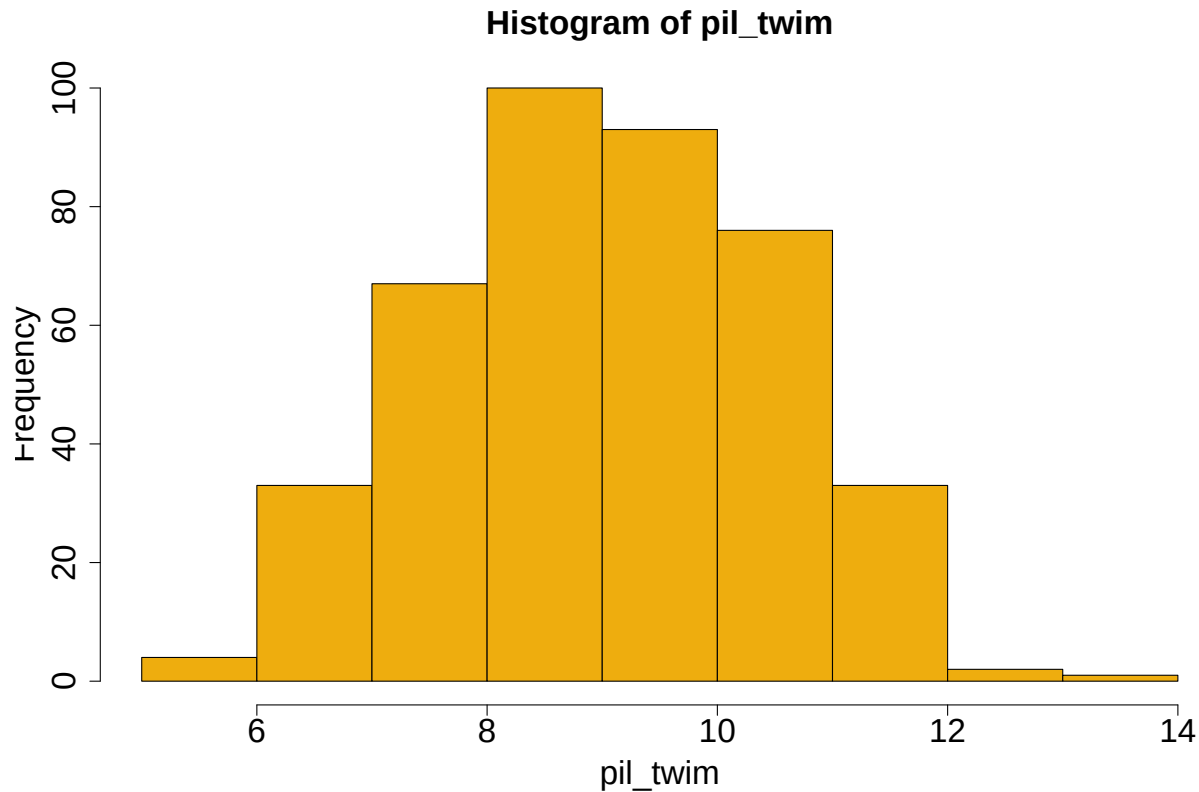
```
pd_pil_twim <- partial(rf_mtry, pred.var = "pil_twim", type = "classification", prob=TRUE, which.class="X")
plot(pd_pil_twim, main = "Area_01 - Mean of topographic wetness index", ylab = expression("Predicted ou
      cex.lab = 1.9,      # Font size for axis titles
      cex.axis = 1.9,    # Font size for axis numbers
      cex.main = 1.9,    # Font size for the title
      cex = 1.9)         # Font size for any other text)
```



```
#Gráfico com contorno. Para centralizar o título use theme()
#https://stackoverflow.com/questions/60678369/center-align-the-title-in-autoplot
plot.pd_pil_twim <- autoplot(pd_pil_twim, main = "Area_01 - Mean of topographic wetness index", contour = FALSE)
plot.pd_pil_twim+theme(plot.title = element_text(hjust = 0.5))
```



```
# Histogram of pil_twim
hist(data2$pil_twim,
      main = "Histogram of pil_twim",
      xlab = "pil_twim",
      border = "black",
      cex.lab = 1.9,      # Font size for axis titles
      cex.axis = 1.9,    # Font size for axis numbers
      cex.main = 1.9,    # Font size for the title
      cex = 1.9,         # Font size for any other text
      col = "darkgoldenrod2") # Set color for histogram bars
```



Export

```
# Export to PDF
pdf("C:/Users/23276776/OneDrive - UWA/DARE/9 - Paper 1/1_Figures/Arrumando_Figuras_LaTeXxxX/vara_1.pdf")

# Set up a 5x2 layout
par(mfrow = c(5, 2),      # 5 rows and 2 columns
    mar = c(4.5, 5, 2, 1), # Margins for each plot (bottom, left, top, right)
    oma = c(2, 2, 2, 2)) # Outer margins

# 1. mrrtf
plot(pd_mrrtf, main = "Area 1 - Multi-resolution ridgetop flatness index", ylab = expression("Predicted outcome (" * hat(y) * ")"),
     cex.lab = 1.9, cex.axis = 1.9, cex.main = 1.9, cex = 1.9)
rug(data_train$mrrtf)
hist(data2$mrrtf,
     main = "Histogram of mrrtf", xlab = "mrrtf",
     border = "black", cex.lab = 1.9, cex.axis = 1.9, cex.main = 1.9, cex = 1.9, col = "darkgoldenrod2")
rug(data_train$mrrtf)

# 2. elevationm
plot(pd_elevationm, main = "Area 1 - Mean elevation", ylab = expression("Predicted outcome (" * hat(y) * ")"),
     cex.lab = 1.9, cex.axis = 1.9, cex.main = 1.9, cex = 1.2)
rug(data_train$elevationm)
hist(data2$elevationm,
```

```

    main = "Histogram of elevationm", xlab = "elevationm",
    border = "black", cex.lab = 1.9, cex.axis = 1.9, cex.main = 1.9, cex = 1.9, col = "darkgoldenrod2")
rug(data_train$elevationm)

# 3. rugg500s
plot(pd_rugg500s, main = "Area 1 - Standard deviation of terrain ruggedness", ylab = expression("Predicted outcome (Standard deviation of terrain ruggedness)"),
     cex.lab = 1.9, cex.axis = 1.9, cex.main = 1.9, cex = 1.9)
rug(data_train$rugg500s)
hist(data2$rugg500s,
     main = "Histogram of rugg500s", xlab = "rugg500s",
     border = "black", cex.lab = 1.9, cex.axis = 1.9, cex.main = 1.9, cex = 1.9, col = "darkgoldenrod2")
rug(data_train$rugg500s)

# 4. bio14
plot(pd_bio14, main = "Area 1 - Precipitation of Driest Period", ylab = expression("Predicted outcome (Precipitation of Driest Period)"),
     cex.lab = 1.9, cex.axis = 1.9, cex.main = 1.9, cex = 1.9)
rug(data_train$bio14)
hist(data2$bio14,
     main = "Histogram of bio14", xlab = "bio14",
     border = "black", cex.lab = 1.9, cex.axis = 1.9, cex.main = 1.9, cex = 1.9, col = "darkgoldenrod2")
rug(data_train$bio14)

# 5. pil_twim
plot(pd_pil_twim, main = "Area 1 - Mean of topographic wetness index", ylab = expression("Predicted outcome (Mean of topographic wetness index)"),
     cex.lab = 1.9, cex.axis = 1.9, cex.main = 1.9, cex = 1.9)
rug(data_train$pil_twim)
hist(data2$pil_twim,
     main = "Histogram of pil_twim", xlab = "pil_twim",
     border = "black", cex.lab = 1.9, cex.axis = 1.9, cex.main = 1.9, cex = 1.9, col = "darkgoldenrod2")
rug(data_train$pil_twim)

# Close the PDF device
dev.off()

```

```
## pdf
## 2
```

```
dim(data_test)
```

```
## [1] 123 27
```

```
colnames(data_test)
```

```
## [1] "trngcv"      "bio14"      "slope"      "rugg500s"
## [5] "pil_twim"    "pil_twicv"  "pil_topocv" "pil_slps"
## [9] "pil_slpcv"   "pil_elr3cv" "mrvbf"      "mrrtf"
## [13] "minfertif"   "lf7rup"     "hstructn"   "geolrngaggn"
## [17] "elevationm"  "wr_unrn"    "solpawhcn"  "slopern"
```

```
## [21] "MIN_AGE_MA"      "HubDist"          "month_collection" "ROCKTYPE1"
## [25] "FORMATION"       "HubName"          "true_troglofauna"
```

```
selectcol_data_test <- data_test[, -c((ncol(data_test) - 0):ncol(data_test))]
dim(selectcol_data_test)
```

```
## [1] 123 26
```

```
names(selectcol_data_test)
```

```
## [1] "trngcv"      "bio14"      "slope"      "rugg500s"
## [5] "pil_twim"    "pil_twicv"  "pil_topocv" "pil_slps"
## [9] "pil_slpcv"   "pil_elr3cv" "mrvbf"      "mrrtf"
## [13] "minfertf"   "lf7rup"     "hstructn"   "geolrngaggn"
## [17] "elevationm" "wr_unrn"    "solpawhcn"  "slopern"
## [21] "MIN_AGE_MA"  "HubDist"    "month_collection" "ROCKTYPE1"
## [25] "FORMATION"   "HubName"
```

```
set.seed(78945)
```

```
predictions <- predict(rf_mtry, newdata = selectcol_data_test, type = "prob")
head(predictions, 5)
```

```
##      X0      X1
## 538 0.678 0.322
## 580 0.312 0.688
## 449 0.526 0.474
## 149 0.088 0.912
## 304 0.078 0.922
```

```
dim(predictions)
```

```
## [1] 123 2
```

```
sapply(predictions, class)
```

```
##      X0      X1
## "numeric" "numeric"
```

```
summary(predictions)
```

```
##      X0      X1
## Min.   :0.0420 Min.   :0.1940
## 1st Qu.:0.1770 1st Qu.:0.5390
## Median :0.3100 Median :0.6900
## Mean   :0.3353 Mean   :0.6647
## 3rd Qu.:0.4610 3rd Qu.:0.8230
## Max.   :0.8060 Max.   :0.9580
```

```
set.seed(78945)
predictions_raw <- predict(rf_mtry, newdata = selectcol_data_test, type = "raw") #The number/class predic
head(predictions_raw, 5)
```

```
## [1] X0 X1 X0 X1 X1
## Levels: X0 X1
```

```
length(predictions_raw)
```

```
## [1] 123
```

```
head(sapply(predictions_raw, class))
```

```
## [1] "factor" "factor" "factor" "factor" "factor" "factor"
```

```
summary(predictions_raw)
```

```
## X0 X1
## 26 97
```

```
set.seed(78945)
predictions1 <- predict(rf_mtry, newdata = selectcol_data_test)
head(predictions1, 5)
```

```
## [1] X0 X1 X0 X1 X1
## Levels: X0 X1
```

```
length(predictions1)
```

```
## [1] 123
```

```
head(sapply(predictions1, class))
```

```
## [1] "factor" "factor" "factor" "factor" "factor" "factor"
```

```
summary(predictions1)
```

```
## X0 X1
## 26 97
```

```
levels(predictions1) <- c(0,1)
head(predictions1, 5)
```

```
## [1] 0 1 0 1 1
## Levels: 0 1
```



```
summary(predictions1)
```

```
## 0 1  
## 26 97
```

```
set.seed(78945)  
CM_predictions1<-confusionMatrix(predictions1, ContraProva,positive="1")  
CM_predictions1
```

```
## Confusion Matrix and Statistics  
##  
##           Reference  
## Prediction 0 1  
##           0 19 7  
##           1 25 72  
##  
##           Accuracy : 0.7398  
##           95% CI : (0.653, 0.8148)  
##       No Information Rate : 0.6423  
##       P-Value [Acc > NIR] : 0.013751  
##  
##           Kappa : 0.3774  
##  
## Mcnemar's Test P-Value : 0.002654  
##  
##           Sensitivity : 0.9114  
##           Specificity : 0.4318  
##       Pos Pred Value : 0.7423  
##       Neg Pred Value : 0.7308  
##           Prevalence : 0.6423  
##       Detection Rate : 0.5854  
##       Detection Prevalence : 0.7886  
##       Balanced Accuracy : 0.6716  
##  
##       'Positive' Class : 1  
##
```

```
str(CM_predictions1)
```

```
## List of 6  
## $ positive: chr "1"  
## $ table : 'table' int [1:2, 1:2] 19 25 7 72  
## ..- attr(*, "dimnames")=List of 2  
## .. ..$ Prediction: chr [1:2] "0" "1"  
## .. ..$ Reference : chr [1:2] "0" "1"  
## $ overall : Named num [1:7] 0.74 0.377 0.653 0.815 0.642 ...  
## ..- attr(*, "names")= chr [1:7] "Accuracy" "Kappa" "AccuracyLower" "AccuracyUpper" ...  
## $ byClass : Named num [1:11] 0.911 0.432 0.742 0.731 0.742 ...  
## ..- attr(*, "names")= chr [1:11] "Sensitivity" "Specificity" "Pos Pred Value" "Neg Pred Value" ...  
## $ mode : chr "sens_spec"  
## $ dots : list()  
## - attr(*, "class")= chr "confusionMatrix"
```

```
CM_predictions1$byClass[1]#Sensitivity de CM_predictions1
```

```
## Sensitivity
## 0.9113924
```

```
CM_predictions1$byClass[2]#Specificity de CM_predictions1
```

```
## Specificity
## 0.4318182
```

```
CM_predictions1$byClass
```

##	Sensitivity	Specificity	Pos Pred Value
##	0.9113924	0.4318182	0.7422680
##	Neg Pred Value	Precision	Recall
##	0.7307692	0.7422680	0.9113924
##	F1	Prevalence	Detection Rate
##	0.8181818	0.6422764	0.5853659
##	Detection Prevalence	Balanced Accuracy	
##	0.7886179	0.6716053	

```
CM_predictions1$byClass["Sensitivity"]
```

```
## Sensitivity
## 0.9113924
```

```
CM_predictions1$byClass[1]
```

```
## Sensitivity
## 0.9113924
```

```
CM_predictions1$byClass["Balanced Accuracy"]
```

```
## Balanced Accuracy
## 0.6716053
```

```
CM_predictions1$byClass[11]
```

```
## Balanced Accuracy
## 0.6716053
```

```
CM_predictions1$overall
```

##	Accuracy	Kappa	AccuracyLower	AccuracyUpper	AccuracyNull
##	0.739837398	0.377412211	0.653020483	0.814750920	0.642276423
##	AccuracyPValue	McNemarPValue			
##	0.013750865	0.002654029			

```
CM_predictions1$overall["McNemarPValue"]
```

```
## McNemarPValue  
## 0.002654029
```

```
CM_predictions1$overall[7]
```

```
## McNemarPValue  
## 0.002654029
```

```
CM_predictions1$table
```

```
##           Reference  
## Prediction 0  1  
##           0 19  7  
##           1 25 72
```

```
tn<-CM_predictions1$table[1] #TN  
fp<-CM_predictions1$table[2] #FP  
fn<-CM_predictions1$table[3] #FN  
tp<-CM_predictions1$table[4] #TP
```

```
#Youden's J statistic
```

```
J_CM_predictions1<-(CM_predictions1$byClass[1] + CM_predictions1$byClass[2]) - 1  
J_CM_predictions1
```

```
## Sensitivity  
## 0.3432106
```

```
J_CM_predictions1<-as.numeric(CM_predictions1$byClass[1] + CM_predictions1$byClass[2]) - 1  
J_CM_predictions1
```

```
## [1] 0.3432106
```

```
rf_mtry$finalModel$forest$cutoff
```

```
## [1] 0.5 0.5
```

```
mcc(predictions1, ContraProva)
```

```
## [1] 0.4029285
```

```
get_confusion_elements(CM_predictions1)
```

```
## [1] 72 25 19 7
```

```
get_confusion_elements(CM_predictions1)[1]
```

```
## [1] 72
```

```
calculate_mcc(tp, fp, tn, fn)
```

```
## [1] 0.4029285
```

```
calculate_mcc1(CM_predictions1)
```

```
## [1] 0.4029285
```

```
calculate_F2(CM_predictions1)
```

```
## [1] 0.8716707
```

```
model_pred_class <- ifelse(predictions < 0.5, "X0", "X1")
```

```
head(model_pred_class,5)
```

```
##      X0   X1  
## 538 "X1" "X0"  
## 580 "X0" "X1"  
## 449 "X1" "X0"  
## 149 "X0" "X1"  
## 304 "X0" "X1"
```

```
dim(model_pred_class)
```

```
## [1] 123  2
```

```
length(ContraProva)
```

```
## [1] 123
```

```
summary(model_pred_class)
```

```
##      X0      X1  
## Length:123    Length:123  
## Class :character Class :character  
## Mode  :character Mode  :character
```

```
Test1<-as.factor(model_pred_class[,2])  
head(Test1)
```

```
## 538 580 449 149 304 15  
## X0 X1 X0 X1 X1 X1  
## Levels: X0 X1
```

```
summary(Test1)
```

```
## X0 X1  
## 25 98
```

```
levels(Test1) <- c(0,1)  
summary(Test1)
```

```
## 0 1  
## 25 98
```

```
head(Test1)
```

```
## 538 580 449 149 304 15  
## 0 1 0 1 1 1  
## Levels: 0 1
```

```
confusionMatrix(Test1, ContraProva, positive="1")
```

```
## Confusion Matrix and Statistics  
##  
##           Reference  
## Prediction 0 1  
##           0 18 7  
##           1 26 72  
##  
##           Accuracy : 0.7317  
##           95% CI : (0.6443, 0.8076)  
##           No Information Rate : 0.6423  
##           P-Value [Acc > NIR] : 0.022378  
##  
##           Kappa : 0.3544  
##  
##           McNemar's Test P-Value : 0.001728  
##  
##           Sensitivity : 0.9114  
##           Specificity : 0.4091  
##           Pos Pred Value : 0.7347  
##           Neg Pred Value : 0.7200  
##           Prevalence : 0.6423  
##           Detection Rate : 0.5854  
##           Detection Prevalence : 0.7967  
##           Balanced Accuracy : 0.6602  
##  
##           'Positive' Class : 1  
##
```

```
summary(Test1); summary(predictions1)
```

```
## 0 1  
## 25 98
```

```
## 0 1
## 26 97
```

```
model_pred_class <- ifelse(predictions < 0.7, "X0", "X1")
head(model_pred_class,20)
```

```
##      X0   X1
## 538 "X0" "X0"
## 580 "X0" "X0"
## 449 "X0" "X0"
## 149 "X0" "X1"
## 304 "X0" "X1"
## 15  "X0" "X1"
## 104 "X0" "X1"
## 394 "X0" "X0"
## 392 "X0" "X0"
## 503 "X0" "X0"
## 99  "X0" "X1"
## 198 "X0" "X1"
## 629 "X0" "X1"
## 505 "X0" "X1"
## 442 "X1" "X0"
## 32  "X0" "X0"
## 195 "X0" "X0"
## 30  "X0" "X1"
## 332 "X0" "X1"
## 327 "X0" "X0"
```

```
dim(model_pred_class)
```

```
## [1] 123  2
```

```
length(ContraProva)
```

```
## [1] 123
```

```
head(sapply(model_pred_class, class))
```

```
##      X0      X0      X0      X0      X0      X0
## "character" "character" "character" "character" "character" "character"
```

```
Test11<-as.factor(model_pred_class[,2])
summary(Test11)
```

```
## X0 X1
## 63 60
```

```
head(Test11)
```

```
## 538 580 449 149 304 15
## X0 X0 X0 X1 X1 X1
## Levels: X0 X1
```

```
levels(Test11) <- c(0,1)
summary(Test11)
```

```
## 0 1
## 63 60
```

```
summary(predictions1)
```

```
## 0 1
## 26 97
```

```
CM_Test11<-confusionMatrix(Test11, ContraProva,positive="1")
CM_Test11
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  0  1
##           0 32 31
##           1 12 48
##
##           Accuracy : 0.6504
##           95% CI : (0.5592, 0.7342)
##           No Information Rate : 0.6423
##           P-Value [Acc > NIR] : 0.466089
##
##           Kappa : 0.3056
##
##           McNemar's Test P-Value : 0.006052
##
##           Sensitivity : 0.6076
##           Specificity : 0.7273
##           Pos Pred Value : 0.8000
##           Neg Pred Value : 0.5079
##           Prevalence : 0.6423
##           Detection Rate : 0.3902
##           Detection Prevalence : 0.4878
##           Balanced Accuracy : 0.6674
##
##           'Positive' Class : 1
##
```

```
str(CM_Test11)
```

```
## List of 6
## $ positive: chr "1"
## $ table : 'table' int [1:2, 1:2] 32 12 31 48
## ..- attr(*, "dimnames")=List of 2
## .. ..$ Prediction: chr [1:2] "0" "1"
## .. ..$ Reference : chr [1:2] "0" "1"
## $ overall : Named num [1:7] 0.65 0.306 0.559 0.734 0.642 ...
## ..- attr(*, "names")= chr [1:7] "Accuracy" "Kappa" "AccuracyLower" "AccuracyUpper" ...
## $ byClass : Named num [1:11] 0.608 0.727 0.8 0.508 0.8 ...
## ..- attr(*, "names")= chr [1:11] "Sensitivity" "Specificity" "Pos Pred Value" "Neg Pred Value" ...
## $ mode : chr "sens_spec"
## $ dots : list()
## - attr(*, "class")= chr "confusionMatrix"
```

```
CM_Test11$byClass[1]#Sensitivity de CM_predictions1
```

```
## Sensitivity
## 0.6075949
```

```
CM_Test11$byClass[2]#Specificity de CM_predictions1
```

```
## Specificity
## 0.7272727
```

```
J_CM_Test11<-as.numeric(CM_Test11$byClass[1] + CM_Test11$byClass[2] - 1) ; J_CM_Test11
```

```
## [1] 0.3348677
```

```
confusionMatrix(predictions1, ContraProva,positive="1")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  0  1
##           0 19  7
##           1 25 72
##
##           Accuracy : 0.7398
##           95% CI : (0.653, 0.8148)
##           No Information Rate : 0.6423
##           P-Value [Acc > NIR] : 0.013751
##
##           Kappa : 0.3774
##
##           Mcnemar's Test P-Value : 0.002654
##
##           Sensitivity : 0.9114
##           Specificity : 0.4318
##           Pos Pred Value : 0.7423
##           Neg Pred Value : 0.7308
##           Prevalence : 0.6423
```



```
##          Detection Rate : 0.5854
##    Detection Prevalence : 0.7886
##      Balanced Accuracy : 0.6716
##
##      'Positive' Class : 1
##
```

```
dim(finalTest)
```

```
## [1] 81 27
```

```
names(finalTest)
```

```
## [1] "trngcv"      "bio14"      "slope"      "rugg500s"
## [5] "pil_twim"    "pil_twicv"  "pil_topocv" "pil_slps"
## [9] "pil_slpcv"    "pil_elr3cv" "mrvbf"      "mrrtf"
## [13] "minfertf"    "lf7rup"     "hstructn"   "geolrngaggn"
## [17] "elevationm"  "wr_unrn"    "solpawhcn"  "slopern"
## [21] "MIN_AGE_MA"  "HubDist"    "month_collection" "ROCKTYPE1"
## [25] "FORMATION"   "HubName"    "true_troglofauna"
```

```
selectcol_finalTest <- finalTest[, -c((ncol(finalTest) - 0):ncol(finalTest))]
dim(selectcol_finalTest)
```

```
## [1] 81 26
```

```
names(selectcol_finalTest)
```

```
## [1] "trngcv"      "bio14"      "slope"      "rugg500s"
## [5] "pil_twim"    "pil_twicv"  "pil_topocv" "pil_slps"
## [9] "pil_slpcv"    "pil_elr3cv" "mrvbf"      "mrrtf"
## [13] "minfertf"    "lf7rup"     "hstructn"   "geolrngaggn"
## [17] "elevationm"  "wr_unrn"    "solpawhcn"  "slopern"
## [21] "MIN_AGE_MA"  "HubDist"    "month_collection" "ROCKTYPE1"
## [25] "FORMATION"   "HubName"
```

```
predictions_2 <- predict(rf_mtry, newdata = selectcol_finalTest, type = "raw")
```

```
summary(predictions_2)
```

```
## X0 X1
## 10 71
```

```
summary(finalTest[, ncol(finalTest)])
```

```
## 0 1
## 29 52
```

```
levels(predictions_2) <- c(0,1)
summary(predictions_2)
```

```
## 0 1
## 10 71
```

```
CM_predictions_2<-confusionMatrix(predictions_2, finalTest[, ncol(finalTest)],positive="1")
CM_predictions_2
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction 0  1
##           0  9  1
##           1 20 51
##
##           Accuracy : 0.7407
##           95% CI : (0.6314, 0.8318)
##           No Information Rate : 0.642
##           P-Value [Acc > NIR] : 0.0387
##
##           Kappa : 0.3404
##
##           Mcnemar's Test P-Value : 8.568e-05
##
##           Sensitivity : 0.9808
##           Specificity : 0.3103
##           Pos Pred Value : 0.7183
##           Neg Pred Value : 0.9000
##           Prevalence : 0.6420
##           Detection Rate : 0.6296
##           Detection Prevalence : 0.8765
##           Balanced Accuracy : 0.6456
##
##           'Positive' Class : 1
##
```

```
mcc(predictions_2, finalTest[, ncol(finalTest)])
```

```
## [1] 0.4242625
```

```
calculate_mcc1(CM_predictions_2)
```

```
## [1] 0.4242625
```

```
calculate_F2(CM_predictions_2)
```

```
## [1] 0.9139785
```

```
predictions_3<- predict(rf_mtry, newdata = selectcol_finalTest,type = "prob")

summary(predictions_3)
```

```
##           X0           X1
## Min.      :0.078   Min.   :0.180
## 1st Qu.:0.204   1st Qu.:0.570
## Median :0.324   Median :0.676
## Mean      :0.327   Mean    :0.673
## 3rd Qu.:0.430   3rd Qu.:0.796
## Max.      :0.820   Max.    :0.922
```

```
sapply(predictions_3, class)
```

```
##           X0           X1
## "numeric" "numeric"
```

```
head(predictions_3)
```

```
##           X0    X1
## 95  0.374 0.626
## 626 0.110 0.890
## 607 0.328 0.672
## 623 0.168 0.832
## 18  0.538 0.462
## 470 0.278 0.722
```

```
model_pred_class_3 <- ifelse(predictions_3 < 0.5, "X0", "X1")
head(sapply(model_pred_class_3, class))
```

```
##           X0           X0           X0           X0           X1           X0
## "character" "character" "character" "character" "character" "character"
```

```
Test_3<-as.factor(model_pred_class_3[,2])
head(sapply(Test_3, class))
```

```
##           95           626           607           623           18           470
## "factor" "factor" "factor" "factor" "factor" "factor"
```

```
levels(Test_3)
```

```
## [1] "X0" "X1"
```

```
head(Test_3)
```

```
## 95 626 607 623 18 470
## X1 X1 X1 X1 X0 X1
## Levels: X0 X1
```

```
levels(Test_3) <- c(0,1)
head(sapply(Test_3, class))
```

```
##      95      626      607      623      18      470
## "factor" "factor" "factor" "factor" "factor" "factor"
```

```
levels(Test_3)
```

```
## [1] "0" "1"
```

```
head(Test_3)
```

```
## 95 626 607 623 18 470
##  1  1  1  1  0  1
## Levels: 0 1
```

```
summary(Test_3)
```

```
## 0 1
## 10 71
```

```
CM_predictions_3<-confusionMatrix(Test_3, finalTest[, ncol(finalTest)],positive="1")
CM_predictions_3
```

```
## Confusion Matrix and Statistics
##
##              Reference
## Prediction  0  1
##           0  9  1
##           1 20 51
##
##              Accuracy : 0.7407
##              95% CI : (0.6314, 0.8318)
##      No Information Rate : 0.642
##      P-Value [Acc > NIR] : 0.0387
##
##              Kappa : 0.3404
##
##  Mcnemar's Test P-Value : 8.568e-05
##
##              Sensitivity : 0.9808
##              Specificity : 0.3103
##      Pos Pred Value : 0.7183
##      Neg Pred Value : 0.9000
##      Prevalence : 0.6420
##      Detection Rate : 0.6296
##      Detection Prevalence : 0.8765
##      Balanced Accuracy : 0.6456
##
##      'Positive' Class : 1
##
```

```
calculate_mcc1(CM_predictions_3)
```

```
## [1] 0.4242625
```

```
calculate_F2(CM_predictions_3)
```

```
## [1] 0.9139785
```

```
testrf_mtry<- predict(rf_mtry)
summary(testrf_mtry)
```

```
## X0 X1
## 69 136
```

```
levels(testrf_mtry) <- c(0,1);
summary(testrf_mtry)
```

```
## 0 1
## 69 136
```

```
CM_testrf_mtry<-confusionMatrix(testrf_mtry, data_train[, ncol(data_train)],positive="1");
CM_testrf_mtry
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  0    1
##           0 69    0
##           1  5 131
##
##           Accuracy : 0.9756
##           95% CI : (0.944, 0.992)
##           No Information Rate : 0.639
##           P-Value [Acc > NIR] : < 2e-16
##
##           Kappa : 0.9463
##
##           McNemar's Test P-Value : 0.07364
##
##           Sensitivity : 1.0000
##           Specificity : 0.9324
##           Pos Pred Value : 0.9632
##           Neg Pred Value : 1.0000
##           Prevalence : 0.6390
##           Detection Rate : 0.6390
##           Detection Prevalence : 0.6634
##           Balanced Accuracy : 0.9662
##
##           'Positive' Class : 1
##
```

```
calculate_mcc1(CM_testrf_mtry)
```

```
## [1] 0.9477087
```

```
calculate_F2(CM_testrf_mtry)
```

```
## [1] 0.9924242
```

```
testrf_mtry1<- predict(rf_mtry,type="prob")
summary(testrf_mtry1)
```

```
##           X0           X1
## Min.      :0.0220   Min.   :0.0580
## 1st Qu.:0.1160   1st Qu.:0.4040
## Median :0.2300   Median :0.7700
## Mean     :0.3413   Mean    :0.6587
## 3rd Qu.:0.5960   3rd Qu.:0.8840
## Max.     :0.9420   Max.    :0.9780
```

```
model_pred_class_4 <- ifelse(testrf_mtry1 < 0.5, "X0", "X1")
summary(model_pred_class_4)
```

```
##           X0           X1
## Length:205      Length:205
## Class :character Class :character
## Mode  :character Mode  :character
```

```
Test_4<-as.factor(model_pred_class_4[,2])
head(Test_4)
```

```
## 173  50 458 383  24 246
## X0 X1 X0 X1 X0 X1
## Levels: X0 X1
```

```
levels(Test_4) <- c(0,1);
summary(Test_4)
```

```
##    0    1
## 69 136
```

```
CM_testrf_mtry1<-confusionMatrix(Test_4, data_train[, ncol(data_train)],positive="1");
CM_testrf_mtry1
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##           0 69   0
```

```
##          1    5 131
##
##          Accuracy : 0.9756
##          95% CI : (0.944, 0.992)
##    No Information Rate : 0.639
##    P-Value [Acc > NIR] : < 2e-16
##
##          Kappa : 0.9463
##
##    McNemar's Test P-Value : 0.07364
##
##          Sensitivity : 1.0000
##          Specificity : 0.9324
##    Pos Pred Value : 0.9632
##    Neg Pred Value : 1.0000
##          Prevalence : 0.6390
##    Detection Rate : 0.6390
##    Detection Prevalence : 0.6634
##    Balanced Accuracy : 0.9662
##
##    'Positive' Class : 1
##
```

```
calculate_mcc1(CM_testrf_mtry1)
```

```
## [1] 0.9477087
```

```
calculate_F2(CM_testrf_mtry1)
```

```
## [1] 0.9924242
```

```
prediction.proBABILITIES <- predictions[, "X1"]
head(prediction.proBABILITIES, 5)
```

```
## [1] 0.322 0.688 0.474 0.912 0.922
```

```
res.roc <- roc(ContraProva, prediction.proBABILITIES)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
res.roc
```

```
##
## Call:
## roc.default(response = ContraProva, predictor = prediction.proBABILITIES)
##
## Data: prediction.proBABILITIES in 44 controls (ContraProva 0) < 79 cases (ContraProva 1).
## Area under the curve: 0.744
```

```
str(res.roc)
```

```
## List of 15
## $ percent          : logi FALSE
## $ sensitivities     : num [1:101] 1 1 1 0.987 0.987 ...
## $ specificities     : num [1:101] 0 0.0227 0.0455 0.0455 0.0682 ...
## $ thresholds        : num [1:101] -Inf 0.199 0.218 0.235 0.239 ...
## $ direction         : chr "<"
## $ cases             : num [1:79] 0.688 0.474 0.912 0.922 0.858 0.87 0.67 0.762 0.728 0.792 ...
## $ controls          : num [1:44] 0.322 0.474 0.69 0.876 0.36 0.48 0.556 0.468 0.204 0.538 ...
## $ fun.sesp          :function (thresholds, controls, cases, direction)
## $ auc               : 'auc' num 0.744
## ..- attr(*, "partial.auc")= logi FALSE
## ..- attr(*, "percent")= logi FALSE
## ..- attr(*, "roc")=List of 15
## .. ..$ percent          : logi FALSE
## .. ..$ sensitivities     : num [1:101] 1 1 1 0.987 0.987 ...
## .. ..$ specificities     : num [1:101] 0 0.0227 0.0455 0.0455 0.0682 ...
## .. ..$ thresholds        : num [1:101] -Inf 0.199 0.218 0.235 0.239 ...
## .. ..$ direction         : chr "<"
## .. ..$ cases             : num [1:79] 0.688 0.474 0.912 0.922 0.858 0.87 0.67 0.762 0.728 0.792 ...
## .. ..$ controls          : num [1:44] 0.322 0.474 0.69 0.876 0.36 0.48 0.556 0.468 0.204 0.538 ...
## .. ..$ fun.sesp          :function (thresholds, controls, cases, direction)
## .. ..$ auc               : 'auc' num 0.744
## .. ..- attr(*, "partial.auc")= logi FALSE
## .. ..- attr(*, "percent")= logi FALSE
## .. ..- attr(*, "roc")=List of 8
## .. .. ..$ percent          : logi FALSE
## .. .. ..$ sensitivities: num [1:101] 1 1 1 0.987 0.987 ...
## .. .. ..$ specificities: num [1:101] 0 0.0227 0.0455 0.0455 0.0682 ...
## .. .. ..$ thresholds    : num [1:101] -Inf 0.199 0.218 0.235 0.239 ...
## .. .. ..$ direction      : chr "<"
## .. .. ..$ cases          : num [1:79] 0.688 0.474 0.912 0.922 0.858 0.87 0.67 0.762 0.728 0.792 ...
## .. .. ..$ controls        : num [1:44] 0.322 0.474 0.69 0.876 0.36 0.48 0.556 0.468 0.204 0.538 ...
## .. .. ..$ fun.sesp        :function (thresholds, controls, cases, direction)
## .. .. ..- attr(*, "class")= chr "roc"
## .. ..$ call              : language roc.default(response = ContraProva, predictor = prediction.proba
## .. ..$ original.predictor: num [1:123] 0.322 0.688 0.474 0.912 0.922 0.858 0.87 0.474 0.67 0.69 ...
## .. ..$ original.response : Factor w/ 2 levels "0","1": 1 2 2 2 2 2 2 1 2 1 ...
## .. ..$ predictor         : num [1:123] 0.322 0.688 0.474 0.912 0.922 0.858 0.87 0.474 0.67 0.69 ...
## .. ..$ response          : Factor w/ 2 levels "0","1": 1 2 2 2 2 2 2 1 2 1 ...
## .. ..$ levels            : chr [1:2] "0" "1"
## .. ..- attr(*, "class")= chr "roc"
## $ call                  : language roc.default(response = ContraProva, predictor = prediction.proba
## $ original.predictor: num [1:123] 0.322 0.688 0.474 0.912 0.922 0.858 0.87 0.474 0.67 0.69 ...
## $ original.response : Factor w/ 2 levels "0","1": 1 2 2 2 2 2 2 1 2 1 ...
## $ predictor           : num [1:123] 0.322 0.688 0.474 0.912 0.922 0.858 0.87 0.474 0.67 0.69 ...
## $ response            : Factor w/ 2 levels "0","1": 1 2 2 2 2 2 2 1 2 1 ...
## $ levels              : chr [1:2] "0" "1"
## - attr(*, "class")= chr "roc"
```

```
head(res.roc$cases,5)
```



```
## [1] 0.688 0.474 0.912 0.922 0.858
```

```
length(res.roc$cases)
```

```
## [1] 79
```

```
summary(res.roc$cases)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2320  0.6300  0.7520  0.7218  0.8320  0.9400
```

```
head(res.roc$controls,5)
```

```
## [1] 0.322 0.474 0.690 0.876 0.360
```

```
length(res.roc$controls)
```

```
## [1] 44
```

```
summary(res.roc$controls)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.1940  0.4560  0.5470  0.5620  0.7055  0.9580
```

```
head(res.roc$thresholds,5)
```

```
## [1] -Inf 0.199 0.218 0.235 0.239
```

```
tail(res.roc$thresholds,5)
```

```
## [1] 0.917 0.930 0.939 0.949    Inf
```

```
length(res.roc$thresholds)
```

```
## [1] 101
```

```
summary(res.roc$thresholds)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      -Inf  0.533   0.684         0.809      Inf
```

```
auc(res.roc)#auc() compute the area under the ROC curve.
```

```
## Area under the curve: 0.744
```

```
ci.auc(res.roc)#ci.auc() Compute the confidence interval of the AUC, default method = "delong".
```

```
## 95% CI: 0.65-0.8379 (DeLong)
```

```
ci.auc(res.roc,method = "bootstrap", boot.n = 10000)#ci.auc() Compute the confidence interval of the AU
```

```
## 95% CI: 0.6441-0.832 (10000 stratified bootstrap replicates)
```

```
#A value of x = "all" will return the coordinates for the curve and their associated cutoffs.  
rfThresh_all <- coords(res.roc, x = "all", best.method = "youden")  
head(rfThresh_all)
```

```
##   threshold specificity sensitivity  
## 1      -Inf  0.00000000  1.0000000  
## 2    0.199  0.02272727  1.0000000  
## 3    0.218  0.04545455  1.0000000  
## 4    0.235  0.04545455  0.9873418  
## 5    0.239  0.06818182  0.9873418  
## 6    0.274  0.09090909  0.9873418
```

```
summary(rfThresh_all)
```

```
##   threshold      specificity      sensitivity  
## Min.   : -Inf   Min.   :0.0000   Min.   :0.0000  
## 1st Qu.:0.533   1st Qu.:0.4773   1st Qu.:0.3797  
## Median :0.684   Median :0.7045   Median :0.6582  
## Mean   :  NaN   Mean   :0.6447   Mean   :0.6011  
## 3rd Qu.:0.809   3rd Qu.:0.9091   3rd Qu.:0.8861  
## Max.   :  Inf   Max.   :1.0000   Max.   :1.0000
```

```
dim(rfThresh_all)
```

```
## [1] 101  3
```

```
#"local maximas"(the local maximas of the ROC curve).  
rfThresh_max <- coords(res.roc, x = "local maximas", best.method = "youden")  
head(rfThresh_max)
```

```
##   threshold specificity sensitivity  
## 1    0.218  0.04545455  1.0000000  
## 2    0.274  0.09090909  0.9873418  
## 3    0.350  0.13636364  0.9746835  
## 4    0.427  0.20454545  0.9620253  
## 5    0.471  0.34090909  0.9240506  
## 6    0.501  0.43181818  0.9113924
```

```
summary(rfThresh_max)
```

```
##      threshold      specificity      sensitivity
## Min.      :0.2180   Min.      :0.04545   Min.      :0.0000
## 1st Qu.:0.5292   1st Qu.:0.48295   1st Qu.:0.4747
## Median :0.6685   Median :0.68182   Median :0.6709
## Mean    :  Inf     Mean    :0.63112   Mean     :0.6431
## 3rd Qu.:0.7642   3rd Qu.:0.85795   3rd Qu.:0.8956
## Max.     :  Inf     Max.     :1.00000   Max.     :1.0000
```

```
dim(rfThresh_max)
```

```
## [1] 26 3
```

```
#Best thresholds: If x="best" e best.method = "youden", Youden's J statistic (Youden, 1950) is employed
rfThresh_youden <- coords(res.roc, x = "best", best.method = "youden")
rfThresh_youden
```

```
##      threshold specificity sensitivity
## 1      0.574    0.5681818    0.8481013
```

```
rfThresh_youden[1,1]
```

```
## [1] 0.574
```

```
#Best thresholds: If x="best" e best.method = "closest.topleft", The optimal threshold is the point clo
rfThresh_topleft <- coords(res.roc, x = "best", best.method = "closest.topleft")
rfThresh_topleft
```

```
##      threshold specificity sensitivity
## 1      0.594    0.5909091    0.8227848
```

```
rfThresh_topleft[1,1]
```

```
## [1] 0.594
```

```
coords(res.roc, "best", ret="all", transpose = FALSE,
       best.method="youden")
```

```
##      threshold specificity sensitivity accuracy tn tp fn fp      npv
## threshold      0.574    0.5681818    0.8481013 0.7479675 25 67 12 19 0.6756757
##      ppv      fdr      fpr      tpr      tnr      fnr
## threshold 0.7790698 0.2209302 0.4318182 0.8481013 0.5681818 0.1518987
##      1-specificity 1-sensitivity 1-accuracy      1-npv      1-ppv precision
## threshold      0.4318182      0.1518987 0.2520325 0.3243243 0.2209302 0.7790698
##      recall      youden closest.topleft
## threshold 0.8481013 1.416283      0.2095402
```

```
coords(res.roc, "best", ret="all", transpose = FALSE,
       best.method="closest.topleft")
```

```
##          threshold specificity sensitivity accuracy tn tp fn fp npv
## threshold      0.594   0.5909091   0.8227848 0.7398374 26 65 14 18 0.65
##          ppv      fdr      fpr      tpr      tnr      fnr
## threshold 0.7831325 0.2168675 0.4090909 0.8227848 0.5909091 0.1772152
##          1-specificity 1-sensitivity 1-accuracy 1-npv      1-ppv precision
## threshold      0.4090909      0.1772152 0.2601626 0.35 0.2168675 0.7831325
##          recall  youden closest.topleft
## threshold 0.8227848 1.413694      0.1987606
```

```
coords(res.roc, "best", ret="threshold", transpose = FALSE,
        best.method="youden") # default
```

```
## threshold
## 1      0.574
```

```
coords(res.roc, "best", ret="threshold", transpose = FALSE,
        best.method="closest.topleft")
```

```
## threshold
## 1      0.594
```

```
coords(res.roc, "best", ret="tn", transpose = FALSE)
```

```
## tn
## threshold 25
```

```
coords(res.roc, "best", ret="tn", transpose = FALSE,
        best.method="closest.topleft")
```

```
## tn
## threshold 26
```

```
coords(res.roc, "best", ret="tp", transpose = FALSE)
```

```
## tp
## threshold 67
```

```
coords(res.roc, "best", ret="tp", transpose = FALSE,
        best.method="closest.topleft")
```

```
## tp
## threshold 65
```

```
coords(res.roc, "best", ret="youden", transpose = FALSE)
```

```
## youden
## 1 1.416283
```

```
coords(res.roc, "best", ret="closest.topleft", transpose = FALSE,best.method="closest.topleft")
```

```
## closest.topleft  
## 1 0.1987606
```

```
rfThresh_youden <- coords(res.roc, x = "best", best.method = "youden")  
rfThresh_youden
```

```
## threshold specificity sensitivity  
## 1 0.574 0.5681818 0.8481013
```

```
rfThresh_youden[1,1]
```

```
## [1] 0.574
```

```
model_pred_class <- ifelse(predictions < rfThresh_youden[1,1], "X0", "X1")  
head(model_pred_class)
```

```
## X0 X1  
## 538 "X1" "X0"  
## 580 "X0" "X1"  
## 449 "X0" "X0"  
## 149 "X0" "X1"  
## 304 "X0" "X1"  
## 15 "X0" "X1"
```

```
summary(model_pred_class)
```

```
## X0 X1  
## Length:123 Length:123  
## Class :character Class :character  
## Mode :character Mode :character
```

```
Test2<-as.factor(model_pred_class[,2])
```

```
levels(Test2) <- c(0,1)  
summary(Test2)
```

```
## 0 1  
## 37 86
```

```
CM_Test2<-confusionMatrix(Test2, ContraProva,positive="1")  
CM_Test2
```

```
## Confusion Matrix and Statistics  
##  
## Reference
```

```
## Prediction 0 1
##           0 25 12
##           1 19 67
##
##           Accuracy : 0.748
##           95% CI : (0.6617, 0.8219)
##           No Information Rate : 0.6423
##           P-Value [Acc > NIR] : 0.008134
##
##           Kappa : 0.4315
##
## Mcnemar's Test P-Value : 0.281198
##
##           Sensitivity : 0.8481
##           Specificity : 0.5682
##           Pos Pred Value : 0.7791
##           Neg Pred Value : 0.6757
##           Prevalence : 0.6423
##           Detection Rate : 0.5447
##           Detection Prevalence : 0.6992
##           Balanced Accuracy : 0.7081
##
##           'Positive' Class : 1
##
```

```
str(CM_Test2)
```

```
## List of 6
## $ positive: chr "1"
## $ table : 'table' int [1:2, 1:2] 25 19 12 67
## .. attr(*, "dimnames")=List of 2
## .. ..$ Prediction: chr [1:2] "0" "1"
## .. ..$ Reference : chr [1:2] "0" "1"
## $ overall : Named num [1:7] 0.748 0.431 0.662 0.822 0.642 ...
## .. attr(*, "names")= chr [1:7] "Accuracy" "Kappa" "AccuracyLower" "AccuracyUpper" ...
## $ byClass : Named num [1:11] 0.848 0.568 0.779 0.676 0.779 ...
## .. attr(*, "names")= chr [1:11] "Sensitivity" "Specificity" "Pos Pred Value" "Neg Pred Value" ...
## $ mode : chr "sens_spec"
## $ dots : list()
## - attr(*, "class")= chr "confusionMatrix"
```

```
CM_Test2$byClass[1]#Sensitivity de CM_predictions1
```

```
## Sensitivity
## 0.8481013
```

```
CM_Test2$byClass[2]#Specificity de CM_predictions1
```

```
## Specificity
## 0.5681818
```

```
J_CM_Test2<-as.numeric(CM_Test2$byClass[1] + CM_Test2$byClass[2] - 1) ; J_CM_Test2
```

```
## [1] 0.4162831
```

```
calculate_mcc1(CM_Test2)
```

```
## [1] 0.4350895
```

```
calculate_F2(CM_Test2)
```

```
## [1] 0.8333333
```

```
rfThresh_youden <- coords(res.roc, x = "best", best.method = "youden")  
rfThresh_youden
```

```
## threshold specificity sensitivity  
## 1 0.574 0.5681818 0.8481013
```

```
rfThresh_youden[1,1]
```

```
## [1] 0.574
```

```
model_pred_class_31 <- ifelse(predictions_3 < rfThresh_youden[1,1], "X0", "X1")  
head(sapply(model_pred_class_31, class))
```

```
## X0 X0 X0 X0 X0 X0  
## "character" "character" "character" "character" "character" "character"
```

```
Test_31<-as.factor(model_pred_class_31[,2])  
head(sapply(Test_31, class))
```

```
## 95 626 607 623 18 470  
## "factor" "factor" "factor" "factor" "factor" "factor"
```

```
levels(Test_31) <- c(0,1)  
head(sapply(Test_31, class))
```

```
## 95 626 607 623 18 470  
## "factor" "factor" "factor" "factor" "factor" "factor"
```

```
levels(Test_31)
```

```
## [1] "0" "1"
```

```
head(Test_31)
```

```
## 95 626 607 623 18 470
## 1 1 1 1 0 1
## Levels: 0 1
```

```
summary(Test_31)
```

```
## 0 1
## 21 60
```

```
CM_predictions_31<-confusionMatrix(Test_31, finalTest$true_troglofauna,positive="1")
CM_predictions_31
```

```
## Confusion Matrix and Statistics
##
##              Reference
## Prediction  0  1
##           0 17  4
##           1 12 48
##
##              Accuracy : 0.8025
##              95% CI : (0.6991, 0.8827)
##      No Information Rate : 0.642
##      P-Value [Acc > NIR] : 0.001286
##
##              Kappa : 0.5424
##
##  Mcnemar's Test P-Value : 0.080118
##
##      Sensitivity : 0.9231
##      Specificity : 0.5862
##      Pos Pred Value : 0.8000
##      Neg Pred Value : 0.8095
##      Prevalence : 0.6420
##      Detection Rate : 0.5926
##      Detection Prevalence : 0.7407
##      Balanced Accuracy : 0.7546
##
##      'Positive' Class : 1
##
```

```
calculate_mcc1(CM_predictions_31)
```

```
## [1] 0.557154
```

```
calculate_F2(CM_predictions_31)
```

```
## [1] 0.8955224
```



```
rfThresh_topleft <- coords(res.roc, x = "best", best.method = "closest.topleft")
rfThresh_topleft
```

```
## threshold specificity sensitivity
## 1      0.594  0.5909091  0.8227848
```

```
rfThresh_topleft[1,1]
```

```
## [1] 0.594
```

```
model_pred_class <- ifelse(predictions < rfThresh_topleft[1,1], "X0", "X1")
Test3<-as.factor(model_pred_class[,2])
```

```
levels(Test3) <- c(0,1)
summary(Test3)
```

```
## 0 1
## 40 83
```

```
CM_Test3<-confusionMatrix(Test3, ContraProva,positive="1")
CM_Test3
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  0  1
##           0 26 14
##           1 18 65
##
##           Accuracy : 0.7398
##           95% CI : (0.653, 0.8148)
##           No Information Rate : 0.6423
##           P-Value [Acc > NIR] : 0.01375
##
##           Kappa : 0.4222
##
##           Mcnemar's Test P-Value : 0.59588
##
##           Sensitivity : 0.8228
##           Specificity : 0.5909
##           Pos Pred Value : 0.7831
##           Neg Pred Value : 0.6500
##           Prevalence : 0.6423
##           Detection Rate : 0.5285
##           Detection Prevalence : 0.6748
##           Balanced Accuracy : 0.7068
##
##           'Positive' Class : 1
##
```

```
str(CM_Test3)
```

```
## List of 6
## $ positive: chr "1"
## $ table : 'table' int [1:2, 1:2] 26 18 14 65
## ..- attr(*, "dimnames")=List of 2
## .. ..$ Prediction: chr [1:2] "0" "1"
## .. ..$ Reference : chr [1:2] "0" "1"
## $ overall : Named num [1:7] 0.74 0.422 0.653 0.815 0.642 ...
## ..- attr(*, "names")= chr [1:7] "Accuracy" "Kappa" "AccuracyLower" "AccuracyUpper" ...
## $ byClass : Named num [1:11] 0.823 0.591 0.783 0.65 0.783 ...
## ..- attr(*, "names")= chr [1:11] "Sensitivity" "Specificity" "Pos Pred Value" "Neg Pred Value" ...
## $ mode : chr "sens_spec"
## $ dots : list()
## - attr(*, "class")= chr "confusionMatrix"
```

```
CM_Test3$byClass[1]#Sensitivity de CM_predictions1
```

```
## Sensitivity
## 0.8227848
```

```
CM_Test3$byClass[2]#Specificity de CM_predictions1
```

```
## Specificity
## 0.5909091
```

```
J_CM_Test3<-as.numeric(CM_Test3$byClass[1] + CM_Test3$byClass[2] - 1) ; J_CM_Test3
```

```
## [1] 0.4136939
```

```
calculate_mcc1(CM_Test3)
```

```
## [1] 0.4233016
```

```
calculate_F2(CM_Test3)
```

```
## [1] 0.8145363
```

```
rfThresh_topleft <- coords(res.roc, x = "best", best.method = "closest.topleft")
rfThresh_topleft
```

```
## threshold specificity sensitivity
## 1 0.594 0.5909091 0.8227848
```

```
rfThresh_topleft[1,1]
```

```
## [1] 0.594
```

```
model_pred_class_41 <- ifelse(predictions_3 < rfThresh_topleft[1,1], "X0", "X1")
head(sapply(model_pred_class_41, class))
```

```
##           X0           X0           X0           X0           X0           X0
## "character" "character" "character" "character" "character" "character"
```

```
Test_41<-as.factor(model_pred_class_41[,2])#
head(sapply(Test_41, class))
```

```
##          95          626          607          623          18          470
## "factor" "factor" "factor" "factor" "factor" "factor"
```

```
levels(Test_41) <- c(0,1)
head(sapply(Test_41, class))
```

```
##          95          626          607          623          18          470
## "factor" "factor" "factor" "factor" "factor" "factor"
```

```
levels(Test_41)
```

```
## [1] "0" "1"
```

```
head(Test_41)
```

```
##  95 626 607 623  18 470
##   1  1  1  1  0  1
## Levels: 0 1
```

```
summary(Test_41)
```

```
##  0  1
## 25 56
```

```
CM_predictions_41<-confusionMatrix(Test_41, finalTest$true_troglofauna,positive="1")
CM_predictions_41
```

```
## Confusion Matrix and Statistics
```

```
##
```

```
##           Reference
```

```
## Prediction  0  1
```

```
##           0 20  5
```

```
##           1  9 47
```

```
##
```

```
##           Accuracy : 0.8272
```

```
##           95% CI : (0.727, 0.9022)
```

```
## No Information Rate : 0.642
```

```
## P-Value [Acc > NIR] : 0.0002035
```

```
##
```

```
##                Kappa : 0.6122
##
## Mcnemar's Test P-Value : 0.4226781
##
##          Sensitivity : 0.9038
##          Specificity : 0.6897
##          Pos Pred Value : 0.8393
##          Neg Pred Value : 0.8000
##          Prevalence : 0.6420
##          Detection Rate : 0.5802
##          Detection Prevalence : 0.6914
##          Balanced Accuracy : 0.7968
##
##          'Positive' Class : 1
##
```

```
calculate_mcc1(CM_predictions_41)
```

```
## [1] 0.6159683
```

```
calculate_F2(CM_predictions_41)
```

```
## [1] 0.8901515
```

```
roc.data <- data_frame( # Create a dataframe 'roc.data' for ROC analysis results
  thresholds = res.roc$thresholds, # Column for ROC curve thresholds
  sensitivity = res.roc$sensitivities, # Column for sensitivity (true positive rate)
  specificity = res.roc$specificities # Column for specificity (true negative rate)
)
```

```
## Warning: 'data_frame()' was deprecated in tibble 1.1.0.
## i Please use 'tibble()' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
roc.data # Display the 'roc.data' dataframe in the R console
```

```
## # A tibble: 101 x 3
##   thresholds sensitivity specificity
##   <dbl>         <dbl>         <dbl>
## 1  -Inf             1             0
## 2    0.199          1          0.0227
## 3    0.218          1          0.0455
## 4    0.235          0.987        0.0455
## 5    0.239          0.987        0.0682
## 6    0.274          0.987        0.0909
## 7    0.315          0.975        0.0909
## 8    0.332          0.975        0.114
## 9    0.35           0.975        0.136
## 10   0.359          0.962        0.136
## # i 91 more rows
```

```
tibble(roc.data)# Convert the 'roc.data' dataframe into a tibble for further analysis (if needed)
```

```
## # A tibble: 101 x 3
##   thresholds sensitivity specificity
##   <dbl>         <dbl>         <dbl>
## 1  -Inf           1           0
## 2    0.199        1          0.0227
## 3    0.218        1          0.0455
## 4    0.235        0.987        0.0455
## 5    0.239        0.987        0.0682
## 6    0.274        0.987        0.0909
## 7    0.315        0.975        0.0909
## 8    0.332        0.975        0.114
## 9    0.35         0.975        0.136
## 10   0.359        0.962        0.136
## # i 91 more rows
```

```
#Get the probability threshold for specificity >= 0.6 e sensitivity >= 0.6
roc.data %>% filter(specificity >= 0.6)#comando com %>% "forward pipe operator"
```

```
## # A tibble: 59 x 3
##   thresholds sensitivity specificity
##   <dbl>         <dbl>         <dbl>
## 1    0.64         0.734        0.614
## 2    0.643        0.734        0.636
## 3    0.645        0.709        0.636
## 4    0.649        0.696        0.636
## 5    0.653        0.684        0.659
## 6    0.662        0.671        0.659
## 7    0.672        0.658        0.659
## 8    0.677        0.658        0.682
## 9    0.684        0.658        0.705
## 10   0.689        0.633        0.705
## # i 49 more rows
```

```
roc.data %>% filter(sensitivity >= 0.6)#comando com %>% "forward pipe operator"
```

```
## # A tibble: 54 x 3
##   thresholds sensitivity specificity
##   <dbl>         <dbl>         <dbl>
## 1  -Inf           1           0
## 2    0.199        1          0.0227
## 3    0.218        1          0.0455
## 4    0.235        0.987        0.0455
## 5    0.239        0.987        0.0682
## 6    0.274        0.987        0.0909
## 7    0.315        0.975        0.0909
## 8    0.332        0.975        0.114
## 9    0.35         0.975        0.136
## 10   0.359        0.962        0.136
## # i 44 more rows
```

```

par(pty="s")

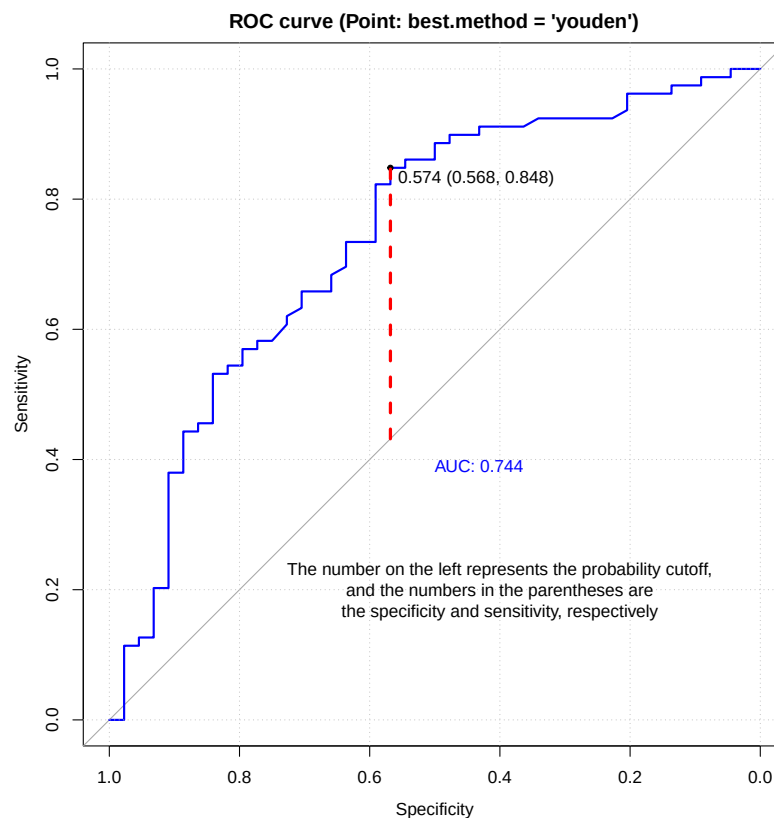
plot.roc(res.roc, print.auc =TRUE,col="blue", print.thres =
        "best",print.auc.y=0.4,main="ROC curve (Point: best.method = 'youden')")

grid(nx = NULL, ny = NULL,
      lty = 3,
      lwd = 1,
      col = "gray")

segments(rfThresh_youden[1,2],1-rfThresh_youden[1,2],rfThresh_youden[1,2], rfThresh_youden[1,3],lwd = 3)

text(0.4,0.2,"The number on the left represents the probability cutoff,\nand the numbers in the parentheses")

```



```

par(pty="s")

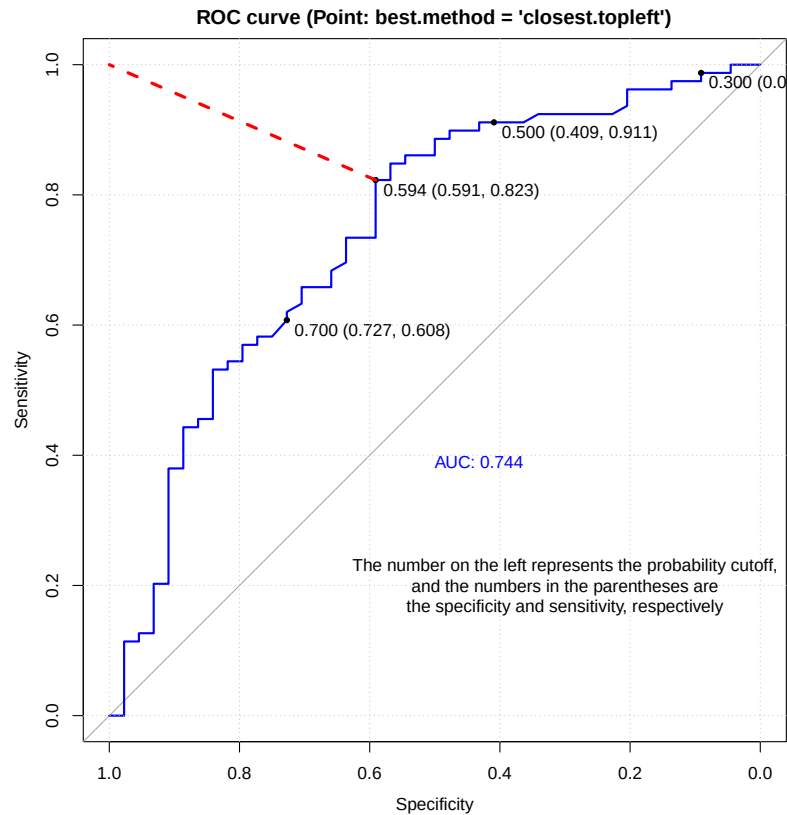
plot.roc(res.roc, print.auc =TRUE,col="blue", print.thres =
        c(0.3, 0.5, 0.7,rfThresh_topleft[1,1]),print.auc.y=0.4,main="ROC curve (Point: best.method =

grid(nx = NULL, ny = NULL,
      lty = 3,
      lwd = 1,
      col = "gray")

```

```
segments(rfThresh_topleft[1,2],rfThresh_topleft[1,3],1, 1,lwd = 3, col = "red",lty=2)
```

```
text(0.3,0.2,"The number on the left represents the probability cutoff,\nand the numbers in the parenth
```



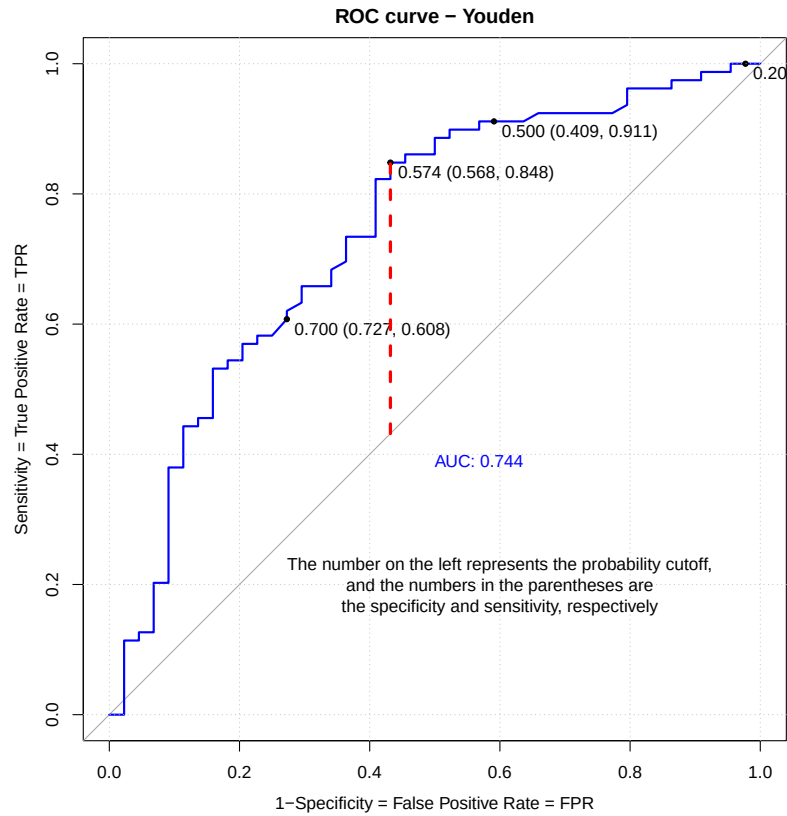
```
par(pty="s")
```

```
plot.roc(res.roc, print.auc = TRUE, print.auc.y=0.4, legacy.axes = TRUE, col="blue", print.thres = c(0.2, 0.3))
```

```
segments(rfThresh_youden[1,2], 1-rfThresh_youden[1,2], rfThresh_youden[1,2], rfThresh_youden[1,3], lwd = 3, col = "red", lty = 2)
```

```
grid(nx = NULL, ny = NULL,
      lty = 3,
      lwd = 1,
      col = "gray")
```

```
text(0.4,0.2,"The number on the left represents the probability cutoff,\nand the numbers in the parenth
```



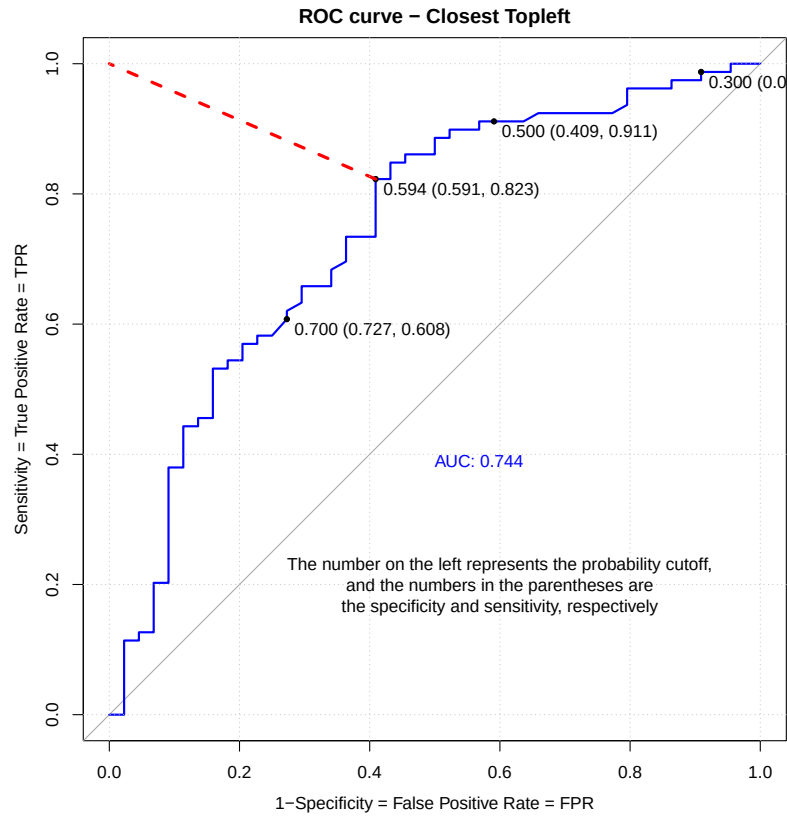
```
par(pty="s")
```

```
plot.roc(res.roc, print.auc = TRUE, print.auc.y=0.4, legacy.axes = TRUE, col="blue", print.thres = c(0.3,
```

```
segments(rfThresh_topleft[1,2], rfThresh_topleft[1,3], 1, 1, lwd = 3, col = "red", lty=2)
```

```
grid(nx = NULL, ny = NULL,  
      lty = 3,  
      lwd = 1,  
      col = "gray")
```

```
text(0.4, 0.2, "The number on the left represents the probability cutoff, \nand the numbers in the parentheses
```

```
par(pty="s")
```

```
rocCurve <- roc(ContraProva,prediction.proBABILITIES, plot=TRUE,legacy.axes = TRUE, col="blue",main="ROC  
"best",print.auc =TRUE,cex.main=0.9)
```

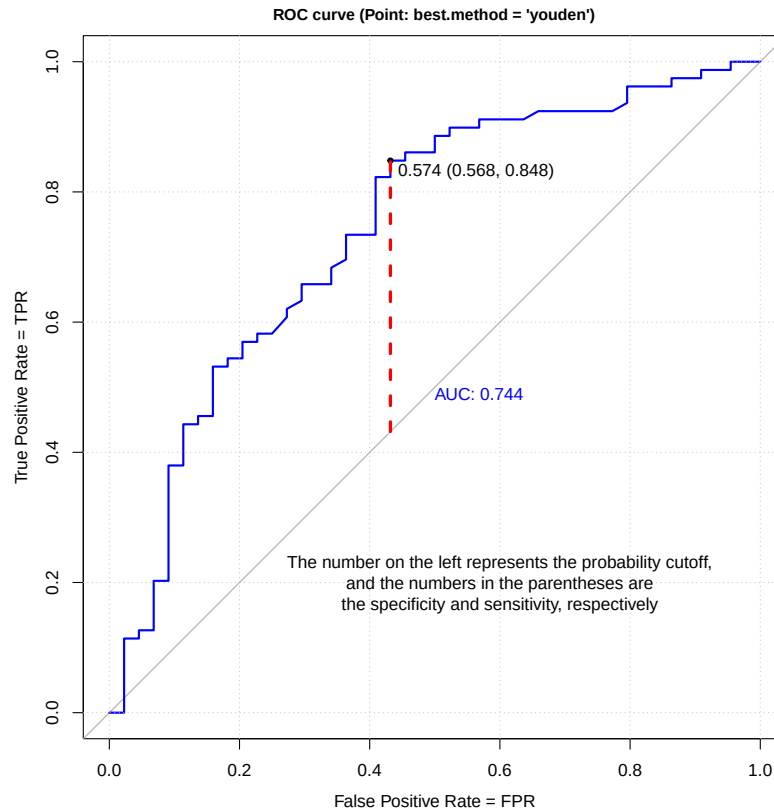
```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
segments(rfThresh_youden[1,2], 1-rfThresh_youden[1,2],rfThresh_youden[1,2],rfThresh_youden[1,3],lwd = 3)
```

```
grid(nx = NULL, ny = NULL,  
      lty = 3,  
      lwd = 1,  
      col = "gray")
```

```
text(0.4,0.2,"The number on the left represents the probability cutoff,\nand the numbers in the parentheses
```



```
dim(finalTest)
```

```
## [1] 81 27
```

```
names(finalTest)
```

```
## [1] "trngcv"      "bio14"      "slope"      "rugg500s"
## [5] "pil_twim"    "pil_twicv"  "pil_topocv" "pil_slps"
## [9] "pil_slpcv"   "pil_elr3cv" "mrvbf"      "mrrtf"
## [13] "minfertf"    "lf7rup"     "hstructn"   "geolrngaggn"
## [17] "elevationm"  "wr_unrn"    "solpawhcn"  "slopern"
## [21] "MIN_AGE_MA"  "HubDist"    "month_collection" "ROCKTYPE1"
## [25] "FORMATION"   "HubName"    "true_troglofauna"
```

```
dim(selectcol_finalTest)
```

```
## [1] 81 26
```

```
names(selectcol_finalTest)
```

```
## [1] "trngcv"      "bio14"      "slope"      "rugg500s"
## [5] "pil_twim"    "pil_twicv"  "pil_topocv" "pil_slps"
## [9] "pil_slpcv"   "pil_elr3cv" "mrvbf"      "mrrtf"
## [13] "minfertf"    "lf7rup"     "hstructn"   "geolrngaggn"
```

```
## [17] "elevationm"      "wr_unrn"          "solpawhcn"        "slopern"
## [21] "MIN_AGE_MA"      "HubDist"          "month_collection" "ROCKTYPE1"
## [25] "FORMATION"       "HubName"
```

```
CM_predictions_2$byClass
```

```
##          Sensitivity      Specificity      Pos Pred Value
##          0.9807692      0.3103448      0.7183099
##      Neg Pred Value      Precision      Recall
##          0.9000000      0.7183099      0.9807692
##          F1      Prevalence      Detection Rate
##          0.8292683      0.6419753      0.6296296
## Detection Prevalence      Balanced Accuracy
##          0.8765432      0.6455570
```

```
beta<-2
```

```
dbF1_11<-CM_predictions_2$byClass
dbF1_11["F1"]
```

```
##          F1
## 0.8292683
```

```
dbF2_11<-((1+beta^2)*dbF1_11["Precision"]*dbF1_11["Sensitivity"])/(beta^2*dbF1_11["Precision"] + dbF1_11["Sensitivity"])
dbF2_11<-as.numeric(dbF2_11)
dbF2_11
```

```
## [1] 0.9139785
```

```
dbF1_11["Precision"];dbF1_11["Sensitivity"]
```

```
## Precision
## 0.7183099
```

```
## Sensitivity
## 0.9807692
```

```
dbF1_22<-CM_predictions_31$byClass
dbF1_22["F1"]
```

```
##          F1
## 0.8571429
```

```
dbF2_22<-((1+beta^2)*dbF1_22["Precision"]*dbF1_22["Sensitivity"])/(beta^2*dbF1_22["Precision"] + dbF1_22["Sensitivity"])
dbF2_22<-as.numeric(dbF2_22)
dbF2_22
```

```
## [1] 0.8955224
```

```
dbF1_22["Precision"];dbF1_22["Sensitivity"]
```

```
## Precision  
##      0.8
```

```
## Sensitivity  
##    0.9230769
```

```
dbF1_33<-CM_predictions_41$byClass  
dbF1_33["F1"]
```

```
##      F1  
## 0.8703704
```

```
dbF2_33<-((1+beta^2)*dbF1_33["Precision"]*dbF1_33["Sensitivity"])/(beta^2*dbF1_33["Precision"] + dbF1_33["Sensitivity"])  
dbF2_33<-as.numeric(dbF2_33)  
dbF2_33
```

```
## [1] 0.8901515
```

```
dbF1_33["Precision"];dbF1_33["Sensitivity"]
```

```
## Precision  
## 0.8392857
```

```
## Sensitivity  
##    0.9038462
```

```
dim(finalTest)
```

```
## [1] 81 27
```

```
basmmcc<-mcc(predictions_2, finalTest[, ncol(finalTest)])  
basmmcc
```

```
## [1] 0.4242625
```

```
yoummcc<-mcc(Test_31, finalTest[, ncol(finalTest)])  
yoummcc
```

```
## [1] 0.557154
```

```
topmmcc<-mcc(Test_41, finalTest[, ncol(finalTest)])  
topmmcc
```

```
## [1] 0.6159683
```

```
#Results
```

```

cat("\n\n", "TABLE 1 - RESULTS OF THE 'finalTest' DATABASE WITH THE MODELS USING THE train() FUNCTION OF
, "\n\n\n", "1 - Basic model"
, "\n\n", "Area Under ROC (AUC)- Basic model =", auc(res.roc)
, "\n\n\n", "Accuracy =", CM_predictions_2$overall[1]
, "\n\n", "Sensitivity =", CM_predictions_2$byClass[1]
, "\n\n", "Specificity =", CM_predictions_2$byClass[2]
, "\n\n", "Balanced Accuracy =", CM_predictions_2$byClass[11]
, "\n\n", "F1 =", dbF1_11["F1"]
, "\n\n", "F2 =", dbF2_11
, "\n\n", "matthews correlation coefficient =", calculate_mcc1(CM_predictions_2)
, "\n\n", "Threshold =", 0.5

, "\n\n\n", "2 - Youden's J statistic Threshold Method"
, "\n\n", "Accuracy =", CM_predictions_31$overall[1]
, "\n\n", "Sensitivity =", CM_predictions_31$byClass[1]
, "\n\n", "Specificity =", CM_predictions_31$byClass[2]
, "\n\n", "Balanced Accuracy =", CM_predictions_31$byClass[11]
, "\n\n", "F1 =", dbF1_22["F1"]
, "\n\n", "F2 =", dbF2_22
, "\n\n", "matthews correlation coefficient =", calculate_mcc1(CM_predictions_31)
, "\n\n", "Threshold =", rfThresh_youden[1,1]

, "\n\n\n", "3 - Closest Top-left Threshold Method"
, "\n\n", "Accuracy =", CM_predictions_41$overall[1]
, "\n\n", "Sensitivity =", CM_predictions_41$byClass[1]
, "\n\n", "Specificity =", CM_predictions_41$byClass[2]
, "\n\n", "Balanced Accuracy =", CM_predictions_41$byClass[11]
, "\n\n", "F1 =", dbF1_33["F1"]

```

```
, "\n\n", "F2 =", dbF2_33

, "\n\n", "matthews correlation coefficient =", calculate_mcc1(CM_predictions_41)

, "\n\n", "Threshold =", rfThresh_topleft[1,1]

)
```

```
##
##
## TABLE 1 - RESULTS OF THE 'finalTest' DATABASE WITH THE MODELS USING THE train() FUNCTION OF THE car
##
##
## 1 - Basic model
##
## Area Under ROC (AUC)- Basic model = 0.7439586
##
##
## Accuracy = 0.7407407
##
## Sensitivity = 0.9807692
##
## Specificity = 0.3103448
##
## Balanced Accuracy = 0.645557
##
## F1 = 0.8292683
##
## F2 = 0.9139785
##
## matthews correlation coefficient = 0.4242625
##
## Threshold = 0.5
##
##
## 2 - Youden's J statistic Threshold Method
##
## Accuracy = 0.8024691
##
## Sensitivity = 0.9230769
##
## Specificity = 0.5862069
##
## Balanced Accuracy = 0.7546419
##
## F1 = 0.8571429
##
## F2 = 0.8955224
##
## matthews correlation coefficient = 0.557154
##
## Threshold = 0.574
##
```

```
##
## 3 - Closest Top-left Threshold Method
##
## Accuracy = 0.8271605
##
## Sensitivity = 0.9038462
##
## Specificity = 0.6896552
##
## Balanced Accuracy = 0.7967507
##
## F1 = 0.8703704
##
## F2 = 0.8901515
##
## matthews correlation coefficient = 0.6159683
##
## Threshold = 0.594
```

Predicting presence

```
# Identify rows with presence of troglofauna
gaa <- which(finalTest1$true_troglofauna == 1)
geo1 <- finalTest1[gaa,]

# Identify rows with absence of troglofauna
gaaAu <- which(finalTest1$true_troglofauna == 0)
geoAu <- finalTest1[gaaAu,]

# Plot the data
Min_Lg <- min(finalTest1$LONGITUDE) - 0.05
Max_Lg <- max(finalTest1$LONGITUDE) + 0.05
Min_Lt <- min(finalTest1$LATITUDE) - 0.05
Max_Lt <- max(finalTest1$LATITUDE) + 0.05

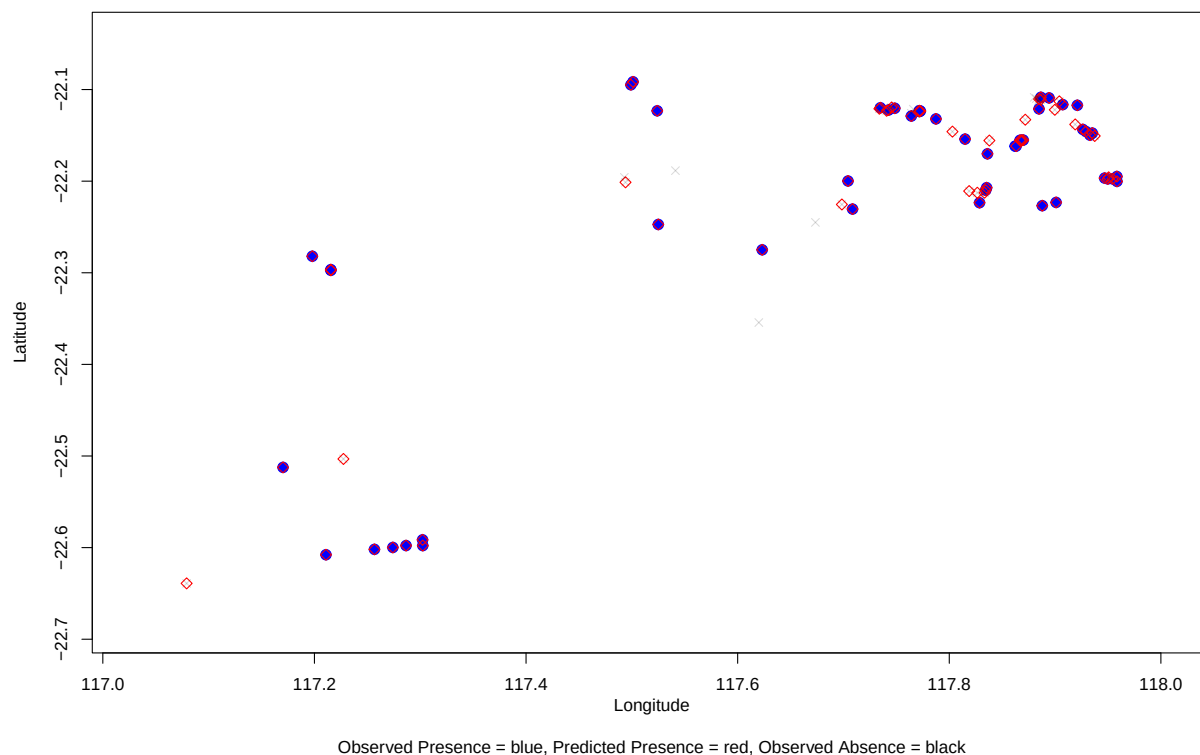
plot(geo1$LONGITUDE, geo1$LATITUDE, xlim = c(Min_Lg, Max_Lg), ylim = c(Min_Lt, Max_Lt),
     cex = 2, pch = 20, col = "blue",
     xlab = 'Longitude\n',
     ylab = 'Latitude',
     main = 'Basic Model. Coordinates of troglofauna presence: Observed and Predicted',
     sub = "\nObserved Presence = blue, Predicted Presence = red, Observed Absence = black")

# Find lines in predictions_3 with presence of troglofauna
gbb <- which(predictions_2 == 1)
geo2 <- finalTest1[gbb,]

# Add predicted presence points to the plot
points(geo2$LONGITUDE, geo2$LATITUDE, pch = 5, col = "red")

# Add absence points to the plot
points(geoAu$LONGITUDE, geoAu$LATITUDE, pch = 4, col = rgb(0, 0, 0, 0.15))
```

Basic Model. Coordinates of troglofauna presence: Observed and Predicted



```
gaa<-which(finalTest1$true_troglofauna==1);gaa
```

```
## [1] 1 2 3 4 6 7 9 10 11 12 16 17 18 19 21 22 23 26 27 29 32 33 34 35 36
## [26] 37 38 40 42 43 45 46 47 50 51 52 53 54 55 57 59 61 65 69 70 71 72 73 76 77
## [51] 78 80
```

```
geol<-finalTest1[gaa,]
summary(geol)
```

```
##      trngcv      bio14      slope      rugg500s
## Min.   :8.808   Min.   :0.01801   Min.   : 0.5627   Min.   :0.2331
## 1st Qu.:9.244   1st Qu.:0.13297   1st Qu.: 2.2721   1st Qu.:1.5072
## Median :9.312   Median :0.17503   Median : 3.4064   Median :2.6873
## Mean   :9.336   Mean   :0.18598   Mean   : 5.3447   Mean   :3.5658
## 3rd Qu.:9.424   3rd Qu.:0.24458   3rd Qu.: 8.0816   3rd Qu.:5.4057
## Max.   :9.889   Max.   :0.34399   Max.   :15.8156   Max.   :9.9756
##
##      pil_twim      pil_twicv      pil_topocv      pil_slps
## Min.   : 5.831   Min.   :0.04324   Min.   :0.00000   Min.   : 0.5535
## 1st Qu.: 7.543   1st Qu.:0.11117   1st Qu.:0.00000   1st Qu.: 1.5083
## Median : 8.304   Median :0.16271   Median :0.01835   Median : 4.3828
## Mean   : 8.594   Mean   :0.16939   Mean   :0.06513   Mean   : 5.1668
## 3rd Qu.: 9.460   3rd Qu.:0.23025   3rd Qu.:0.12912   3rd Qu.: 8.0398
## Max.   :11.582   Max.   :0.30956   Max.   :0.20161   Max.   :12.7276
##
```



```

##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
## Min.      :0.2188    Min.      :0.04501    Min.      :0.0000    Min.      :0.0000
## 1st Qu.:0.3770    1st Qu.:0.12937    1st Qu.:0.0000    1st Qu.:0.0000
## Median :0.4892    Median :0.20425    Median :0.0000    Median :0.0000
## Mean   :0.5491    Mean   :0.23279    Mean   :0.2087    Mean   :0.0733
## 3rd Qu.:0.6604    3rd Qu.:0.31706    3rd Qu.:0.0000    3rd Qu.:0.0000
## Max.   :1.6098    Max.   :0.56449    Max.   :1.8239    Max.   :1.8897
##
##      minfertf      lf7rup      hstructn      geolrngaggn
## Min.      :2.00    Min.      :1.000    Min.      :0.000    Min.      : 0.0115
## 1st Qu.:2.00    1st Qu.:2.000    1st Qu.:1.043    1st Qu.: 2.5880
## Median :2.00    Median :3.000    Median :1.043    Median : 2.5880
## Mean   :2.25    Mean   :2.808    Mean   :1.515    Mean   :175.7587
## 3rd Qu.:2.25    3rd Qu.:4.000    3rd Qu.:1.043    3rd Qu.:300.0000
## Max.   :3.00    Max.   :6.000    Max.   :5.143    Max.   :900.0000
##
##      elevationm      wr_unrn      solpawhcn      slopern
## Min.      :518.3    Min.      : 0.00    Min.      : 96.0    Min.      :0.00179
## 1st Qu.:561.1    1st Qu.:17.86    1st Qu.: 96.0    1st Qu.:0.01874
## Median :570.5    Median :17.86    Median : 96.0    Median :0.04360
## Mean   :585.6    Mean   :15.43    Mean   :101.2    Mean   :0.09140
## 3rd Qu.:611.1    3rd Qu.:17.86    3rd Qu.: 96.0    3rd Qu.:0.09728
## Max.   :703.9    Max.   :17.86    Max.   :157.0    Max.   :0.70049
##
##      MIN_AGE_MA      HubDist      month_collection
## Min.      :2454    Min.      : 3.218    May      : 8
## 1st Qu.:2454    1st Qu.:16.263    September: 8
## Median :2454    Median :26.703    March    : 7
## Mean   :2487    Mean   :43.441    June     : 6
## 3rd Qu.:2494    3rd Qu.:76.671    November : 6
## Max.   :2597    Max.   :86.707    January  : 5
##                                     (Other) :12
##                                     ROCKTYPE1
## sedimentary carbonate      : 0
## sedimentary other chemical or biochemical:41
## sedimentary siliciclastic      :11
##
##
##
##                                     FORMATION
## Brockman Iron Formation      :32
## Marra Mamba Iron Formation    : 9
## Mount McRae Shale and Mount Sylvia Formation:11
## Wittenoom Formation          : 0
##
##
##
##                                     HubName
## anticline, exposed      :23
## exposed      : 6
## strike-slip, exposed, showing relative dextral displacement: 1
## syncline, exposed      :22
##

```

```
##
##
## true_troglofauna    LATITUDE    LONGITUDE
## 0: 0                Min.      :-22.61    Min.      :117.2
## 1:52                1st Qu.: -22.23    1st Qu.: 117.5
##                    Median    :-22.16    Median    :117.8
##                    Mean      :-22.23    Mean      :117.7
##                    3rd Qu.: -22.12    3rd Qu.: 117.9
##                    Max.      :-22.09    Max.      :118.0
##
```

```
gaaAu<-which(finalTest1$true_troglofauna==0);gaaAu
```

```
## [1] 5 8 13 14 15 20 24 25 28 30 31 39 41 44 48 49 56 58 60 62 63 64 66 67 68
## [26] 74 75 79 81
```

```
geoAu<-finalTest1[gaaAu,]
summary(geoAu)
```

```
##      trngcv      bio14      slope      rugg500s
## Min.      :8.894    Min.      :0.02573    Min.      : 0.2666    Min.      :0.0407
## 1st Qu.: 9.244    1st Qu.: 0.11380    1st Qu.: 1.4436    1st Qu.: 1.2771
## Median : 9.337    Median : 0.14993    Median : 3.0893    Median : 2.7764
## Mean      : 9.335    Mean      : 0.16881    Mean      : 4.5309    Mean      : 3.0913
## 3rd Qu.: 9.394    3rd Qu.: 0.20636    3rd Qu.: 6.2448    3rd Qu.: 4.1029
## Max.      : 9.887    Max.      : 0.42288    Max.      :14.3589    Max.      : 8.0468
##
##      pil_twim      pil_twicv      pil_topocv      pil_slps
## Min.      : 5.749    Min.      :0.04954    Min.      :0.00000    Min.      : 0.1344
## 1st Qu.: 7.956    1st Qu.: 0.12372    1st Qu.: 0.00000    1st Qu.: 0.8647
## Median : 9.267    Median : 0.21281    Median : 0.02834    Median : 3.2229
## Mean      : 9.061    Mean      : 0.18747    Mean      : 0.07876    Mean      : 3.5669
## 3rd Qu.:10.169    3rd Qu.: 0.25139    3rd Qu.: 0.14605    3rd Qu.: 5.3619
## Max.      :11.442    Max.      : 0.36698    Max.      : 0.32611    Max.      :10.9676
##
##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
## Min.      :0.1797    Min.      :0.0359    Min.      :0.0000    Min.      :0.0000
## 1st Qu.: 0.3397    1st Qu.: 0.1528    1st Qu.: 0.0000    1st Qu.: 0.0000
## Median : 0.4196    Median : 0.2261    Median : 0.0000    Median : 0.0000
## Mean      : 0.4479    Mean      : 0.2242    Mean      : 0.2452    Mean      : 0.1054
## 3rd Qu.: 0.5034    3rd Qu.: 0.2841    3rd Qu.: 0.0000    3rd Qu.: 0.0000
## Max.      : 0.7925    Max.      : 0.4060    Max.      : 2.8219    Max.      : 0.6614
##
##      minfertf      lf7rup      hstructn      geolrngaggn
## Min.      :2.000    Min.      :2.000    Min.      : 0.000    Min.      : 0.0115
## 1st Qu.: 2.000    1st Qu.: 3.000    1st Qu.: 1.043    1st Qu.: 2.5880
## Median : 2.000    Median : 3.000    Median : 1.043    Median : 2.5880
## Mean      : 2.207    Mean      : 3.621    Mean      : 1.572    Mean      :147.1018
## 3rd Qu.: 2.000    3rd Qu.: 4.000    3rd Qu.: 1.043    3rd Qu.: 2.5880
## Max.      : 3.000    Max.      : 7.000    Max.      :15.429    Max.      :900.0000
##
##      elevationm      wr_unrn      solpawhcn      slopern
## Min.      :515.2    Min.      : 0.00    Min.      : 96.0    Min.      :0.00341
```

```

## 1st Qu.:536.0 1st Qu.:17.86 1st Qu.: 96.0 1st Qu.:0.02670
## Median :568.4 Median :17.86 Median : 96.0 Median :0.03745
## Mean :577.2 Mean :15.41 Mean :103.7 Mean :0.52368
## 3rd Qu.:601.6 3rd Qu.:17.86 3rd Qu.: 96.0 3rd Qu.:0.12267
## Max. :726.1 Max. :17.86 Max. :157.0 Max. :4.28009
##
## MIN_AGE_MA HubDist month_collection
## Min. :2454 Min. : 1.931 December :8
## 1st Qu.:2454 1st Qu.:17.066 January :6
## Median :2454 Median :21.429 September:5
## Mean :2472 Mean :34.474 April :4
## 3rd Qu.:2494 3rd Qu.:54.509 March :2
## Max. :2597 Max. :88.221 February :1
## (Other) :3
## ROCKTYPE1
## sedimentary carbonate : 1
## sedimentary other chemical or biochemical:20
## sedimentary siliciclastic : 8
##
##
##
## FORMATION
## Brockman Iron Formation :19
## Marra Mamba Iron Formation : 1
## Mount McRae Shale and Mount Sylvia Formation: 8
## Wittenoom Formation : 1
##
##
##
## HubName
## anticline, exposed :20
## exposed : 0
## strike-slip, exposed, showing relative dextral displacement: 1
## syncline, exposed : 8
##
##
## true_troglofauna LATITUDE LONGITUDE
## 0:29 Min. :-22.64 Min. :117.1
## 1: 0 1st Qu.: -22.21 1st Qu.:117.7
## Median :-22.15 Median :117.8
## Mean :-22.19 Mean :117.7
## 3rd Qu.: -22.12 3rd Qu.:117.9
## Max. :-22.11 Max. :118.0
##

```

```

Min_Lg<-min(finalTest1$LONGITUDE)-0.05
Max_Lg<-max(finalTest1$LONGITUDE)+0.05

```

```

Min_Lt<-min(finalTest1$LATITUDE)-0.05
Max_Lt<-max(finalTest1$LATITUDE)+0.05

```

```
plot(geo1$LONGITUDE,geo1$LATITUDE,xlim = c(Min_Lg,Max_Lg),ylim = c(Min_Lt,Max_Lt), cex=2, pch=20, col="black",
     sub = "\nObserved Presence = blue, Predicted Presence = red, Observed Absence = black")
```

```
gbb<-which(predictions_2==1);gbb
```

```
## [1] 1 2 3 4 6 7 8 10 11 12 13 15 16 17 18 19 20 21 22 23 25 26 27 28 29
## [26] 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 49 50 51 52 53 54 55 56
## [51] 57 58 59 60 61 65 66 67 68 69 70 71 72 73 75 76 77 78 79 80 81
```

```
geo2<-finalTest1[gbb,]
summary(geo2)
```

```
##      trngcv      bio14      slope      rugg500s
## Min.   :8.808   Min.   :0.01801   Min.   : 0.2666   Min.   :0.1508
## 1st Qu.:9.243   1st Qu.:0.11960   1st Qu.: 2.1535   1st Qu.:1.5066
## Median :9.325   Median :0.17200   Median : 3.3626   Median :2.7764
## Mean   :9.344   Mean   :0.18270   Mean   : 5.1586   Mean   :3.4517
## 3rd Qu.:9.417   3rd Qu.:0.23916   3rd Qu.: 7.5055   3rd Qu.:5.2003
## Max.   :9.889   Max.   :0.42288   Max.   :15.8156   Max.   :9.9756
##
##      pil_twim      pil_twicv      pil_topocv      pil_slps
## Min.   : 5.749   Min.   :0.04324   Min.   :0.00000   Min.   : 0.4874
## 1st Qu.: 7.693   1st Qu.:0.11822   1st Qu.:0.00000   1st Qu.: 1.4048
## Median : 8.453   Median :0.16354   Median :0.02168   Median : 3.9729
## Mean   : 8.645   Mean   :0.17513   Mean   :0.06882   Mean   : 4.7409
## 3rd Qu.: 9.662   3rd Qu.:0.23356   3rd Qu.:0.13549   3rd Qu.: 7.0271
## Max.   :11.582   Max.   :0.36698   Max.   :0.32611   Max.   :12.7276
##
##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
## Min.   :0.2188   Min.   :0.0359   Min.   :0.000   Min.   :0.00000
## 1st Qu.:0.3624   1st Qu.:0.1301   1st Qu.:0.000   1st Qu.:0.00000
## Median :0.4713   Median :0.2056   Median :0.000   Median :0.00000
## Mean   :0.5201   Mean   :0.2294   Mean   :0.204   Mean   :0.07929
## 3rd Qu.:0.6220   3rd Qu.:0.3147   3rd Qu.:0.000   3rd Qu.:0.00000
## Max.   :1.6098   Max.   :0.5645   Max.   :2.806   Max.   :1.88966
##
##      minfertf      lf7rup      hstructn      geolrngaggn
## Min.   :2.000   Min.   :1.000   Min.   :0.000   Min.   : 0.0115
## 1st Qu.:2.000   1st Qu.:2.000   1st Qu.:1.043   1st Qu.: 2.5880
## Median :2.000   Median :3.000   Median :1.043   Median : 2.5880
## Mean   :2.239   Mean   :2.986   Mean   :1.431   Mean   :175.8770
## 3rd Qu.:2.000   3rd Qu.:4.000   3rd Qu.:1.043   3rd Qu.:300.0000
## Max.   :3.000   Max.   :7.000   Max.   :5.143   Max.   :900.0000
##
##      elevationm      wr_unrn      solpawhcn      slopern
## Min.   :515.2   Min.   : 0.00   Min.   : 96.0   Min.   :0.00179
## 1st Qu.:560.1   1st Qu.:17.86   1st Qu.: 96.0   1st Qu.:0.01985
## Median :570.5   Median :17.86   Median : 96.0   Median :0.04257
## Mean   :584.8   Mean   :15.71   Mean   :100.7   Mean   :0.15044
## 3rd Qu.:613.0   3rd Qu.:17.86   3rd Qu.: 96.0   3rd Qu.:0.09939
## Max.   :726.1   Max.   :17.86   Max.   :157.0   Max.   :4.24974
```

```

##
##      MIN_AGE_MA      HubDist      month_collection
##  Min.      :2454      Min.      : 2.183      September:10
##  1st Qu.:2454      1st Qu.:16.549      December :10
##  Median :2454      Median :22.459      January  : 9
##  Mean   :2482      Mean   :42.584      March    : 9
##  3rd Qu.:2494      3rd Qu.:76.502      May      : 9
##  Max.    :2597      Max.    :88.221      June     : 6
##                                     (Other) :18
##                                     ROCKTYPE1
##  sedimentary carbonate                : 1
##  sedimentary other chemical or biochemical:54
##  sedimentary siliciclastic            :16
##
##
##
##                                     FORMATION
##  Brockman Iron Formation                :45
##  Marra Mamba Iron Formation            : 9
##  Mount McRae Shale and Mount Sylvia Formation:16
##  Wittenoom Formation                   : 1
##
##
##
##                                     HubName
##  anticline, exposed                    :34
##  exposed                              : 6
##  strike-slip, exposed, showing relative dextral displacement: 2
##  syncline, exposed                    :29
##
##
##
##  true_troglofauna      LATITUDE      LONGITUDE
##  0:20                  Min.      :-22.64      Min.      :117.1
##  1:51                  1st Qu.: -22.22      1st Qu.:117.7
##                      Median : -22.16      Median :117.8
##                      Mean   : -22.22      Mean   :117.7
##                      3rd Qu.: -22.12      3rd Qu.:117.9
##                      Max.    : -22.09      Max.    :118.0
##
##

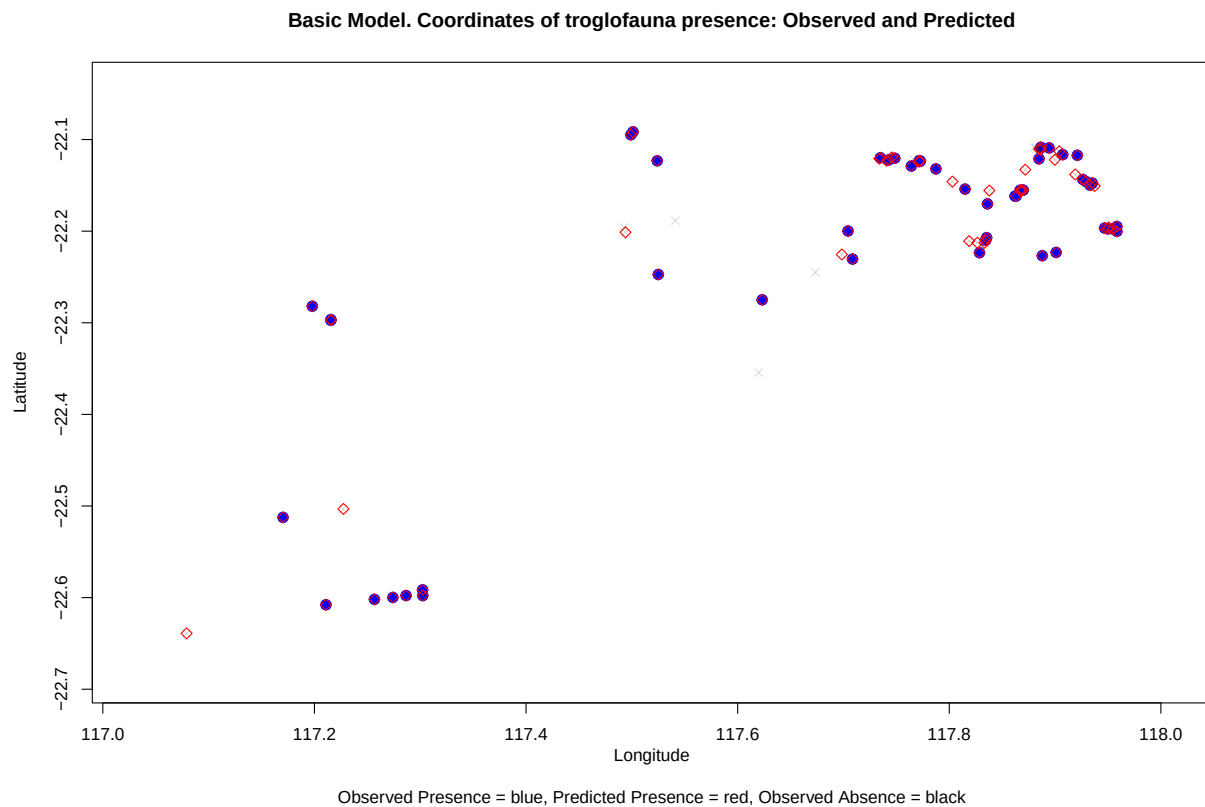
```

```

points(geo2$LONGITUDE,geo2$LATITUDE,xlim = c(Min_Lg,Max_Lg),ylim = c(Min_Lt,Max_Lt),pch=5, col="red")

points(geoAu$LONGITUDE,geoAu$LATITUDE,xlim = c(115,122),ylim = c(-26,-21),pch = 4, col = rgb(0, 0, 0, 0

```



Closest Top-left Threshold Method

```
gaa<-which(finalTest1$true_troglofauna==1);gaa
```

```
## [1] 1 2 3 4 6 7 9 10 11 12 16 17 18 19 21 22 23 26 27 29 32 33 34 35 36
## [26] 37 38 40 42 43 45 46 47 50 51 52 53 54 55 57 59 61 65 69 70 71 72 73 76 77
## [51] 78 80
```

```
geol<-finalTest1[gaa,]
summary(geol)
```

```
##      trngcv      bio14      slope      rugg500s
## Min.   :8.808   Min.   :0.01801   Min.   : 0.5627   Min.   :0.2331
## 1st Qu.:9.244   1st Qu.:0.13297   1st Qu.: 2.2721   1st Qu.:1.5072
## Median :9.312   Median :0.17503   Median : 3.4064   Median :2.6873
## Mean   :9.336   Mean   :0.18598   Mean   : 5.3447   Mean   :3.5658
## 3rd Qu.:9.424   3rd Qu.:0.24458   3rd Qu.: 8.0816   3rd Qu.:5.4057
## Max.   :9.889   Max.   :0.34399   Max.   :15.8156   Max.   :9.9756
##
##      pil_twim      pil_twicv      pil_topocv      pil_slps
## Min.   : 5.831   Min.   :0.04324   Min.   :0.00000   Min.   : 0.5535
## 1st Qu.: 7.543   1st Qu.:0.11117   1st Qu.:0.00000   1st Qu.: 1.5083
## Median : 8.304   Median :0.16271   Median :0.01835   Median : 4.3828
## Mean   : 8.594   Mean   :0.16939   Mean   :0.06513   Mean   : 5.1668
```

```

## 3rd Qu.: 9.460    3rd Qu.:0.23025    3rd Qu.:0.12912    3rd Qu.: 8.0398
## Max.    :11.582    Max.    :0.30956    Max.    :0.20161    Max.    :12.7276
##
##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
## Min.    :0.2188    Min.    :0.04501    Min.    :0.0000    Min.    :0.0000
## 1st Qu.:0.3770    1st Qu.:0.12937    1st Qu.:0.0000    1st Qu.:0.0000
## Median :0.4892    Median :0.20425    Median :0.0000    Median :0.0000
## Mean    :0.5491    Mean    :0.23279    Mean    :0.2087    Mean    :0.0733
## 3rd Qu.:0.6604    3rd Qu.:0.31706    3rd Qu.:0.0000    3rd Qu.:0.0000
## Max.    :1.6098    Max.    :0.56449    Max.    :1.8239    Max.    :1.8897
##
##      minfertf      lf7rup      hstructn      geolrngaggn
## Min.    :2.00      Min.    :1.000    Min.    :0.000    Min.    : 0.0115
## 1st Qu.:2.00      1st Qu.:2.000    1st Qu.:1.043    1st Qu.: 2.5880
## Median :2.00      Median :3.000    Median :1.043    Median : 2.5880
## Mean    :2.25      Mean    :2.808    Mean    :1.515    Mean    :175.7587
## 3rd Qu.:2.25      3rd Qu.:4.000    3rd Qu.:1.043    3rd Qu.:300.0000
## Max.    :3.00      Max.    :6.000    Max.    :5.143    Max.    :900.0000
##
##      elevationm      wr_unrn      solpawhcn      slopern
## Min.    :518.3      Min.    : 0.00      Min.    : 96.0      Min.    :0.00179
## 1st Qu.:561.1      1st Qu.:17.86      1st Qu.: 96.0      1st Qu.:0.01874
## Median :570.5      Median :17.86      Median : 96.0      Median :0.04360
## Mean    :585.6      Mean    :15.43      Mean    :101.2      Mean    :0.09140
## 3rd Qu.:611.1      3rd Qu.:17.86      3rd Qu.: 96.0      3rd Qu.:0.09728
## Max.    :703.9      Max.    :17.86      Max.    :157.0      Max.    :0.70049
##
##      MIN_AGE_MA      HubDist      month_collection
## Min.    :2454      Min.    : 3.218      May      : 8
## 1st Qu.:2454      1st Qu.:16.263      September: 8
## Median :2454      Median :26.703      March    : 7
## Mean    :2487      Mean    :43.441      June     : 6
## 3rd Qu.:2494      3rd Qu.:76.671      November : 6
## Max.    :2597      Max.    :86.707      January  : 5
##                                     (Other) :12
##                                     ROCKTYPE1
## sedimentary carbonate      : 0
## sedimentary other chemical or biochemical:41
## sedimentary siliciclastic      :11
##
##
##
##                                     FORMATION
## Brockman Iron Formation      :32
## Marra Mamba Iron Formation    : 9
## Mount McRae Shale and Mount Sylvia Formation:11
## Wittenoom Formation          : 0
##
##
##
##                                     HubName
## anticline, exposed          :23
## exposed                      : 6

```

```
## strike-slip, exposed, showing relative dextral displacement: 1
## syncline, exposed :22
##
##
##
## true_troglofauna LATITUDE LONGITUDE
## 0: 0 Min. :-22.61 Min. :117.2
## 1:52 1st Qu.: -22.23 1st Qu.:117.5
## Median :-22.16 Median :117.8
## Mean :-22.23 Mean :117.7
## 3rd Qu.: -22.12 3rd Qu.:117.9
## Max. :-22.09 Max. :118.0
##
```

```
gaaAu<-which(finalTest1$true_troglofauna==0);gaaAu
```

```
## [1] 5 8 13 14 15 20 24 25 28 30 31 39 41 44 48 49 56 58 60 62 63 64 66 67 68
## [26] 74 75 79 81
```

```
geoAu<-finalTest1[gaaAu,]
summary(geoAu)
```

```
##      trngcv      bio14      slope      rugg500s
## Min.   :8.894   Min.   :0.02573   Min.   : 0.2666   Min.   :0.0407
## 1st Qu.:9.244   1st Qu.:0.11380   1st Qu.: 1.4436   1st Qu.:1.2771
## Median :9.337   Median :0.14993   Median : 3.0893   Median :2.7764
## Mean   :9.335   Mean   :0.16881   Mean   : 4.5309   Mean   :3.0913
## 3rd Qu.:9.394   3rd Qu.:0.20636   3rd Qu.: 6.2448   3rd Qu.:4.1029
## Max.   :9.887   Max.   :0.42288   Max.   :14.3589   Max.   :8.0468
##
##      pil_twim      pil_twicv      pil_topocv      pil_slps
## Min.   : 5.749   Min.   :0.04954   Min.   :0.00000   Min.   : 0.1344
## 1st Qu.: 7.956   1st Qu.:0.12372   1st Qu.:0.00000   1st Qu.: 0.8647
## Median : 9.267   Median :0.21281   Median :0.02834   Median : 3.2229
## Mean   : 9.061   Mean   :0.18747   Mean   :0.07876   Mean   : 3.5669
## 3rd Qu.:10.169   3rd Qu.:0.25139   3rd Qu.:0.14605   3rd Qu.: 5.3619
## Max.   :11.442   Max.   :0.36698   Max.   :0.32611   Max.   :10.9676
##
##      pil_slpcv      pil_elr3cv      mrvbf      mrrtf
## Min.   :0.1797   Min.   :0.0359   Min.   :0.0000   Min.   :0.0000
## 1st Qu.:0.3397   1st Qu.:0.1528   1st Qu.:0.0000   1st Qu.:0.0000
## Median :0.4196   Median :0.2261   Median :0.0000   Median :0.0000
## Mean   :0.4479   Mean   :0.2242   Mean   :0.2452   Mean   :0.1054
## 3rd Qu.:0.5034   3rd Qu.:0.2841   3rd Qu.:0.0000   3rd Qu.:0.0000
## Max.   :0.7925   Max.   :0.4060   Max.   :2.8219   Max.   :0.6614
##
##      minfertf      lf7rup      hstructn      geolrngaggn
## Min.   :2.000   Min.   :2.000   Min.   : 0.000   Min.   : 0.0115
## 1st Qu.:2.000   1st Qu.:3.000   1st Qu.: 1.043   1st Qu.: 2.5880
## Median :2.000   Median :3.000   Median : 1.043   Median : 2.5880
## Mean   :2.207   Mean   :3.621   Mean   : 1.572   Mean   :147.1018
## 3rd Qu.:2.000   3rd Qu.:4.000   3rd Qu.: 1.043   3rd Qu.: 2.5880
## Max.   :3.000   Max.   :7.000   Max.   :15.429   Max.   :900.0000
```



```

##
##      elevationm      wr_unrn      solpawhcn      slopern
## Min.      :515.2    Min.      : 0.00    Min.      : 96.0    Min.      :0.00341
## 1st Qu.:536.0    1st Qu.:17.86    1st Qu.: 96.0    1st Qu.:0.02670
## Median :568.4    Median :17.86    Median : 96.0    Median :0.03745
## Mean      :577.2    Mean      :15.41    Mean      :103.7    Mean      :0.52368
## 3rd Qu.:601.6    3rd Qu.:17.86    3rd Qu.: 96.0    3rd Qu.:0.12267
## Max.      :726.1    Max.      :17.86    Max.      :157.0    Max.      :4.28009
##
##      MIN_AGE_MA      HubDist      month_collection
## Min.      :2454      Min.      : 1.931    December :8
## 1st Qu.:2454      1st Qu.:17.066    January  :6
## Median :2454      Median :21.429    September:5
## Mean      :2472      Mean      :34.474    April     :4
## 3rd Qu.:2494      3rd Qu.:54.509    March     :2
## Max.      :2597      Max.      :88.221    February  :1
##                                     (Other)   :3
##                                     ROCKTYPE1
## sedimentary carbonate                : 1
## sedimentary other chemical or biochemical:20
## sedimentary siliciclastic            : 8
##
##
##                                     FORMATION
## Brockman Iron Formation                :19
## Marra Mamba Iron Formation             : 1
## Mount McRae Shale and Mount Sylvia Formation: 8
## Wittenoom Formation                   : 1
##
##
##                                     HubName
## anticline, exposed                    :20
## exposed                               : 0
## strike-slip, exposed, showing relative dextral displacement: 1
## syncline, exposed                     : 8
##
##
## true_troglofauna      LATITUDE      LONGITUDE
## 0:29                  Min.      :-22.64    Min.      :117.1
## 1: 0                  1st Qu.: -22.21    1st Qu.:117.7
##                      Median : -22.15    Median :117.8
##                      Mean      :-22.19    Mean      :117.7
##                      3rd Qu.: -22.12    3rd Qu.:117.9
##                      Max.      :-22.11    Max.      :118.0
##

```

```

plot(geo1$LONGITUDE,geo1$LATITUDE,xlim = c(Min_Lg,Max_Lg),ylim = c(Min_Lt,Max_Lt), cex=2, pch=20, col="l",
      sub = "\nObserved Presence = blue, Predicted Presence = red, Observed Absence = black")

```

```
gbb<-which(Test_41==1);gbb
```

```
## [1] 1 2 3 4 6 7 11 12 15 16 17 18 19 21 23 25 26 27 28 32 33 34 35 36 37
## [26] 38 40 42 43 45 47 49 50 51 52 53 54 55 57 58 59 61 65 66 69 70 71 72 73 75
## [51] 76 77 78 79 80 81
```

```
geo2<-finalTest1[gbb,]
summary(geo2)
```

```
##          trngcv          bio14          slope          rugg500s
## Min.      :8.808   Min.      :0.01801   Min.      : 0.2666   Min.      :0.1508
## 1st Qu.:9.244   1st Qu.:0.12940   1st Qu.: 2.3069   1st Qu.:1.5008
## Median :9.323   Median :0.17818   Median : 3.4572   Median :2.8021
## Mean      :9.347   Mean      :0.18819   Mean      : 5.2976   Mean      :3.5312
## 3rd Qu.:9.424   3rd Qu.:0.25360   3rd Qu.: 8.1336   3rd Qu.:5.3938
## Max.      :9.889   Max.      :0.42288   Max.      :15.8156   Max.      :9.9756
##
##          pil_twim          pil_twicv          pil_topocv          pil_slps
## Min.      : 5.749   Min.      :0.04324   Min.      :0.00000   Min.      : 0.6456
## 1st Qu.: 7.646   1st Qu.:0.11117   1st Qu.:0.00000   1st Qu.: 1.5083
## Median : 8.304   Median :0.16351   Median :0.01835   Median : 3.8509
## Mean      : 8.569   Mean      :0.17155   Mean      :0.06325   Mean      : 4.9431
## 3rd Qu.: 9.460   3rd Qu.:0.23075   3rd Qu.:0.12472   3rd Qu.: 7.4627
## Max.     :11.582   Max.      :0.30956   Max.      :0.20161   Max.     :12.7276
##
##          pil_slpcv          pil_elr3cv          mrvbf          mrrtf
## Min.      :0.2188   Min.      :0.04501   Min.      :0.000   Min.      :0.00000
## 1st Qu.:0.3720   1st Qu.:0.12937   1st Qu.:0.000   1st Qu.:0.00000
## Median :0.4791   Median :0.19643   Median :0.000   Median :0.00000
## Mean      :0.5324   Mean      :0.22579   Mean      :0.201   Mean      :0.06806
## 3rd Qu.:0.6313   3rd Qu.:0.29446   3rd Qu.:0.000   3rd Qu.:0.00000
## Max.      :1.6098   Max.      :0.56449   Max.      :2.806   Max.      :1.88966
##
##          minfertf          lf7rup          hstructn          geolrngaggn
## Min.      :2.000   Min.      :1.000   Min.      :0.000   Min.      : 0.0115
## 1st Qu.:2.000   1st Qu.:2.000   1st Qu.:1.043   1st Qu.: 2.5880
## Median :2.000   Median :3.000   Median :1.043   Median : 2.5880
## Mean      :2.232   Mean      :2.804   Mean      :1.591   Mean      :169.0912
## 3rd Qu.:2.000   3rd Qu.:4.000   3rd Qu.:1.043   3rd Qu.:300.0000
## Max.      :3.000   Max.      :7.000   Max.      :5.143   Max.      :900.0000
##
##          elevationm          wr_unrn          solpawhcn          slopern
## Min.      :518.3   Min.      : 0.00   Min.      : 96.00   Min.      :0.00179
## 1st Qu.:561.2   1st Qu.:17.86   1st Qu.: 96.00   1st Qu.:0.01965
## Median :569.4   Median :17.86   Median : 96.00   Median :0.04045
## Mean      :586.4   Mean      :16.10   Mean      : 98.75   Mean      :0.07946
## 3rd Qu.:605.6   3rd Qu.:17.86   3rd Qu.: 96.00   3rd Qu.:0.09517
## Max.      :726.1   Max.      :17.86   Max.      :157.00   Max.      :0.51337
##
##          MIN_AGE_MA          HubDist          month_collection
## Min.      :2454   Min.      : 3.218   May      : 9
## 1st Qu.:2454   1st Qu.:16.443   March    : 8
```

```

## Median :2454    Median :28.650    September: 7
## Mean   :2488    Mean   :44.277    June      : 6
## 3rd Qu.:2494    3rd Qu.:76.490    November : 6
## Max.   :2597    Max.   :84.554    January  : 5
##                                     (Other)  :15
##                                     ROCKTYPE1
## sedimentary carbonate                : 1
## sedimentary other chemical or biochemical:41
## sedimentary siliciclastic            :14
##
##
##
##                                     FORMATION
## Brockman Iron Formation                :32
## Marra Mamba Iron Formation            : 9
## Mount McRae Shale and Mount Sylvia Formation:14
## Wittenoom Formation                   : 1
##
##
##
##                                     HubName
## anticline, exposed                    :23
## exposed                              : 6
## strike-slip, exposed, showing relative dextral displacement: 2
## syncline, exposed                     :25
##
##
##
## true_troglofauna    LATITUDE    LONGITUDE
## 0: 9                Min.    :-22.64    Min.    :117.1
## 1:47                1st Qu.:-22.23    1st Qu.:117.6
##                    Median   :-22.17    Median  :117.8
##                    Mean     :-22.24    Mean    :117.7
##                    3rd Qu.:-22.13    3rd Qu.:117.9
##                    Max.     :-22.09    Max.    :118.0
##
##

```

```

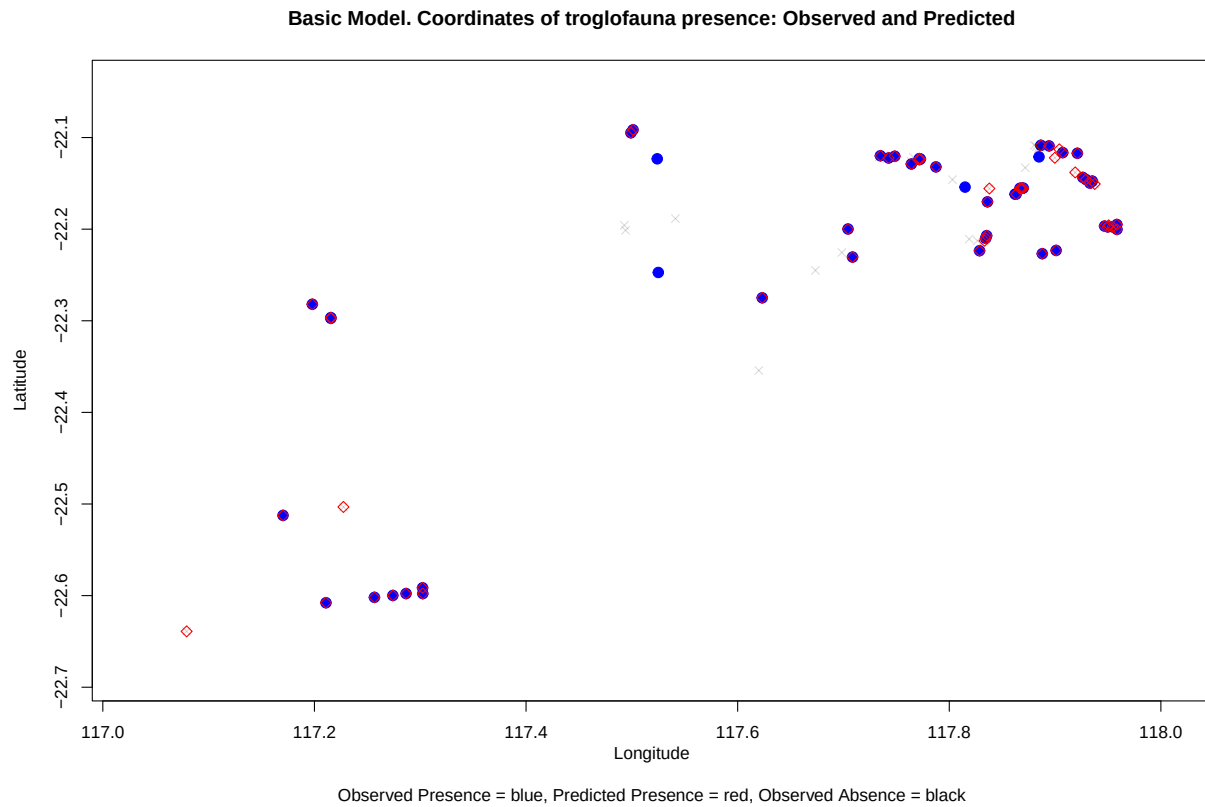
points(geo2$LONGITUDE,geo2$LATITUDE,xlim = c(Min_Lg,Max_Lg),ylim = c(Min_Lt,Max_Lt),pch=5, col="red")

```

```

points(geoAu$LONGITUDE,geoAu$LATITUDE,xlim = c(115,122),ylim = c(-26,-21),pch = 4, col = rgb(0, 0, 0, 0

```



```
write.csv(geo2, file = "Area1__predicted_presence_points_troglofauna_coordinates__.csv", row.names = F)
```